

内分泌2026文献翻译

制作：云梦山

文献汇编目录

TABLE OF CONTENTS

1型糖尿病

1型糖尿病 (type 1 diabetes)

Insulin Resistance in Type 1 Diabetes: Pathophysiological, Clinical, and Therapeutic Relevance.
1型糖尿病中的胰岛素抵抗：病理生理、临床及治疗相关性。

New advances in type 1 diabetes.
1型糖尿病的新进展

成人隐匿性自身免疫性糖尿病 (latent autoimmune diabetes in adults)

Searching for LADA in patients with Hashimoto's thyroiditis: A tool for the prevention of LADA.
在桥本甲状腺炎患者中筛查LADA：预防LADA的一种工具

2型糖尿病

2型糖尿病 (type 2 diabetes)

Exploring the Gut Microbiota: Key Insights Into Its Role in Obesity, Metabolic Syndrome, and T...
探索肠道微生物群：关于其在肥胖、代谢综合征和2型糖尿病中作用的关键见解

青少年发病的成年型糖尿病 (maturity onset diabetes of the young)

Maturity onset diabetes of the young and beyond: the changing face of single-gene diabetes.
青少年起病的成人型糖尿病及其扩展：单基因糖尿病的演变面貌。

Prevalence and Clinical Characteristics of NEUROD1-MODY in Chinese Early-Onset Type 2 D...
中国早发型2型糖尿病中NEUROD1-MODY的患病率与临床特征及文献综述

Phenotypic features, prevalence of KCNJ11-MODY in Chinese patients with early-onset diabet...
中国早发糖尿病患者中KCNJ11-MODY的表型特征、患病率及文献综述。

下丘脑-垂体疾病

垂体腺瘤 (pituitary adenoma)

Revised European Society of Endocrinology Clinical Practice Guideline for the management of...
修订版欧洲内分泌学会临床实践指南：侵袭性垂体肿瘤与垂体癌的管理

泌乳素瘤 (prolactinoma)

Advances in Drug Treatments for Male Patients with Prolactinomas.
男性泌乳素瘤患者药物治疗进展

空泡蝶鞍综合征 (empty sella syndrome)

A Comprehensive Review of Empty Sella and Empty Sella Syndrome.
空蝶鞍与空蝶鞍综合征的综合综述

肢端肥大症 (acromegaly)

The socioeconomic burden of acromegaly.
肢端肥大症的社会经济负担。

内分泌功能试验

地塞米松抑制试验 (dexamethasone suppression test)

An Overlooked Disease: Minimal Autonomous Cortisol Secretion (MACS). A Narrative Review.
被忽视的疾病：微小自主性皮质醇分泌（MACS）。一篇叙述性综述。

Current challenges in Cushing's syndrome testing: blood, saliva, urine, or hair?
库欣综合征检测的当前挑战：血液、唾液、尿液还是头发？

嘌呤代谢疾病

高尿酸血症 (hyperuricemia)

Hyperuricemia and Gout Reduction by SGLT2 Inhibitors in Diabetes and Heart Failure: JACC ...
SGLT2抑制剂在糖尿病和心力衰竭中降低高尿酸血症和痛风的作用：JACC本周评论主题

性腺疾病

Klinefelter综合征 (klinefelter syndrome)

Update on Physical, Psychological, and Quality of Life Management in Klinefelter Syndrome.
克氏综合征患者身体、心理及生活质量管理的最新进展

Turner综合征 (turner syndrome)

Navigating fertility dilemmas across the lifespan in girls with Turner syndrome-a scoping review.
特纳综合征女孩一生中面临的生育困境——一项范围综述。

多囊卵巢综合征 (polycystic ovary syndrome)

Lifestyle management in polycystic ovary syndrome - beyond diet and physical activity.

多囊卵巢综合征的生活方式管理——超越饮食与体育锻炼

性早熟 (precocious puberty)

Precocious Puberty in Hypothyroidism: Mini-Review of Van Wyk-Grumbach Syndrome.

甲状腺功能减退症中的性早熟：Van Wyk-Grumbach综合征的迷你综述

男性性腺功能减退 (male hypogonadism)

Plasma metabolomics in male primary and functional hypogonadism.

男性原发性和功能性性腺功能减退症的血浆代谢组学研究

青春期延迟 (delayed puberty)

Puberty and Growth in Cystic Fibrosis: A Review.

囊性纤维化中的青春期与生长：综述。

甲状旁腺疾病

假性甲状旁腺功能减退症 (pseudohypoparathyroidism)

A Comprehensive Review of Syndromic Forms of Obesity: Genetic Etiology, Clinical Features ...

综合征性肥胖的综合综述：遗传病因学、临床特征与分子诊断

原发性甲状旁腺功能亢进症 (primary hyperparathyroidism)

Advances in Parathyroid Hormone-based medicines.

基于甲状旁腺激素的药物的进展。

甲状旁腺功能减退症 (hypoparathyroidism)

Revised European Society of Endocrinology Clinical Practice Guideline: Treatment of Chronic ...

修订版欧洲内分泌学会临床实践指南：成人慢性甲状旁腺功能减退症的治疗

Endocrine features of primary mitochondrial diseases.

原发性线粒体疾病的内分泌特征。

Hypoparathyroidism and assisted reproductive technology: considerations while undergoing o...

甲状旁腺功能减退与辅助生殖技术：卵巢过度刺激过程中的注意事项

甲状腺疾病

Graves病 (graves disease)

Epidemiologic changes in thyroid disease.
甲状腺疾病的流行病学变化。

Management Aspects of Medical Therapy in Graves Disease.
格雷夫斯病药物治疗的管理方面

The genetics of Graves' disease.
格雷夫斯病的遗传学

甲状腺功能亢进 (hyperthyroidism)

Considerations in the Diagnosis and Management of Thyroid Dysfunction in Older Adults.
老年人甲状腺功能障碍诊断与管理的考量因素

Thyroid dysfunction and female infertility. A comprehensive review.
甲状腺功能障碍与女性不孕症：一项全面综述

甲状腺功能减退 (hypothyroidism)

Hypothyroidism in Older Adults: A Narrative Review.
老年人甲状腺功能减退症：叙述性综述。

甲状腺癌 (thyroid cancer)

Reimagining the Therapeutic Approach for Anaplastic Thyroid Cancer: The Roadmap to a Cure.
重新构想间变性甲状腺癌的治疗方法：治愈之路

Thyroid cancer and insulin resistance.
甲状腺癌与胰岛素抵抗。

Dermatologic manifestations of thyroid disease: a literature review.
甲状腺疾病的皮肤表现：文献综述。

Progress in Thyroid Cancer Genomics: A 40-Year Journey.
甲状腺癌基因组学进展：40年历程

甲状腺相关性眼病 (thyroid associated ophthalmopathy)

Sirolimus for the treatment of Graves' orbitopathy.
西罗莫司治疗格雷夫斯眼病

Advances in artificial intelligence in thyroid-associated ophthalmopathy.
人工智能在甲状腺相关眼病中的进展。

Role of genetics and epigenetics in Graves' orbitopathy.
遗传学与表观遗传学在格雷夫斯眼病中的作用。

甲状腺结节 (thyroid nodule)

2023 European Thyroid Association Clinical Practice Guidelines for thyroid nodule management.
2023年欧洲甲状腺协会甲状腺结节管理临床实践指南

Thyroid nodules: Global, economic, and personal burdens.
甲状腺结节：全球、经济和个人负担。

Artificial Intelligence in Thyroidology: A Narrative Review of the Current Applications, Associate...
甲状腺学中的人工智能：当前应用、相关挑战及未来方向的叙述性综述

神经内分泌肿瘤

胰岛素瘤 (insulinoma)

Endogenous hyperinsulinemic hypoglycemia: case series and literature review.
内源性高胰岛素血症性低血糖：病例系列与文献综述。

胰高血糖素瘤 (glucagonoma)

Glucagonoma and Glucagonoma Syndrome: An Updated Review.
胰高血糖素瘤与胰高血糖素瘤综合征：最新综述。

Practical therapeutic approach in the management of diabetes mellitus secondary to Cushing's...
库欣综合征、肢端肥大症及神经内分泌肿瘤继发糖尿病的实用治疗策略

神经内分泌肿瘤前沿

PRRT (prrt)

Nuclear medicine and perspective thoughts in the diagnosis and treatment of pheochromocyto...
嗜铬细胞瘤和副神经节瘤诊断与治疗中的核医学及前瞻性思考。

糖尿病前沿

SGLT2抑制剂 (sglt2 inhibitor)

A review of the safety of sodium-glucose co-transporter-2 inhibitors.
钠-葡萄糖共转运蛋白2抑制剂的安全性综述。

Treatment recommendations for glycogen storage disease type IB- associated neutropenia an...
恩格列净治疗糖原贮积症IB型相关中性粒细胞减少症和中性粒细胞功能障碍的建议：国际研讨会共识。

口服司美格鲁肽 (oral semaglutide)

Rewriting Diabetes Therapy: How Incretin Modulation is Transforming Cardiovascular and Ren...
改写糖尿病治疗：肠促胰岛素调节如何改变心血管和肾脏结局

Semaglutide and obesity: beyond the nutritional and lifestyle intervention?
司美格鲁肽与肥胖：超越营养与生活方式干预？

替尔泊肽 (tirzepatide)

Strategies for minimizing muscle loss during use of incretin-mimetic drugs for treatment of obe...
使用肠促胰岛素类似物治疗肥胖期间减少肌肉流失的策略

Incretin-Based Weight Loss Pharmacotherapy: Can Resistance Exercise Optimize Changes in...
基于肠促胰岛素的减重药物治疗：抗阻运动能否优化身体成分变化？

胰岛素周制剂 (once weekly insulin)

Emerging perspectives on once-weekly insulins in type 1 and type 2 diabetes: a mini-review.
关于每周一次胰岛素在1型和2型糖尿病中的新兴观点：小型综述。

Once-weekly Insulin Icodec Versus Once-daily Long-acting Insulin for Type II Diabetes: A Meta...
每周一次胰岛素Icodec对比每日一次长效胰岛素治疗2型糖尿病：随机对照试验的荟萃分析。

糖尿病并发症

低血糖症 (hypoglycemia)

Advances in Cell Replacement Therapies for Diabetes.
糖尿病细胞替代疗法的最新进展

Clinical Presentation and Diagnostic Approach to Hypoglycemia in Adults Without Diabetes Me...
非糖尿病成人低血糖的临床表现与诊断方法

糖尿病视网膜病变 (diabetic retinopathy)

A Review on Diabetic Retinopathy.
糖尿病视网膜病变综述

糖尿病足 (diabetic foot)

Inflammation, diabetic foot and related treatments.
炎症、糖尿病足及相关治疗。

Current status and progress in research on dressing management for diabetic foot ulcer.
糖尿病足溃疡敷料管理的研究现状与进展

Mechanisms of diabetic foot ulceration: A review.
糖尿病足溃疡的机制：综述。

糖尿病酮症酸中毒 (diabetic ketoacidosis)

Developing a Protocol for Management of Euglycemic Diabetic Ketoacidosis.
制定正常血糖糖尿病酮症酸中毒管理方案

Euglycemic diabetic ketoacidosis in the era of SGLT-2 inhibitors.
SGLT-2抑制剂时代的正常血糖性糖尿病酮症酸中毒。

高血糖高渗状态 (hyperosmolar hyperglycemic state)

Diabetic ketoacidosis and hyperosmolar hyperglycemic state in patients with end-stage renal d...
终末期肾病透析患者中的糖尿病酮症酸中毒和高渗性高血糖状态：综述。

糖尿病管理技术

人工胰腺 (artificial pancreas)

The role of automated insulin delivery technology in diabetes.
自动化胰岛素输送技术在糖尿病中的作用

持续葡萄糖监测 (continuous glucose monitoring)

Is It Time to Move Beyond TIR to TITR? Real-World Data from Over 20,000 Users of Continuo...
是时候从TIR转向TITR了吗？来自超过20,000名1型和2型糖尿病患者持续血糖监测的真实世界数据。

Once-weekly insulin efsitora alfa: Design and rationale for the QWINT phase 3 clinical develop...
每周一次胰岛素efsitora alfa：QWINT 3期临床开发项目的设计与原理

混合闭环系统 (hybrid closed loop system)

Safety and glycaemic control during exercise with the MiniMed 780G: A narrative review of evi...
MiniMed 780G运动期间的安全性与血糖控制：从实验室到现实世界证据的叙述性综述

胰岛素泵 (insulin pump)

Diabetes Technology in People with Type 2 Diabetes: Novel Indications.
2型糖尿病患者中的糖尿病技术：新适应症。

肥胖与代谢综合征

代谢综合征 (metabolic syndrome)

Obesity, Metabolic Syndrome, and Osteoarthritis-An Updated Review.
肥胖、代谢综合征与骨关节炎——最新综述

肥胖症 (obesity)

Obesity Paradox: Fact or Fiction?
肥胖悖论：事实还是虚构？

非酒精性脂肪性肝病 (nonalcoholic fatty liver disease)

Resmetirom: The First Disease-Specific Treatment for MASH.
瑞司美替罗：首个针对MASH的疾病特异性治疗药物

Metabolic dysfunction-associated steatotic liver disease and its link to cancer.
代谢功能障碍相关脂肪性肝病及其与癌症的关联。

Osteokines in Nonalcoholic Fatty Liver Disease.
非酒精性脂肪性肝病中的骨因子。

MAFLD: a multisystem disease.
MAFLD: 一种多系统疾病

高脂血症 (dyslipidemia)

Obesity and Dyslipidemia: A Review of Current Evidence.
肥胖与血脂异常: 当前证据综述。

肥胖前沿

减重新药 (anti obesity drug)

Semaglutide for the treatment of overweight and obesity: A review.
司美格鲁肽治疗超重和肥胖的综述。

肾上腺疾病

Addison病 (addison disease)

Female fertility and pregnancy in autoimmune Addison's disease - a mini review.
自身免疫性艾迪生病的女性生育能力与妊娠——小型综述。

原发性醛固酮增多症 (primary aldosteronism)

Primary Aldosteronism: An Endocrine Society Clinical Practice Guideline.
原发性醛固酮增多症: 美国内分泌学会临床实践指南。

Outcomes after medical treatment for primary aldosteronism: an international consensus and a...
原发性醛固酮增多症药物治疗后的结果: 一项国际共识及国际队列中治疗反应的分析。

嗜铬细胞瘤 (pheochromocytoma)

Pheochromocytoma: an updated scoping review from clinical presentation to management and...
嗜铬细胞瘤: 从临床表现到管理与治疗的最新范围综述

Clinical consensus guideline on the management of phaeochromocytoma and paraganglioma i...
携带种系SDHD致病性变异患者的嗜铬细胞瘤和副神经节瘤管理临床共识指南

库欣综合征 (cushing syndrome)

Cushing Syndrome, Hypercortisolism, and Glucose Homeostasis: A Review.

库欣综合征、高皮质醇血症与葡萄糖稳态：综述。

肾上腺意外瘤 (adrenal incidentaloma)

Update on clinical characteristics in the evaluation of pheochromocytoma and paraganglioma.
嗜铬细胞瘤和副神经节瘤评估中临床特征的更新

肾上腺皮质功能减退 (adrenal insufficiency)

European Society of Endocrinology and Endocrine Society Joint Clinical Guideline: Diagnosis ...
欧洲内分泌学会与内分泌学会联合临床指南：糖皮质激素诱导的肾上腺功能不全的诊断与治疗。

European Society of Endocrinology and Endocrine Society Joint Clinical Guideline: Diagnosis ...
欧洲内分泌学会与内分泌学会联合临床指南：糖皮质激素诱导的肾上腺功能不全的诊断与治疗。

A Contemporary Approach to the Diagnosis and Management of Adrenal Insufficiency.
肾上腺功能不全诊断与治疗的当代方法

骨质疏松与骨代谢疾病

佝偻病 (rickets)

Updates on renal phosphate transport.
肾磷酸盐转运的最新进展。

成骨不全 (osteogenesis imperfecta)

Update on the Genetics of Osteogenesis Imperfecta.
成骨不全症遗传学的最新进展

骨质疏松症 (osteoporosis)

Impact of osteoporosis and osteoporosis medications on fracture healing: a narrative review.
骨质疏松症及骨质疏松药物对骨折愈合的影响：一项叙述性综述。

Pharmacologic Treatment of Primary Osteoporosis or Low Bone Mass to Prevent Fractures in ...
成人原发性骨质疏松或低骨量的药物治疗以预防骨折：美国医师学会的活体临床指南

骨软化症 (osteomalacia)

Recent advances in fibroblast growth factor 23-related hypophosphatemic disorders.
成纤维细胞生长因子23相关低磷血症疾病的最新进展。

Insulin Resistance in Type 1 Diabetes: Pathophysiological, Clinical, and Therapeutic Relevance.

1型糖尿病中的胰岛素抵抗：病理生理、临床及治疗相关性。

期刊: Endocrine reviews 日期: 2025

Abstract

People with type 1 diabetes (T1D) are usually considered to exclusively exhibit beta-cell failure, but they frequently also feature **insulin resistance**. This review discusses the mechanisms, clinical features, and therapeutic relevance of **insulin resistance** by focusing mainly on human studies using gold-standard techniques (euglycemic-hyperinsulinemic clamp). In T1D, tissue-specific **insulin resistance** can develop early and sustain throughout disease progression. The underlying **pathophysiology** is complex, involving both metabolic- and **autoimmune**-related factors operating synergistically. Insulin treatment may play an important pathogenic role in predisposing individuals with T1D to **insulin resistance**. However, the established lifestyle-related risk factors and peripheral insulin administration inducing **glucolipotoxicity**, **hyperinsulinemia**, **hyperglucagonemia**, **inflammation**, mitochondrial abnormalities, and **oxidative stress** cannot always fully explain **insulin resistance** in T1D, suggesting a phenotype distinct from type 2 diabetes. The mutual interaction between **insulin resistance** and impaired **endothelial function** further contributes to diabetes-related complications. **insulin resistance** should therefore be considered a treatment target in T1D. Aside from lifestyle modifications, **continuous subcutaneous insulin infusion** can ameliorate **insulin resistance** and **hyperinsulinemia**, thereby improving glucose toxicity compared with multiple injection insulin treatment. Among other concepts, metformin, pioglitazone, incretin-based drugs such as GLP-1 receptor agonists, sodium-glucose cotransporter inhibitors, and pramlintide can improve **insulin resistance**, either directly or indirectly. However, considering the current issues of high cost, side effects, limited efficacy, and their off-label status, these agents in people with T1D are not widely used in routine clinical care at present.

中文翻译

1型糖尿病（T1D）患者通常被认为仅存在β细胞功能衰竭，但常伴有胰岛素抵抗。本综述通过聚焦采用金标准技术（高胰岛素-正葡萄糖钳夹）的人体研究，探讨胰岛素抵抗的机制、临床特征及治疗意义。在T1D中，组织特异性胰岛素抵抗可早期出现并贯穿疾病全程。其病理生理机制复杂，涉及代谢与自身免疫相关因素的协同作用。胰岛素治疗可能在T1D患者易发胰岛素抵抗中发挥重要致病作用。然而，已确立的生活方式相关风险因素及外周胰岛素给药所诱导的糖脂毒性、高胰岛素血症、高胰高血糖素血症、炎症、线粒体异常和氧化应激，并不能完全解释T1D中的胰岛素抵抗，提示其表型有别于2型糖尿病。胰岛素抵抗与内皮功能受损的相互作用进一步加剧糖尿病相关并发症。因此，胰岛素抵抗应被视为T1D的治疗靶点。除生活方式干预外，与多次胰岛素注射治疗相比，持续皮下胰岛素输注可改善胰岛素抵抗和高胰岛素血症，从而减轻葡萄糖毒性。在其他方案中，二甲双胍、吡格列酮、基于肠促胰素的药物（如GLP-1受体激动剂）、钠-葡萄糖协同转运蛋白抑制剂及普兰林肽可直接或间接改善胰岛素抵抗。然而，考虑到当前存在的高成本、副作用、疗效有限及超说明书用药等问题，这些药物在T1D患者中尚未广泛应用于常规临床治疗。

核心词汇

insulin resistance /ˈɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗	pathophysiology /ˌpæθəʊˌfɪziˈɒlədʒi/ : 病理生理学
autoimmune /ˌɔːtəʊˈɪmjʊːn/ : 自身免疫的	glucolipotoxicity /ˌɡluːkəʊˌlɪpəʊtɒkˈsɪsɪti/ : 糖脂毒性
hyperinsulinemia /ˌhaɪpərɪnsjʊlɪˈniːmiə/ : 高胰岛素血症	hyperglucagonemia /ˌhaɪpərˌɡluːkəɡəˈniːmiə/ : 高胰高血糖素血症
inflammation /ˌɪnfləˈmeɪʃən/ : 炎症	oxidative stress /ˈɒksɪdətɪv stres/ : 氧化应激
endothelial function /ˌendəʊˈθiːliəl ˈfʌŋkʃən/ : 内皮功能	continuous subcutaneous insulin infusion /kənˈtɪnjuəs ˌsʌbkjuːˈteɪniəs ˈɪnsjʊlɪn ɪnˈfjuːʒən/ : 持续皮下胰岛素输注

New advances in type 1 diabetes.

1型糖尿病的新进展

期刊: BMJ (Clinical research ed.) 日期: 2024

Abstract

Type 1 diabetes is an **autoimmune** condition resulting in **insulin** deficiency and eventual loss of **pancreatic beta cell** function requiring lifelong **insulin therapy**. Since the discovery of **insulin** more than 100 years ago, vast advances in treatments have improved care for many people with type 1 diabetes. Ongoing research on the **genetics** and **immunology** of type 1 diabetes and on interventions to modify disease course and preserve beta cell function have expanded our broad understanding of this condition. Biomarkers of type 1 diabetes are detectable months to years before development of overt disease, and three stages of diabetes are now recognized. The advent of **continuous glucose monitoring** and the newer automated **insulin** delivery systems have changed the landscape of type 1 diabetes management and are associated with improved **glycated hemoglobin** and decreased **hypoglycemia**. Adjunctive therapies such as sodium glucose cotransporter-1 inhibitors and glucagon-like peptide 1 receptor agonists may find use in management in the future. Despite these rapid advances in the field, people living in under-resourced parts of the world struggle to obtain necessities such as **insulin**, syringes, and blood glucose monitoring essential for managing this condition. This review covers recent developments in diagnosis and treatment and future directions in the broad field of type 1 diabetes.

中文翻译

1型糖尿病是一种自身免疫性疾病，导致胰岛素缺乏并最终丧失胰腺β细胞功能，需要终身胰岛素治疗。自100多年前发现胰岛素以来，治疗手段的巨大进步改善了许多1型糖尿病患者的护理。针对1型糖尿病的遗传学、免疫学以及干预疾病进程、保护β细胞功能的研究，拓展了我们对这种疾病的广泛认知。1型糖尿病的生物标志物在显性疾病出现前数月甚至数年即可检测到，目前该病被划分为三个阶段。持续葡萄糖监测系统及新型自动化胰岛素输送系统的问世，改变了1型糖尿病的管理格局，并有助于改善糖化血红蛋白水平、减少低血糖发生。辅助疗法如钠-葡萄糖协同转运蛋白1抑制剂和胰高血糖素样肽1受体激动剂未来可能在治疗中发挥作用。尽管该领域进展迅速，但全球资源匮乏地区的人们仍难以获得胰岛素、注射器和血糖监测等管理疾病所必需的物资。本综述涵盖了1型糖尿病诊断与治疗的最新进展，以及该广阔领域的未来方向。

核心词汇

autoimmune /ˌɔːtəʊɪˈmjʊːn/ : 自身免疫的	insulin /ˈɪnsəlɪn/ : 胰岛素
pancreatic beta cell /ˌpæŋkriˈætiːk ˈbeɪtə sel/ : 胰腺β细胞	insulin therapy /ˈɪnsəlɪn ˈθerəpi/ : 胰岛素治疗
genetics /dʒəˈnetɪks/ : 遗传学	immunology /ˌɪmjʊˈnɒlədʒi/ : 免疫学
biomarker /ˈbaɪəʊˌmaːrkər/ : 生物标志物	continuous glucose monitoring /kənˈtɪnjuəs ˈgluːkəʊs ˈmɒnɪtərɪŋ/ : 持续葡萄糖监测
automated insulin delivery system /ˈɔːtəmeɪtɪd ˈɪnsəlɪn dɪˈlɪvəri ˈsɪstəm/ : 自动化胰岛素输注系统	glycated hemoglobin /ˈɡlaɪkeɪtɪd ˈhiːməˈɡlɒbɪn/ : 糖化血红蛋白
hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖	adjunctive therapy /əˈdʒʌŋktɪv ˈθerəpi/ : 辅助治疗

Searching for LADA in patients with Hashimoto's thyroiditis: A tool for the prevention of LADA.

在桥本甲状腺炎患者中筛查LADA：预防LADA的一种工具

期刊: Journal of diabetes investigation 日期: 2026

Abstract

autoimmune thyroid diseases (AITD), such as Hashimoto's thyroiditis (HT), frequently co-occur with autoimmune diabetes, including latent autoimmune diabetes in adults (LADA). While guidelines recommend thyroid autoantibody screening in patients with autoimmune diabetes, the reverse approach—screening for diabetes-related autoantibodies in AITD patients—is not routinely practiced. Given the substantial link between glutamic acid decarboxylase antibodies (GADA) and both thyroid and beta-cell autoimmunity, we recommend frequent GADA testing in AITD patients, especially those with HT. This screening technique may discover LADA early, leading to timely glycaemic monitoring and treatment measures to protect beta-cell function and reduce diabetes complications.

中文翻译

自身免疫性甲状腺疾病（AITD），如桥本甲状腺炎（HT），常与自身免疫性糖尿病（包括成人隐匿性自身免疫性糖尿病（LADA））共存。尽管指南建议对自身免疫性糖尿病患者进行甲状腺自身抗体筛查，但反向方法——即在AITD患者中筛查糖尿病相关自身抗体——并未常规实施。鉴于谷氨酸脱羧酶抗体（GADA）与甲状腺及β细胞自身免疫之间存在显著关联，我们建议对AITD患者（尤其是HT患者）进行频繁的GADA检测。该筛查技术可能早期发现LADA，从而及时采取血糖监测和治疗措施，以保护β细胞功能并减少糖尿病并发症。

核心词汇

autoimmune thyroid diseases /ˌɔːtəʊˈmjuːn ˈθaɪrɔɪd dɪˈziːzɪz/ : 自身免疫性甲状腺疾病	Hashimoto's thyroiditis /hɑːʃɪˈmɔʊtəʊz ˌθaɪrɔɪˈdaɪtɪs/ : 桥本甲状腺炎
autoimmune diabetes /ˌɔːtəʊˈmjuːn ˌdaɪəˈbiːtɪz/ : 自身免疫性糖尿病	latent autoimmune diabetes in adults /ˈleɪtənt ˌɔːtəʊˈmjuːn ˌdaɪəˈbiːtɪz ɪn ˈædʌlts/ : 成人隐匿性自身免疫性糖尿病
thyroid autoantibody /ˈθaɪrɔɪd ˌɔːtəʊˈæntɪbɒdi/ : 甲状腺自身抗体	screening /ˈskriːnɪŋ/ : 筛查
glutamic acid decarboxylase antibodies /ɡluːˈtæmɪk ˈæsɪd ˌdiːkɑːˈbɒksɪleɪz ˈæntɪbɒdɪz/ : 谷氨酸脱羧酶抗体	autoimmunity /ˌɔːtəʊˈmjuːnɪti/ : 自身免疫
glycaemic monitoring /ɡlaɪˈsiːmɪk ˈmɒnɪtərɪŋ/ : 血糖监测	complication /ˌkɒmplɪˈkeɪʃən/ : 并发症

Exploring the Gut Microbiota: Key Insights Into Its Role in Obesity, Metabolic Syndrome, and Type 2 Diabetes.

探索肠道微生物群：关于其在肥胖、代谢综合征和2型糖尿病中作用的关键见解

期刊: The Journal of clinical endocrinology and metabolism 日期: 2024

Abstract

The **gut microbiota** (GM), comprising trillions of microorganisms in the gastrointestinal tract, is a key player in the development of **obesity** and related metabolic disorders, such as type 2 diabetes (T2D), **metabolic syndrome** (MS), and cardiovascular diseases. This mini-review delves into the intricate roles and mechanisms of the GM in these conditions, offering insights into potential therapeutic strategies targeting the microbiota. The review elucidates the diversity and development of the human GM, highlighting its pivotal functions in host physiology, including nutrient absorption, **immune regulation**, and **energy metabolism**. Studies show that GM **dysbiosis** is linked to increased energy extraction, altered metabolic pathways, and **inflammation**, contributing to **obesity**, MS, and T2D. The interplay between dietary habits and GM composition is explored, underscoring the influence of diet on microbial diversity and metabolic functions. Additionally, the review addresses the impact of common medications and therapeutic interventions like **fecal microbiota transplantation** on GM composition. The evidence so far advocates for further research to delineate the therapeutic potential of GM modulation in mitigating **obesity** and metabolic diseases, emphasizing the necessity of clinical trials to establish effective and sustainable treatment protocols.

中文翻译

肠道微生物群 (GM) 由胃肠道中数万亿微生物组成，在肥胖及2型糖尿病 (T2D)、代谢综合征 (MS) 和心血管疾病等相关代谢紊乱的发展中扮演关键角色。本综述深入探讨了GM在这些疾病中的复杂作用与机制，为靶向微生物群的潜在治疗策略提供了见解。文章阐述了人类GM的多样性与发育过程，强调其在宿主生理功能中的核心作用，包括营养吸收、免疫调节和能量代谢。研究表明，GM失调与能量摄取增加、代谢通路改变及炎症反应相关，进而促进肥胖、MS和T2D的发生。本文还探讨了饮食习惯与GM组成之间的相互作用，强调饮食对微生物多样性和代谢功能的影响。此外，综述分析了常见药物及粪菌移植等治疗干预措施对GM组成的影响。现有证据支持进一步研究GM调节在缓解肥胖和代谢疾病中的治疗潜力，并强调开展临床试验以建立有效且可持续治疗方案的必要性。

核心词汇

gut microbiota /gʌt ˌmaɪkroʊbaɪˈoʊtə/ : 肠道微生物群

metabolic disorder /ˌmetəˈbɒlɪk dɪsˈɔːdə/ : 代谢紊乱

cardiovascular disease /ˌkɑːrdiəvˈæskjələr dɪˈziːz/ : 心血管疾病

inflammation /ˌɪnfləˈmeɪʃən/ : 炎症

energy metabolism /ˈɛnədʒi məˈtæbəlaɪzəm/ : 能量代谢

therapeutic intervention /ˌθerəˈpjʊːtɪk ˌɪntərˈvenʃən/ : 治疗干预

obesity /ouˈbiːsəti/ : 肥胖

metabolic syndrome /ˌmetəˈbɒlɪk ˈsɪndroum/ : 代谢综合征

dysbiosis /dɪsbaɪˈoʊsɪs/ : 菌群失调

immune regulation /ɪˈmjuːn ˌrɛɡjʊˈleɪʃən/ : 免疫调节

fecal microbiota transplantation /ˈfiːkəl ˌmaɪkroʊbaɪˈoʊtə ˌtrænsplænˈteɪʃən/ : 粪菌移植

Maturity onset diabetes of the young and beyond: the changing face of single-gene diabetes.

青少年起病的成人型糖尿病及其扩展：单基因糖尿病的演变面貌。

期刊: European journal of endocrinology 日期: 2025

Abstract

maturity onset diabetes of the young (MODY) is the most common form of monogenic diabetes. A genetic diagnosis can help tailor treatment. However, it is vital to test the right genes. This review discusses the need to test the correct genes and widen the genes that are included on panels while also discussing examples of genes that should be removed from panels due to insufficient evidence. This is important to ensure the maximum number of patients receive a genetic diagnosis while avoiding misdiagnosis and mistreatment. To maximize the cost-benefit of genetic testing, we need to select the right patients for genetic testing-this review discusses criteria that increase the chance of a patient having a monogenic cause of their diabetes. Reduced penetrance, where some individuals have the pathogenic genotype but do not exhibit the phenotype, is now known to be increasingly common in MODY and other monogenic disorders. This has important implications when a variant is identified in an unaffected individual. This review highlights recent work that polygenic risk may play a role in determining penetrance.

中文翻译

青少年起病的成人型糖尿病 (MODY) 是单基因糖尿病中最常见的类型。基因诊断有助于制定个体化治疗方案，但关键在于检测正确的基因。本综述探讨了检测正确基因的必要性，以及扩大检测基因组合范围的必要性，同时讨论了因证据不足应从检测组合中剔除的基因示例。这对于确保尽可能多的患者获得基因诊断，同时避免误诊和误治至关重要。为最大化基因检测的成本效益，需筛选合适的患者进行检测——本综述分析了提高患者糖尿病单基因病因检出率的筛选标准。外显率降低（即携带致病基因型但未表现相应表型）在MODY及其他单基因疾病中日益常见，这一现象对在未患病个体中发现基因变异具有重要临床意义。本综述强调，多基因风险可能在决定外显率中发挥作用。

核心词汇

maturity onset diabetes of the young /məˈtʃʊrɪti ˈɒnsət ˌdaɪəˈbiːtiːz ʌv ðə ˈjʌŋ/ : 青少年发病的成年型糖尿病	monogenic diabetes /ˌmɒnəʊˈdʒenɪk ˌdaɪəˈbiːtiːz/ : 单基因糖尿病
genetic diagnosis /dʒəˈnetɪk ˌdaɪəɡˈnoʊsɪs/ : 遗传诊断	gene panel /dʒiːn ˈpænel/ : 基因面板
misdiagnosis /ˌmɪs ˌdaɪəɡˈnoʊsɪs/ : 误诊	mistreatment /ˌmɪs ˈtriːtmənt/ : 错误治疗
genetic testing /dʒəˈnetɪk ˈtestɪŋ/ : 基因检测	penetrance /ˈpenɪtrəns/ : 外显率
pathogenic genotype /ˌpæθəˈdʒenɪk ˈdʒenəʊtaɪp/ : 致病基因型	phenotype /ˈfiːnəʊtaɪp/ : 表型
polygenic risk /ˌpɒliˈdʒenɪk rɪsk/ : 多基因风险	variant /ˈvərɪənt/ : 变异

Prevalence and Clinical Characteristics of NEUROD1-MODY in Chinese Early-Onset Type 2 Diabetes Mellitus and a Literature Review.

中国早发型2型糖尿病中NEUROD1-MODY的患病率与临床特征及文献综述

期刊: Journal of diabetes 日期: 2025

Abstract

BACKGROUND: Maturity-onset diabetes of the young resulting from mutations of the NEUROD1 gene (NEUROD1-MODY) is a rare form of diabetes and has not been well studied. We aimed to estimate its prevalence in Chinese patients with early-onset type 2 diabetes mellitus (EOD) and summarize its clinical and genetic characteristics. METHODS: We performed next-generation sequencing in 679 patients with EOD to screen rare variants in NEUROD1 exons and evaluated the effects of variants using in vitro experiments. All the reported NEUROD1-MODY cases were reviewed. Patients carrying pathogenic or likely pathogenic variants were diagnosed with NEUROD1-MODY according to the American College of Medical Genetics and Genomics guidelines. RESULTS: Four rare variants were identified in 679 patients with EOD, but only P197H decreased the transcriptional activity in in vitro functional assays to an extent comparable to the well-known mutation causing NEUROD1-MODY. Its frequency was pretty higher in the European population (0.024) than that in the East Asian population (0.00034) according to the gnomAD database. Twenty-eight previously reported patients could be confirmed as diagnosed. The patients in Asia had a lower body mass index and a higher rate of ketosis compared with Caucasians, and the mutations present in Asia often occurred in the transactivation domain. Neurological abnormalities were observed in 10.7% of the patients with NEUROD1-MODY. CONCLUSIONS: NEUROD1-MODY in Chinese patients with EOD is not common (penetrance than Caucasian and needs further exploration. The possible differences of phenotypes exist between the two ethnic populations.

中文翻译

背景：由NEUROD1基因突变引起的青少年发病的成年型糖尿病（NEUROD1-MODY）是一种罕见的糖尿病类型，尚未得到充分研究。我们旨在评估其在中国早发2型糖尿病（EOD）患者中的患病率，并总结其临床和遗传特征。方法：我们对679名EOD患者进行了二代测序，以筛选NEUROD1外显子中的罕见变异，并通过体外实验评估变异的影响。回顾了所有已报道的NEUROD1-MODY病例。根据美国医学遗传学与基因组学学会指南，携带致病性或可能致病性变异的患者被诊断为NEUROD1-MODY。结果：在679名EOD患者中发现了四种罕见变异，但只有P197H在体外功能实验中使转录活性降低的程度与已知导致NEUROD1-MODY的突变相当。根据gnomAD数据库，该变异在欧洲人群中的频率（0.024）远高于东亚人群（0.00034）。此前报道的28例患者可被确诊。与白种人相比，亚洲患者的体重指数较低，酮症发生率较高，且亚洲患者中出现的突变常发生在反式激活结构域。10.7%的NEUROD1-MODY患者观察到神经系统异常。结论：NEUROD1-MODY在中国EOD患者中并不常见（≤0.15%）。P197H可能在中国人群中导致MODY，且其外显率高于白种人，需进一步探索。两个种族人群之间可能存在表型差异。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtiːz ˈmɛlɪtəs/ : 糖尿病	mutation /mjuːˈteɪʃən/ : 突变
gene /dʒiːn/ : 基因	prevalence /ˈpreɪvələns/ : 患病率
variant /ˈvəriənt/ : 变异	transcriptional activity /trænˈskrɪpʃənəl ækˈtɪvɪti/ : 转录活性
functional assay /ˈfʌŋkʃənəl əˈseɪ/ : 功能检测	penetrance /ˈpenɪtrəns/ : 外显率
phenotype /ˈfiːnətaɪp/ : 表型	body mass index /ˈbɒdi məs ˈɪndeks/ : 体重指数
ketosis /kɪˈtəʊsɪs/ : 酮症	neurological abnormality /ˌnjuːələˈlɒdʒɪkəl æbnɔːrˈmælɪti/ : 神经系统异常

Phenotypic features, prevalence of KCNJ11-MODY in Chinese patients with early-onset diabetes and a literature review.

中国早发糖尿病患者中KCNJ11-MODY的表型特征、患病率及文献综述。

期刊: Clinical endocrinology 日期: 2024

Abstract

OBJECTIVE: Gain-of-function (GOF) variants of KCNJ11 cause **neonate diabetes** and maturity-onset **diabetes** of the young (KCNJ11-MODY), while loss-of-function (LOF) variants lead to **hyperinsulinemia hypoglycemia** and subsequent **diabetes**. Given the limited research of KCNJ11-MODY, we aimed to analyse its **phenotypic features** and **prevalence** in Chinese patients with early-onset type 2 **diabetes** (EOD). DESIGN, PATIENTS AND MEASUREMENTS: We performed next-generation sequencing on 679 Chinese EOD patients to screen for KCNJ11 exons variants. Bioinformatics prediction and the American College of Medical Genetics and Genomics guidelines was used to determine the **pathogenicity** and diagnosed KCNJ11-MODY. A literature review was conducted to investigate the **phenotypic features** of KCNJ11-MODY. RESULTS: We identified six predicted deleterious rare variants in six EOD patients (0.88%). They were classified as uncertain significance (**variant** of uncertain significance [VUS]), but more common in this EOD cohort than a general Chinese population database, however, without significant difference (53/10,588, 0.50%) ($p = .268$). Among 80 previously reported patients with KCNJ11-MODY, 23.8% (19/80) carried 9 (32.1%) LOF variants, who had significantly older age at diagnosis, higher **birthweight** and higher fasting C-peptide compared to patients with GOF variants. Many patients carrying VUS were not correctly diagnosed. CONCLUSIONS: Some rare variants of KCNJ11 might contribute to the development of Chinese EOD, although available evidence has not enough power to support them as cause of KCNJ11-MODY. The clinical features of LOF variants were different from GOF variants in KCNJ11-MODY patients. It is necessary to evaluate the **pathogenicity** of VUS through function experiments.

中文翻译

目的: KCNJ11的功能获得性变异可导致新生儿糖尿病和青少年的成人起病型糖尿病 (KCNJ11-MODY), 而功能丧失性变异则引发高胰岛素血症性低血糖及后续糖尿病。鉴于KCNJ11-MODY研究有限, 本研究旨在分析中国早发2型糖尿病患者的表型特征及患病率。设计、患者与测量方法: 对679例中国早发2型糖尿病患者进行二代测序, 筛查KCNJ11外显子变异。通过生物信息学预测及美国医学遗传学与基因组学学会指南判定致病性并诊断KCNJ11-MODY。通过文献综述探究KCNJ11-MODY的表型特征。结果: 在6例早发2型糖尿病患者 (0.88%) 中鉴定出6种预测有害的罕见变异。这些变异被归类为意义不明确变异, 但在本早发2型糖尿病队列中的发生率高于中国普通人群数据库 (53/10,588, 0.50%), 但差异无统计学意义 ($p=0.268$)。在既往报道的80例KCNJ11-MODY患者中, 23.8% (19/80) 携带9种 (32.1%) 功能丧失性变异, 与功能获得性变异患者相比, 其诊断年龄显著更大、出生体重更高、空腹C肽水平更高。许多携带意义不明确变异的患者未得到正确诊断。结论: 部分KCNJ11罕见变异可能参与中国早发2型糖尿病的发生, 但现有证据尚不足以支持其为KCNJ11-MODY的致病原因。KCNJ11-MODY患者中功能丧失性变异与功能获得性变异的临床特征存在差异。有必要通过功能实验评估意义不明确变异的致病性。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病	neonate diabetes /ˈniːəneɪt ˈdaɪəˈbiːtiːz/ : 新生儿糖尿病
hyperinsulinemia hypoglycemia /ˌhaɪpərɪnsjʊlɪˈniːmiə ˌhaɪpəʊglɪˈsiːmiə/ : 高胰岛素血症性低血糖	variant /ˈveəriənt/ : 变异
phenotypic features /ˌfiːnəˈtɪpɪk ˈfiːtʃəz/ : 表型特征	prevalence /ˈpreɪvləns/ : 患病率
exon /ˈeksɒn/ : 外显子	pathogenicity /ˌpæθədʒɪˈnɪsɪti/ : 致病性
birthweight /ˈbɜːθweɪt/ : 出生体重	

Revised European Society of Endocrinology Clinical Practice Guideline for the management of aggressive pituitary tumours and pituitary carcinomas.

修订版欧洲内分泌学会临床实践指南：侵袭性垂体肿瘤与垂体癌的管理

期刊: European journal of endocrinology 日期: 2025

Abstract

Pituitary tumours, originating from endocrine cells of the **anterior pituitary**, are quite common, and in most cases well-controlled by surgery or medical treatment. However, a small subset of pituitary tumours presents with multiple local recurrences or tumour progression despite combined surgical, medical or radiotherapeutic treatment. These are known as aggressive pituitary tumours (APT); also called aggressive pituitary neuroendocrine tumours (PitNETs); or, in the rare case of metastases, pituitary carcinomas (PC) or metastatic PitNETs. Early identification of APT is challenging but is of major clinical importance as they are associated with an increased **morbidity** and **mortality** even in the absence of metastases. Here, we provide a revision of the first international, interdisciplinary European Society of Endocrinology (ESE) clinical practice **guideline** on APTs and PC (2018). Since publication of the 2018 **guideline**, results from the second ESE survey on APT and PC were published, and more data on APT treatment, including temozolomide, immune checkpoint inhibitors and bevacizumab, emerged. These data are reviewed in this **guideline** and translated into a practical algorithm to guide APT and PC management. Furthermore, standardized reporting of **imaging** and histopathological investigations of these tumours is proposed, and the role of **molecular analysis** is discussed. Last, a section is dedicated to special circumstances such as APT in **pregnancy**.

中文翻译

垂体肿瘤起源于垂体前叶内分泌细胞，较为常见，多数病例可通过手术或药物治疗得到良好控制。然而，少数垂体肿瘤即使经过手术、药物或放疗等综合治疗，仍会出现多次局部复发或肿瘤进展。这类肿瘤被称为侵袭性垂体肿瘤（APT），亦称侵袭性垂体神经内分泌肿瘤（PitNETs）；在罕见转移病例中则称为垂体癌（PC）或转移性PitNETs。早期识别APT具有挑战性，但具有重要临床意义——即使未发生转移，这类肿瘤仍与发病率和死亡率升高相关。本文对2018年首版国际跨学科欧洲内分泌学会（ESE）APT及PC临床实践指南进行修订。自2018版指南发布以来，第二项关于APT及PC的ESE调查结果已公布，同时涌现出更多关于替莫唑胺、免疫检查点抑制剂和贝伐珠单抗等APT治疗数据。本指南对这些数据进行梳理，并转化为指导APT及PC管理的实用诊疗流程。此外，本文提出此类肿瘤影像学和组织病理学检查的标准化报告规范，探讨分子分析的作用，并专设章节讨论妊娠期APT等特殊情况。

核心词汇

pituitary tumour /piˈtjuːɪtəri ˈtjuːmə/ : 垂体肿瘤

endocrine cell /ˈendəkrɪn sel/ : 内分泌细胞

metastasis /məˈtæstəsis/ : 转移

pituitary carcinoma /piˈtjuːɪtəri ˌkɑːrsɪˈnoumə/ : 垂体癌

mortality /mɔːrˈtæləti/ : 死亡率

imaging /ˈɪmɪdʒɪŋ/ : 影像学

molecular analysis /məˈlekjʊlər əˈnæləsis/ : 分子分析

anterior pituitary /ænˈtɪəriər piˈtjuːɪtəri/ : 垂体前叶

recurrence /rɪˈkʌrəns/ : 复发

aggressive pituitary tumour /əˈɡresɪv piˈtjuːɪtəri ˈtjuːmə/ : 侵袭性垂体肿瘤

morbidity /mɔːrˈbɪdəti/ : 发病率

guideline /ˈɡaɪdlaɪn/ : 指南

histopathology /ˌhɪstəpəˈθɒlədʒi/ : 组织病理学

pregnancy /ˈpregnənsi/ : 妊娠

Advances in Drug Treatments for Male Patients with Prolactinomas.

男性泌乳素瘤患者药物治疗进展

期刊: Neuroendocrinology 日期: 2025

Abstract

BACKGROUND: Pituitary adenomas (PAs), the most common intracranial neuroendocrine tumors, are typically benign. Among them, prolactinomas - tumors that secrete prolactin - account for approximately 60% of all PAs and are characterized by hyperprolactinemia and potential mass effects. Significant epidemiological and clinical differences exist between male and female prolactinoma patients. SUMMARY: Prolactinomas in male patients tend to be larger and more aggressive, presenting unique challenges in treatment and management. Although dopamine agonists (DAs) remain the first-line therapy, men exhibit higher rates of DA resistance compared to women. For refractory prolactinomas in males, alternative treatments include temozolomide (TMZ), immune checkpoint inhibitors (anti-PD-1/PD-L1 and anti-CTLA4), somatostatin receptor analogs, mTOR inhibitors, tyrosine kinase inhibitors, bevacizumab, and aromatase inhibitors. These therapies may provide greater benefits to men with refractory prolactinomas than to women. KEY MESSAGES: Despite advancements, significant challenges and opportunities persist in managing male prolactinoma patients. Key areas requiring attention include early diagnosis, predicting drug responsiveness, understanding sex-specific molecular mechanisms, and developing novel therapeutic strategies. Large-scale clinical trials are crucial to further assess the efficacy and safety of these treatments in men. This review consolidates recent progress in medical therapies and management approaches for male prolactinoma patients.

中文翻译

背景：垂体腺瘤是最常见的颅内神经内分泌肿瘤，通常为良性。其中，分泌催乳素的催乳素瘤约占所有垂体腺瘤的60%，以高催乳素血症和潜在占位效应为特征。男性和女性催乳素瘤患者在流行病学和临床表现上存在显著差异。摘要：男性催乳素瘤患者肿瘤体积更大、侵袭性更强，在治疗和管理方面面临独特挑战。尽管多巴胺激动剂仍是一线疗法，但男性患者对多巴胺激动剂的耐药率高于女性。针对男性难治性催乳素瘤，替代治疗方案包括替莫唑胺、免疫检查点抑制剂（抗PD-1/PD-L1和抗CTLA4）、生长抑素受体类似物、mTOR抑制剂、酪氨酸激酶抑制剂、贝伐珠单抗和芳香化酶抑制剂。这些疗法可能对男性难治性催乳素瘤患者比女性患者带来更大获益。核心信息：尽管取得进展，男性催乳素瘤患者的管理仍面临重大挑战与机遇。需重点关注早期诊断、药物反应预测、性别特异性分子机制解析及新型治疗策略开发。大规模临床试验对进一步评估这些疗法在男性患者中的有效性和安全性至关重要。本综述整合了男性催乳素瘤患者药物治疗和管理策略的最新进展。

核心词汇

pituitary adenoma /pɪˈtjuːɪtəri ˌædɪˈnoʊmə/ : 垂体腺瘤	prolactinoma /prɒˌlæktɪˈnoʊmə/ : 泌乳素瘤
hyperprolactinemia /ˌhaɪpəprɒˌlæktɪˈniːmiə/ : 高催乳素血症	dopamine agonist /ˈdoʊpəmiːn ˈæɡənɪst/ : 多巴胺激动剂
refractory /rɪˈfræktəri/ : 难治性	immune checkpoint inhibitor /ɪˈmjuːn ˈtʃekpɔɪnt ɪnˈhɪbɪtər/ : 免疫检查点抑制剂
somatostatin receptor analog /ˌsoʊmətəˈstetɪn rɪˈseptər ˈænəlɔːɡ/ : 生长抑素受体类似物	mTOR inhibitor /ˈɛm tɔːr ɪnˈhɪbɪtər/ : mTOR抑制剂
tyrosine kinase inhibitor /ˈtaɪrəsiːn ˈkaɪneɪz ɪnˈhɪbɪtər/ : 酪氨酸激酶抑制剂	aromatase inhibitor /əˈrouməteɪz ɪnˈhɪbɪtər/ : 芳香化酶抑制剂
neuroendocrine tumor /ˌnjuːroʊˈendəkrɪn ˈtuːmə/ : 神经内分泌肿瘤	mass effect /mæs ɪˈfekt/ : 占位效应

A Comprehensive Review of Empty Sella and Empty Sella Syndrome.

空蝶鞍与空蝶鞍综合征的综合综述

期刊: Endocrine practice : official journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists

日期: 2024

Abstract

OBJECTIVE: Incidental radiographic findings of an **empty sella** are prevalent in up to 35% of the general population. While **empty sella** was initially considered clinically insignificant, a subset of patients exhibits **endocrine** or neuro-ophthalmologic manifestations which are diagnostic of **empty sella syndrome** (ESS). Recent studies suggest that more patients are affected by ESS than previously recognized, necessitating a deeper understanding of this condition. This comprehensive review describes a practical approach to evaluating and managing ESS. METHODS: Literature review was conducted on etiologies and risk factors associated with primary and secondary **empty sella**, the radiologic features that differentiate **empty sella** from other sellar lesions, and the role of clinical history and **hormone** testing in identifying patients with ESS, as well as treatment modalities. RESULTS: **pituitary** function testing for **somatotroph**, **lactotroph**, **gonadotroph**, **corticotroph**, and **thyrotroph** abnormalities is necessary when suspecting ESS. While an isolated **empty sella** finding does not require treatment, ESS may require pharmacologic or surgical interventions to address **hormone** deficits or **intracranial hypertension**. Targeted **hormone replacement** as directed by the endocrinologist should align with guidelines and patient-specific needs. Treatment may involve a multidisciplinary collaboration with neurology, neurosurgery, or ophthalmology to address patient symptoms. CONCLUSION: This review underscores the evolving understanding of ESS, stressing the significance of accurate diagnosis and tailored management to mitigate potential neurologic and **endocrine** complications in affected individuals.

中文翻译

目的：在普通人群中，空蝶鞍的影像学偶然发现发生率高达35%。虽然空蝶鞍最初被认为无临床意义，但部分患者会出现内分泌或神经眼科表现，符合空蝶鞍综合征（ESS）的诊断标准。近期研究表明，ESS的实际患病人数较以往认知更多，亟需深入理解该疾病。本综述系统阐述了ESS评估与管理的实用方法。方法：通过文献回顾，分析原发性和继发性空蝶鞍的病因及危险因素、空蝶鞍与其他鞍区病变的影像学鉴别特征、临床病史与激素检测在ESS患者识别中的作用，以及治疗策略。结果：疑似ESS时，需对生长激素、催乳素、促性腺激素、促肾上腺皮质激素及促甲状腺激素功能进行检测。单纯空蝶鞍无需治疗，但ESS可能需要药物或手术干预以纠正激素缺乏或颅内高压。内分泌科医生应根据指南及患者个体需求进行靶向激素替代治疗，必要时联合神经内科、神经外科或眼科进行多学科协作以缓解症状。结论：本综述强调了对ESS认知的持续更新，指出精准诊断与个体化管理对减少患者潜在神经及内分泌并发症的重要性。

核心词汇

empty sella /'empti 'sɛlə/ : 空蝶鞍	empty sella syndrome /'empti 'sɛlə 'sɪndroum/ : 空蝶鞍综合征
pituitary /pi 'tu:ɪtəri/ : 垂体	endocrine /'ɛndoukrɪn/ : 内分泌的
intracranial hypertension /,ɪntre 'kreɪniəl 'haɪpər'tɛnfən/ : 颅内高压	hormone /'hɔ:rmon/ : 激素
somatotroph /,soume'toutrə:f/ : 生长激素细胞	lactotroph /'læktətrə:f/ : 泌乳素细胞
gonadotroph /'gounədoutrə:f/ : 促性腺激素细胞	corticotroph /,kɔ:rtɪ'koutrə:f/ : 促肾上腺皮质激素细胞
thyrotroph /'θaɪrətrə:f/ : 促甲状腺激素细胞	hormone replacement /'hɔ:rmon rɪ'pleɪsmənt/ : 激素替代治疗
intracranial hypertension /,ɪntre 'kreɪniəl 'haɪpər'tɛnfən/ : 颅内高压	

The socioeconomic burden of acromegaly.

肢端肥大症的社会经济负担。

期刊: European journal of endocrinology 日期: 2023

Abstract

acromegaly is a rare and insidious disease characterized by chronic excess growth hormone, leading to various morphological changes and systemic complications. Despite its low prevalence, acromegaly poses a significant socioeconomic burden on patients and healthcare systems. This review synthesizes the current state of knowledge on the psychosocial burden, disability, impact on daily life, and cost of acromegaly disease, focusing on the quality of life, partnership, medical care and treatment afflictions, participation in daily activities, professional and leisure impairment, and cost of treatment for acromegaly and its comorbidities. It also examines management strategies, coping mechanisms, and interventions aimed at alleviating this burden. A comprehensive understanding of the extent of the socioeconomic burden in acromegaly is crucial to develop effective strategies to improve treatment and care. Further research is warranted to explore the myriad factors contributing to this burden, as well as the efficacy of interventions to alleviate it, ultimately enhancing the quality of life for patients with acromegaly.

中文翻译

肢端肥大症是一种罕见且隐匿的疾病，其特征是生长激素长期过量，导致多种形态改变和全身性并发症。尽管患病率较低，但肢端肥大症给患者和医疗系统带来了显著的社会经济负担。本综述综合了关于肢端肥大症社会心理负担、残疾、对日常生活的影响以及疾病成本的最新知识，重点关注生活质量、伴侣关系、医疗护理与治疗困扰、日常活动参与、职业与休闲障碍，以及肢端肥大症及其合并症的治疗成本。同时，本文还探讨了旨在减轻这一负担的管理策略、应对机制和干预措施。全面了解肢端肥大症社会经济负担的程度，对于制定改善治疗和护理的有效策略至关重要。未来需进一步研究导致这一负担的多种因素，以及减轻负担的干预措施的有效性，最终提升肢端肥大症患者的生活质量。

核心词汇

acromegaly /ˌækroʊˈmegəli/ : 肢端肥大症	growth hormone /ɡroʊθ ˈhɔːrˌmoʊn/ : 生长激素
morphological changes /ˌmɔːrfoʊˈlɒdʒɪkəl ˈtʃeɪndʒɪz/ : 形态学改变	systemic complications /sɪˈstemɪk ˌkɒmplɪˈkeɪʃənz/ : 全身性并发症
psychosocial burden /ˌsaɪkoʊˈsoʊʃəl ˈbɜːrdən/ : 心理社会负担	disability /ˌdɪsəˈbɪləti/ : 残疾
quality of life /ˈkwɒləti ɒv laɪf/ : 生活质量	comorbidity /ˌkoʊmɔːrˈbɪdəti/ : 合并症
management strategies /ˈmænɪdʒmənt ˈstrætədʒɪz/ : 管理策略	coping mechanisms /ˈkɒpɪŋ ˈmekəˌnɪzəmz/ : 应对机制
intervention /ˌɪntərˈvenʃən/ : 干预	socioeconomic burden /ˌsoʊsiəʊˌekəˈnɒmɪk ˈbɜːrdən/ : 社会经济负担

An Overlooked Disease: Minimal Autonomous Cortisol Secretion (MACS). A Narrative Review.

被忽视的疾病：微小自主性皮质醇分泌（MACS）。一篇叙述性综述。

期刊: Endocrine, metabolic & immune disorders drug targets 日期: 2024

Abstract

BACKGROUND: A far more common disease than Cushing's syndrome is subclinical **hypercortisolism** or mild autonomous **cortisol** secretion (MACS), with an overall prevalence of 0.2-2%. OBJECTIVE: This review aims to shed light on the prevalence, **screening** and diagnostic criteria, comorbidities, and management of Mild Autonomous **cortisol** Secretion (MACS). METHODS: Studies eligible targeted MACS regarding prevalence, **screening**, comorbidities, management, and clinical outcome. This is a narrative-review. IRB approval was not needed. RESULTS: The 1 mg **dexamethasone suppression test** (DST) remains the first **screening** test. MACS is associated with adverse cardiometabolic and renal outcomes, **osteoporosis** and **osteopenia**, **immunodeficiency**, depression, **coagulopathy**, and **sarcopenia**. **surgery** is the gold standard treatment. Medical therapy is recommended when **surgery** is contraindicated or not feasible. Clinically silent **hypercortisolism** is a frequent entity that necessitates early detection and treatment. The production of **cortisol** should be looked at as a spectrum where subtle, undetectable levels can still be produced. They know its association with **adverse health outcomes**. CONCLUSION: MACS is no longer considered an asymptomatic disorder; repeated hormonal and functional tests are crucial to prevent multiorgan damage.

中文翻译

背景：比库欣综合征更为常见的疾病是亚临床高皮质醇血症或轻度自主皮质醇分泌（MACS），总体患病率为0.2-2%。目的：本综述旨在阐明轻度自主皮质醇分泌（MACS）的患病率、筛查与诊断标准、合并症及管理方法。方法：纳入的研究针对MACS的患病率、筛查、合并症、管理及临床结局。本文为叙述性综述，无需机构审查委员会批准。结果：1毫克地塞米松抑制试验仍是首选筛查方法。MACS与不良心血管代谢及肾脏结局、骨质疏松与骨量减少、免疫缺陷、抑郁、凝血功能障碍及肌少症相关。手术是金标准治疗方案，当存在手术禁忌或不可行时推荐药物治疗。临床静默性高皮质醇血症是一种常见情况，需早期发现与治疗。皮质醇的生成应被视为一个连续谱，即使微量且无法检测的水平仍可能产生。已知其与不良健康结局相关。结论：MACS不再被视为无症状疾病；重复激素及功能检查对预防多器官损伤至关重要。

核心词汇

hypercortisolism /ˌhaɪpərˈkɔːrtɪsəˌlɪzəm/ : 高皮质醇血症	cortisol /ˈkɔːrtɪsɒl/ : 皮质醇
screening /ˈskriːnɪŋ/ : 筛查	comorbidity /ˌkɒmɔːrˈbɪdəti/ : 合并症
osteoporosis /ˌɒstiəpəˈrəʊsɪs/ : 骨质疏松症	osteopenia /ˌɒstiəʊˈpiːniə/ : 骨量减少
immunodeficiency /ˌɪmjʊnəʊdɪˈfɪʃənsi/ : 免疫缺陷	coagulopathy /kəʊˌæɡjuːˈləpəθi/ : 凝血功能障碍
sarcopenia /ˌsɑːrkeˈpiːniə/ : 肌少症	surgery /ˈsɜːrdʒəri/ : 手术
dexamethasone suppression test /ˌdekseˈmɛθəsəʊn səˈpreʃən tɛst/ : 地塞米松抑制试验	adverse health outcomes /ˈædvɜːrs hɛlθ ˈaʊtkʌmz/ : 不良健康结局

Current challenges in Cushing's syndrome testing: blood, saliva, urine, or hair?

库欣综合征检测的当前挑战：血液、唾液、尿液还是头发？

期刊: Current opinion in endocrinology, diabetes, and obesity 日期: 2025

Abstract

PURPOSE OF REVIEW: Endogenous **Cushing's syndrome** is one of the most difficult diagnoses in **endocrinology**. It is mainly based on biochemical tests that assess qualitative and quantitative alterations in **cortisol** production. Whether performed on blood, **saliva**, **urine**, or hair, the biochemical tests currently available have limited **diagnostic specificity** and sensitivity. Their performance varies depending on the individual characteristics of the patient, the stage of the disease, and the laboratory techniques used to determine **cortisol** levels. Knowledge of these factors will help physicians choose the most appropriate diagnostic test for each individual. RECENT FINDINGS: Relevant studies and meta-analyses have clarified the actual performance of various biochemical tests in a real-world setting. In addition, the diagnostic arsenal for **Cushing's syndrome** has been expanded with new tools, such as **hair cortisol** and **cortisone** assays, new methods for measuring **cortisol** in **urine** or **saliva**, and new approaches such as **steroid metabolomics** or **multiomics** techniques. SUMMARY: This review is a critical reappraisal of the merits and drawbacks of the various "old" and new biochemical tools. Its purpose is to provide the physician with practical advice on the use of biochemical tests at various stages of **Cushing's syndrome** in order to improve diagnostic accuracy.

中文翻译

综述目的：内源性库欣综合征是内分泌学中最具诊断难度的疾病之一，其诊断主要基于评估皮质醇生成质与量改变的生化检测。无论是血液、唾液、尿液还是毛发检测，现有生化检测方法的诊断特异性和敏感性均有限。检测效能因患者个体特征、疾病分期及皮质醇测定实验室技术而异。掌握这些因素有助于医师为每位患者选择最适宜的诊断检测。近期发现：相关研究与荟萃分析明确了各类生化检测在真实临床环境中的实际效能。此外，库欣综合征的诊断工具库已扩展至新型手段，包括毛发皮质醇与可的松检测、尿液或唾液皮质醇测定新方法，以及类固醇代谢组学或多组学技术等创新方案。总结：本综述对各类“传统”与新型生化检测工具的优缺点进行了批判性再评估，旨在为医师提供库欣综合征不同阶段合理运用生化检测的实用建议，以提升诊断准确性。

核心词汇

Cushing's syndrome /ˈkʊʃɪŋz ˈsɪndroʊm/ : 库欣综合征	endocrinology /ˌendəʊkrɪˈnɒlədʒi/ : 内分泌学
cortisol /ˈkɔːrtɪsɒl/ : 皮质醇	biochemical test /ˌbaɪəʊˈkɛmɪkəl tɛst/ : 生化检测
diagnostic specificity /ˌdaɪəɡˈnɒstɪk ˌspɛsɪˈfɪsɪti/ : 诊断特异性	diagnostic sensitivity /ˌdaɪəɡˈnɒstɪk ˌsɛnsɪˈtɪvɪti/ : 诊断敏感性
saliva /səˈlaɪvə/ : 唾液	urine /ˈjʊrɪn/ : 尿液
hair cortisol /hɛr ˈkɔːrtɪsɒl/ : 毛发皮质醇	cortisone /ˈkɔːrtɪzəʊn/ : 可的松
steroid metabolomics /ˈstɛrɔɪd ˌmɛtəˈbɒləmɪks/ : 类固醇代谢组学	multiomics /ˌmʌltiˈoʊmɪks/ : 多组学

Hyperuricemia and Gout Reduction by SGLT2 Inhibitors in Diabetes and Heart Failure: JACC Review Topic of the Week.

SGLT2抑制剂在糖尿病和心力衰竭中降低高尿酸血症和痛风的作用：JACC本周评论主题

期刊: Journal of the American College of Cardiology 日期: 2024

Abstract

gout is characterized by increased production of purines (through the **pentose phosphate pathway**), which is coupled with reduced renal or intestinal excretion of **urate**. Concurrent upregulation of nutrient surplus signaling (mammalian target of rapamycin and hypoxia-inducible factor-1a) and downregulation of nutrient deprivation signaling (sirtuin-1 and adenosine monophosphate-activated protein kinase) redirects glucose toward anabolic pathways (rather than adenosine triphosphate production), thus promoting heightened **oxidative stress** and cardiomyocyte and proximal tubular dysfunction, leading to **cardiomyopathy** and **kidney disease**. **hyperuricemia** is a marker (rather than a driver) of these cellular stresses. By inducing a state of starvation mimicry in a state of nutrient surplus, sodium-glucose cotransporter-2 inhibitors decrease flux through the **pentose phosphate pathway** (thereby attenuating **purine** and **urate** synthesis) while promoting renal **urate** excretion. These convergent actions exert a meaningful effect to lower serum **uric acid** by approximately 0.6 to 1.5 mg/dL and to reduce the risk of **gout** by 30% to 50% in large-scale clinical trials.

中文翻译

痛风的特点是嘌呤生成增加（通过磷酸戊糖途径），同时伴随肾脏或肠道尿酸排泄减少。营养过剩信号（雷帕霉素靶蛋白和缺氧诱导因子-1α）的上调与营养缺乏信号（去乙酰化酶-1和腺苷酸活化蛋白激酶）的下调，共同将葡萄糖转向合成代谢途径（而非生成三磷酸腺苷），从而加剧氧化应激，导致心肌细胞和近端肾小管功能障碍，进而引发心肌病和肾脏疾病。高尿酸血症是这些细胞应激的标志物（而非驱动因素）。通过模拟营养过剩状态下的饥饿状态，钠-葡萄糖协同转运蛋白2抑制剂可减少磷酸戊糖途径的通量（从而抑制嘌呤和尿酸的合成），同时促进肾脏尿酸排泄。这些协同作用在大型临床试验中显著降低血清尿酸水平约0.6至1.5 mg/dL，并将痛风风险降低30%至50%。

核心词汇

gout /ɡaʊt/ : 痛风	purine /ˈpjʊəriːn/ : 嘌呤
urate /ˈjʊəreɪt/ : 尿酸盐	hyperuricemia /ˌhaɪpərˈjʊərɪˈsiːmiə/ : 高尿酸血症
oxidative stress /ˈɒksɪdeɪtɪv stres/ : 氧化应激	cardiomyopathy /ˌkɑːrdiəʊmaɪˈɒpəθi/ : 心肌病
kidney disease /ˈkɪdni dɪˈziːz/ : 肾脏疾病	pentose phosphate pathway /ˈpentəʊs ˈfɒsfet ˈpɑːθweɪ/ : 磷酸戊糖途径
anabolic pathway /ˌænəˈbɒlɪk ˈpɑːθweɪ/ : 合成代谢途径	renal excretion /ˈriːnəl ɪkˈskriːʃən/ : 肾脏排泄
uric acid /ˈjʊərɪk ˈæsɪd/ : 尿酸	

Update on Physical, Psychological, and Quality of Life Management in Klinefelter Syndrome.

克氏综合征患者身体、心理及生活质量管理最新进展

期刊: The Journal of clinical endocrinology and metabolism 日期: 2025

Abstract

CONTEXT: **Klinefelter syndrome** (KS), most commonly arises from a 47,XXY **karyotype**. While KS affects around 1 in 450 to 600 male births, an estimated 50% to 75% of cases go undiagnosed. Individuals with KS are at increased risk of **metabolic**, **cardiovascular**, and **reproductive** comorbidities, increasing mortality risk, psychological burden, and significantly diminished health-related quality of life (HR-QoL) compared to healthy controls. We provide an updated review of the recent literature on the clinical management for people with KS, associated comorbidities, and the implications on HR-QoL. EVIDENCE ACQUISITION: A comprehensive literature search was conducted of key databases MEDLINE, CINAHL, Cochrane, Psychinfo, and EMBASE, followed by a gray search of relevant key papers in KS with medical **guideline** organizations being searched online and, where available, their publications and proposed guidelines being assessed. All databases were searched from their inception until December 2024, and English-language restrictions applied. EVIDENCE SYNTHESIS: Current evidence highlights the need for early detection, the importance of appropriate medication management, and **multidisciplinary** care to address **infertility**, cancer risks, **neurocognitive** deficits, and adverse **mental health**. Lifespan-specific interventions remain underexplored, necessitating further research to optimize outcomes and refine clinical guidelines. CONCLUSION: KS management is guided by only one endorsed **guideline**. International and interprofessional collaboration is needed to develop consensus documents. Existing guidelines pay minimal attention to the numerous HR-QoL challenges, overlooking the effect and diagnosis of psychological comorbidities. Enhancing HR-QoL and optimizing physical and mental well-being should be prioritized to improve HR-QoL outcomes in people with KS.

中文翻译

背景：克氏综合征（KS）最常见源于47,XXY核型。尽管KS影响约1/450至1/600的男性新生儿，但估计有50%至75%的病例未被确诊。与健康对照组相比，KS患者发生代谢、心血管和生殖系统合并症的风险更高，死亡风险增加，心理负担加重，健康相关生活质量（HR-QoL）显著下降。本文对KS患者临床管理、相关合并症及其对HR-QoL影响的最新文献进行了综述更新。证据获取：对MEDLINE、CINAHL、Cochrane、Psychinfo和EMBASE等主要数据库进行了全面文献检索，随后对KS相关重要论文进行灰色文献检索，并在线检索医学指南组织，评估其已发表文献和拟议指南。所有数据库检索时间均从建库至2024年12月，并限定英文文献。证据综合：现有证据强调早期检测的重要性、合理用药管理的必要性，以及通过多学科协作解决不孕不育、癌症风险、神经认知缺陷和不良心理健康问题。针对不同生命周期的干预措施仍研究不足，需进一步研究以优化结局并完善临床指南。结论：KS管理仅有一项经认可的指南作为指导。需通过国际与跨专业合作制定共识性文件。现有指南对HR-QoL面临的诸多挑战关注甚少，忽视了心理合并症的影响与诊断。应优先提升HR-QoL并优化身心健康，以改善KS患者的HR-QoL结局。

核心词汇

Klinefelter syndrome /ˈklaɪnˌfɛltər ˈsɪndroʊm/ : 克氏综合征	karyotype /ˈkæriəˌtaɪp/ : 核型
comorbidity /ˌkoumɔːrˈbɪdɪti/ : 合并症	metabolic /ˌmetəˈbɒlɪk/ : 代谢的
cardiovascular /ˌkɑːrdiəvˈæskjələr/ : 心血管的	reproductive /ˌriːprəˈdʌktɪv/ : 生殖的
infertility /ˌɪnfərˈtɪlɪti/ : 不育症	neurocognitive /ˌnjʊərəʊˈkɒɡnɪtɪv/ : 神经认知的
mental health /ˈmentəl heɪθ/ : 心理健康	multidisciplinary /ˌmʌltiˈdɪsəplɪnəri/ : 多学科的
guideline /ˈɡaɪdlaɪn/ : 指南	

Navigating fertility dilemmas across the lifespan in girls with Turner syndrome-a scoping review.

特纳综合征女孩一生中面临的生育困境——一项范围综述。

期刊: Human reproduction update 日期: 2024

Abstract

BACKGROUND: Girls with **Turner syndrome** (TS) lack a partial or complete sex chromosome, which causes an accelerated decline of their **ovarian reserve**. Girls have to deal with several dilemmas related to their **fertility**, while only a limited number of them are referred to a **fertility** specialist and counselled about options of family planning on time. **OBJECTIVE AND RATIONALE:** This scoping review provides an update of the literature on **fertility** in girls with TS throughout their lifespan and aims to propose a clinical practice guideline on **fertility** in TS. **SEARCH METHODS:** Databases of PubMed, Embase, and Web of science were searched using the following key terms: **Turner syndrome**, **fertility**, **puberty**, **pregnancy**, sex-hormones, **karyotype**, **fertility preservation**, assisted reproductive techniques, and counselling, alongside relevant subject headings and synonymous terms. English language articles published since 2007 were critically reviewed. Pregnancies after using donated oocytes and data about girls with TS with Y-chromosomal content were excluded. **OUTCOMES:** This search identified 1269 studies of which 120 were extracted for the review. The prevalence of natural conception ranged from 15% to 48% in women with 45,X/46,XX, 1% to 3% in women with 45,X, and 4% to 9% in women with other TS karyotypes. When assessing a girl's **fertility** potential, it was crucial to determine the **karyotype** in two cell lines, because hidden **mosaicism** may exist. In addition to **karyotype**, assessment of anti-Mullerian hormone (AMH) played a significant role in estimating ovarian function. Girls with AMH above the detection limit were most likely to experience spontaneous **thelarche**, **menarche**, and ongoing ovarian function during the reproductive lifespan. **fertility preservation** became more routine practice: **vitrification** of oocytes was reported in 58 girls with TS and a median of five oocytes were preserved per stimulation. **ovarian tissue cryopreservation** has demonstrated the presence of follicles in approximately 30% of girls with TS, mostly in girls with mosaic-TS, spontaneous **puberty**, and AMH above the detection limit. Although girls and their parents appreciated receiving counselling on **fertility** in TS, only one in ten girls with TS received specialized counselling. Unfamiliarity with **fertility preservation** techniques or uncertainties regarding the eligibility of a girl for **fertility preservation** constituted barriers for healthcare professionals when discussing **fertility** with girls with TS. **WIDER IMPLICATIONS:** There currently is a high demand for **fertility preservation** techniques in girls with TS. A reliable prognostic model to determine which girls with TS might benefit from **fertility preservation** is lacking. Only a minority of these girls received comprehensive **fertility** counselling on the full spectrum of **fertility**, including uncertainties of **fertility preservation**, **pregnancy** risks, and alternatives, such as adoption. **fertility preservation** could be a viable option for girls with TS. However, the question remains whether enough oocytes can be obtained for a realistic prospect of a live birth. It is important that girls and parents are empowered with the necessary information to make a well-informed decision.

中文翻译

背景: 特纳综合征 (TS) 女孩因部分或完全缺失一条性染色体, 导致其卵巢储备功能加速衰退。这些女孩需面对与生育能力相关的多重困境, 但仅有少数被转诊至生育专科医生处, 并接受关于计划生育选项的及时咨询。**目的与依据:** 本范围综述更新了TS女孩全生命周期生育能力的文献, 旨在提出TS患者生育管理的临床实践指南。**检索方法:** 在PubMed、Embase和Web of Science数据库中, 使用以下关键词进行检索: 特纳综合征、生育能力、青春期、妊娠、性激素、核型、生育力保存、辅助生殖技术及咨询, 同时结合相关主题词和同义词。对2007年以来发表的英文文献进行严格评审。排除使用捐赠卵母细胞妊娠的案例及含Y染色体成分的TS女孩数据。**结果:** 共检索到1269项研究, 其中120项纳入分析。自然受孕率在45,X/46,XX核型女性中为15%-48%, 45,X核型女性为1%-3%, 其他TS核型女性为4%-9%。评估女孩生育潜力时, 需确定两种细胞系的核型, 因为可能存在隐匿性嵌合体。除核型外, 抗缪勒管激素 (AMH) 检测在评估卵巢功能中起关键作用。AMH高于检测阈值的女孩更可能经历自发性乳房发育、月经初潮, 并在生育年龄维持卵巢功能。生育力保存逐渐成为常规操作: 58例TS女孩报告了卵母细胞玻璃化冷冻, 每次刺激平均保存5枚卵母细胞。卵巢组织冷冻保存显示约30%的TS女孩存在卵泡, 主要为嵌合型TS、自发性青春期及AMH高于检测阈值的女孩。尽管女孩及其父母认可接受TS生育咨询, 但仅十分之一的TS女孩获得专业咨询。医疗专业人员在与TS女孩讨论生育问题时, 因对生育力保存技术不熟悉或对女孩是否符合保存条件存在不确定性而面临障碍。**广泛意义:** 目前TS女孩对生育力保存技术需求极高, 但缺乏可靠的预测模型来确定哪些TS女孩可能从中获益。仅少数女孩接受了涵盖生育能力全貌 (包括生育力保存的不确定性、妊娠风险及领养等替代方案) 的全面生育咨询。对于部分TS女孩而言, 生育力保存可能是可行选择。

核心词汇

Turner syndrome /'tʃ:rnər 'sɪndroum/: 特纳综合征

ovarian reserve /oʊ'veəriən rɪ'zɜ:rv/: 卵巢储备

fertility /fərˈtɪləti/ : 生育力

mosaicism /mouˈzeɪsɪzəm/ : 嵌合体

menarche /məˈnɑːrki/ : 月经初潮

vitrification /ˌvɪtrɪfɪˈkeɪʃən/ : 玻璃化冷冻

ovarian tissue cryopreservation /ouˈvəriən ˈtɪʃuː ˌkraɪouːˌprezərˈveɪʃən/ : 卵巢组织冷冻保存

puberty /ˈpjuːbərti/ : 青春期

karyotype /ˈkəriotaɪp/ : 核型

thelarche /θɪˈləːrki/ : 乳房初发育

fertility preservation /fərˈtɪləti ˌprezərˈveɪʃən/ : 生育力保存

oocyte /ˈoʊəsait/ : 卵母细胞

follicle /ˈfɒlɪkəl/ : 卵泡

pregnancy /ˈpregnənsi/ : 妊娠

云梦山-侵权必究

Lifestyle management in polycystic ovary syndrome - beyond diet and physical activity.

多囊卵巢综合征的生活方式管理——超越饮食与体育锻炼

期刊: BMC endocrine disorders 日期: 2023

Abstract

polycystic ovary syndrome (PCOS) is a common condition affecting reproductive-aged women with reproductive, metabolic and psychological consequences. Weight and lifestyle (diet, physical activity and behavioural) management are first-line therapy in international evidence-based guidelines for PCOS. While these recommend following population-level diet and physical activity guidelines, there is ongoing interest and research in the potential benefit of including psychological and sleep interventions, as well as a range of traditional, complimentary and integrative medicine (TCIM) approaches, for optimal management of PCOS. There is limited evidence to recommend a specific diet composition for PCOS with approaches including modifying protein, carbohydrate or fat quality or quantity generally having similar effects on the presentations of PCOS. With regards to physical activity, promising evidence supports the provision of vigorous **aerobic exercise**, which has been shown to improve **body composition**, **cardiorespiratory fitness** and **insulin resistance**. Psychological and sleep interventions are also important considerations, with women displaying poor emotional wellbeing and higher rates of clinical and subclinical **sleep disturbance**, potentially limiting their ability to make positive lifestyle change. While optimising sleep and emotional wellbeing may aid symptom management in PCOS, research exploring the efficacy of clinical interventions is lacking. Uptake of TCIM approaches, in particular **supplement** and **herbal medicine** use, by women with PCOS is growing. However, there is currently insufficient evidence to support integration into routine clinical practice. Research investigating **inositol** supplementation have produced the most promising findings, showing improved metabolic profiles and reduced **hyperandrogenism**. Findings for other supplements, herbal medicines, **acupuncture** and **yoga** is so far inconsistent, and to reduce heterogeneity more research in specific PCOS populations, (e.g. defined age and BMI ranges) and consistent approaches to intervention delivery, duration and comparators are needed. While there are a range of lifestyle components in addition to population-recommendations for diet and physical activity of potential benefit in PCOS, robust clinical trials are warranted to expand the relatively limited evidence-base regarding holistic **lifestyle management**. With consumer interest in holistic healthcare rising, healthcare providers will be required to broaden their knowledge pertaining to how these therapies can be safely and appropriately utilised as adjuncts to conventional medical management.

中文翻译

多囊卵巢综合征（PCOS）是一种常见疾病，影响育龄女性的生殖、代谢及心理健康。国际循证指南将体重与生活方式（饮食、体力活动及行为）管理作为PCOS的一线疗法。尽管指南建议遵循人群层面的饮食与体力活动指南，但关于心理干预、睡眠干预以及传统、补充与整合医学（TCIM）方法对PCOS优化管理的潜在益处，仍持续引发关注与研究。目前尚无充分证据推荐PCOS的特定饮食构成，调整蛋白质、碳水化合物或脂肪的质量或数量等方法对PCOS临床表现的影响通常相似。在体力活动方面，有前景的证据支持进行高强度有氧运动，该运动已被证实可改善身体成分、心肺功能及胰岛素抵抗。心理与睡眠干预同样重要——PCOS女性常表现出情绪健康水平低下，且临床及亚临床睡眠障碍发生率较高，这可能限制其积极改变生活方式的能力。尽管优化睡眠与情绪健康可能有助于PCOS症状管理，但探索临床干预效果的研究仍较为缺乏。PCOS女性对TCIM方法（尤其是补充剂与草药）的使用率正在上升，但目前尚无充分证据支持将其纳入常规临床实践。关于肌醇补充剂的研究结果最具前景，显示其可改善代谢指标并降低高雄激素血症。其他补充剂、草药、针灸及瑜伽的研究结果尚不一致。为减少异质性，需针对特定PCOS人群（如限定年龄与BMI范围）开展更多研究，并采用一致的干预实施方式、持续时间及对照方案。尽管除人群推荐的饮食与体力活动外，多种生活方式干预对PCOS具有潜在益处，但仍需开展高质量临床试验以扩展当前相对有限的整体生活方式管理证据基础。随着消费者对整体健康关注度的提升，医疗保健提供者需拓宽知识储备，了解如何安全、恰当地将这些疗法作为常规医疗管理的辅助手段。

核心词汇

polycystic ovary syndrome /ˌpɒliˈsɪstɪk ˈəʊvəri ˈsɪndrəm/ : 多囊卵巢综合征	hyperandrogenism /ˌhaɪpəˈræn ˈdrɒdʒɪnɪzəm/ : 高雄激素血症
insulin resistance /ˈɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗	lifestyle management /ˈlaɪfstɑɪl ˈmænɪdʒmənt/ : 生活方式管理
aerobic exercise /eəˈrəʊbɪk ˈeksəsaɪz/ : 有氧运动	body composition /ˈbɒdi ˌkɒmpəˈzɪʃən/ : 身体成分
cardiorespiratory fitness /ˌkɑːdiəʊrɪˈspɪrətəri ˈfɪtnəs/ : 心肺适能	sleep disturbance /sliːp dɪˈstɜːbəns/ : 睡眠障碍

supplement /ˈsʌplɪmənt/ : 补充剂

inositol /ɪˈneʊsɪtəl/ : 肌醇

yoga /ˈjəʊgə/ : 瑜伽

herbal medicine /ˈhɜːbəl ˈmedɪsɪn/ : 草药医学

acupuncture /ˈækjʊpʌŋktʃə/ : 针灸

云梦山-侵权必究

Precocious Puberty in Hypothyroidism: Mini-Review of Van Wyk-Grumbach Syndrome.

甲状腺功能减退症中的性早熟：Van Wyk-Grumbach综合征的迷你综述

期刊: Journal of the Endocrine Society 日期: 2023

Abstract

Severe **hypothyroidism** can affect a variety of organs and can develop atypical manifestations. Peripheral **precocious puberty** may be secondary to other endocrinological diseases, which must be taken into account in the **differential diagnosis** in order to avoid unnecessary additional tests. Van Wyk-Grumbach **syndrome** is an infrequent manifestation characterized by severe **hypothyroidism** and incomplete **precocious puberty**. Diagnosis is made by clinical and complementary tests, and the main treatment goal is to achieve **euthyroidism** through **hormone replacement**. **prognosis** is good once the treatment is established. The aim of this study is to review the available literature about Van Wyk-Grumbach **syndrome** following the PRISMA statement, and to present the first **clinical case** published in Spain. We have included the articles published during the period from 1905 to week 40 of 2022. A total of 68 articles have been selected for study and analysis, within which there are 99 published clinical cases. Girls accounted for 92.1% of cases (median age at the diagnosis 8.5 years). **metrorrhagia** was the most prevalent symptom, present in 80.5% of the girls. **abdominal ultrasound** was performed in 93.3% of the girls and 97.8% of them had at least one **ovarian cyst**. All cases were treated with **levothyroxine**, responding satisfactorily after the first doses of treatment. To conclude, Van Wyk-Grumbach **syndrome** is characterized by severe **hypothyroidism** and incomplete **precocious puberty**, which is important to keep in mind in order to avoid complementary exams and unnecessary surgical interventions.

中文翻译

严重甲状腺功能减退症可影响多种器官，并出现非典型表现。外周性性早熟可能继发于其他内分泌疾病，在鉴别诊断中必须考虑这一点，以避免不必要的额外检查。Van Wyk-Grumbach综合征是一种罕见表现，其特征为严重甲状腺功能减退症和不完全性性早熟。诊断需通过临床及辅助检查确定，主要治疗目标是通过激素替代疗法实现甲状腺功能正常化。一旦开始治疗，预后良好。本研究旨在遵循PRISMA声明回顾关于Van Wyk-Grumbach综合征的现有文献，并介绍西班牙首例临床病例。我们纳入了1905年至2022年第40周期间发表的文章，共筛选出68篇文献进行分析，其中包含99例临床病例。女性患者占92.1%（诊断时中位年龄8.5岁）。月经过多是最常见症状，见于80.5%的女性患者。93.3%的女性患者接受了腹部超声检查，其中97.8%存在至少一个卵巢囊肿。所有病例均使用左甲状腺素治疗，首次给药后反应良好。总之，Van Wyk-Grumbach综合征以严重甲状腺功能减退症和不完全性性早熟为特征，需牢记此特点以避免不必要的辅助检查和手术干预。

核心词汇

hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症	precocious puberty /prɪˈkʊʃəs ˈpjʊːbərti/ : 性早熟
syndrome /ˈsɪndrəm/ : 综合征	differential diagnosis /ˌdɪfəˈrɛnʃəl ˌdaɪəɡˈnoʊsɪs/ : 鉴别诊断
euthyroidism /juːˈθaɪrɔɪdɪzəm/ : 甲状腺功能正常状态	hormone replacement /ˈhɔːrmoun rɪˈpleɪsmənt/ : 激素替代治疗
metrorrhagia /ˌmɛtrəˈreɪdʒiə/ : 子宫不规则出血	abdominal ultrasound /æbˈdɒmɪnəl ˈʌltrəsəʊnd/ : 腹部超声检查
ovarian cyst /oʊˈvɛəriən sɪst/ : 卵巢囊肿	levothyroxine /ˌliːvəʊθaɪˈrɒksɪːn/ : 左甲状腺素
prognosis /ˈprɒɡˈnoʊsɪs/ : 预后	clinical case /ˈklɪnɪkəl keɪs/ : 临床病例

Plasma metabolomics in male primary and functional hypogonadism.

男性原发性和功能性性腺功能减退症的血浆代谢组学研究

期刊: Frontiers in endocrinology 日期: 2023

Abstract

metabolomics proposes to unveil the molecular machinery involved in each specific disease by the comprehensive analysis of low-molecular-weight metabolites in a biological sample. This narrative mini-review analyzes previous studies applying ultra-high-performance liquid chromatography-high-resolution mass spectrometry (HRMS)-based **metabolomics** to highlight different metabolic pathways involved in male **hypogonadism** and **testosterone** replacement therapy, both in the case of **insulin**-sensitive patients with primary **hypogonadism** and in the case of **insulin**-resistant patients with functional **hypogonadism**. In functional **hypogonadism**, **metabolomics** revealed that different biochemical pathways are affected. In detail, **glycolysis** is the most important biochemical process involved in these patients. Glucose metabolism is fueled by amino acid degradation, and **gluconeogenesis** is widely stimulated. Some important pathways, including glycerol, are compromised. Furthermore, **mitochondrial** electron transport is influenced, namely, by a decrease in ATP production. On the contrary, beta-oxidation of short- and medium-chain fatty acids does not represent an energy source in hypogonadal patients. Both lactate and acetyl-CoA are converted into **ketone bodies**, which increased immensely. However, **carnosine** and beta-alanine are greatly reduced. These metabolic changes are associated with increased **fatigue** and mental confusion. After **testosterone** replacement therapy, a complete restoration is achieved for only a part of the metabolites. It is of note that only in patients with functional **hypogonadism** treated with **testosterone** are **ketone bodies** produced at high levels, so the symptoms sometimes reported by these patients after the beginning of the therapy (difficulty in concentrating, depressed mood, brain fog, and memory impairment) might represent a specific "keto flu-like" syndrome, related to the metabolic ketonic state.

中文翻译

代谢组学旨在通过对生物样本中低分子量代谢物的全面分析，揭示每种特定疾病涉及的分子机制。本叙述性小型综述分析了以往基于超高效液相色谱-高分辨质谱（HRMS）的代谢组学研究，以突出男性性腺功能减退症及睾酮替代疗法中涉及的不同代谢通路，研究对象包括胰岛素敏感的原发性性腺功能减退症患者和胰岛素抵抗的功能性性腺功能减退症患者。在功能性性腺功能减退症中，代谢组学揭示多种生化通路受到影响。具体而言，糖酵解是这些患者中最重要的生化过程。葡萄糖代谢由氨基酸降解驱动，糖异生被广泛激活。包括甘油在内的一些重要通路受损。此外，线粒体电子传递受到影响，表现为ATP生成减少。相反，短链和中链脂肪酸的β氧化并不构成性腺功能减退症患者的能量来源。乳酸和乙酰辅酶A均转化为酮体，其水平显著升高。然而，肌肽和β-丙氨酸大幅减少。这些代谢变化与疲劳加剧和精神混乱相关。在睾酮替代疗法后，仅部分代谢物完全恢复。值得注意的是，仅在接受睾酮治疗的功能性性腺功能减退症患者中，酮体以高水平产生，因此这些患者在治疗开始后有时报告的症状（注意力难以集中、情绪低落、脑雾和记忆障碍）可能代表一种与代谢性酮症状态相关的特定“酮流感样”综合征。

核心词汇

metabolomics /ˌmetəbeˈlɒmɪks/ : 代谢组学

testosterone /ˈteːstəstəroun/ : 睾酮

glycolysis /ˈɡlaɪˈkɒlɪsɪs/ : 糖酵解

mitochondrial /ˌmaɪtəʊˈkɒndriəl/ : 线粒体的

carnosine /ˈkɑːrnəʊsiːn/ : 肌肽

hypogonadism /ˌhaɪpəʊˈɡɒnədəɪzəm/ : 性腺功能减退症

insulin /ˈɪnsjʊlɪn/ : 胰岛素

gluconeogenesis /ˌɡluːkəʊˈniːəʊˈdʒenɪsɪs/ : 糖异生

ketone bodies /ˈkiːtəʊnˈbɒdɪz/ : 酮体

fatigue /fəˈtiːɡ/ : 疲劳

Puberty and Growth in Cystic Fibrosis: A Review.

囊性纤维化中的青春期中与生长：综述。

期刊: Endocrine practice : official journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists

日期: 2026

Abstract

cystic fibrosis (CF) is a chronic disease process that has historically been associated with delayed puberty and growth. Mechanisms for delayed growth and puberty in chronic disease include malnutrition, inflammation, alterations of the growth hormone axis, chronic corticosteroid use, and changes to gonadotropin signaling or release. Early evidence suggested that people with CF have delayed puberty, decreased peak height velocity, and decreased final adult height compared to the general population. Management of delayed puberty in CF typically involves addressing the underlying cause of delay, and evidence for the utility of growth hormone to optimize growth in people with CF is inconsistent. In the era of highly effective modulator therapies for CF, early studies are examining changes in the growth and puberty trajectory in youth with CF. The full impact of highly effective modulator therapy on height outcomes remains unclear.

中文翻译

囊性纤维化（CF）是一种慢性疾病过程，历来与青春期延迟和生长迟缓相关。慢性疾病中生长与青春期延迟的机制包括营养不良、炎症、生长激素轴改变、长期使用皮质类固醇以及促性腺激素信号传导或释放的变化。早期证据表明，与普通人群相比，CF患者存在青春期延迟、身高增长峰值速度降低以及最终成年身高减少。CF患者青春期延迟的管理通常涉及解决延迟的根本原因，而生长激素在优化CF患者生长方面的效用证据并不一致。在CF高效调节剂治疗时代，早期研究正在探讨CF青少年生长与青春期轨迹的变化。高效调节剂治疗对身高结局的全面影响仍不明确。

核心词汇

cystic fibrosis /ˈsɪstɪk faɪˈbrɒsɪs/ : 囊性纤维化	puberty /ˈpjuːbərti/ : 青春期
growth /ɡrouθ/ : 生长	malnutrition /ˌmælnuːˈtrɪʃən/ : 营养不良
inflammation /ˌɪnfləˈmeɪʃən/ : 炎症	growth hormone /ɡrouθ ˈhɔːrmoun/ : 生长激素
corticosteroid /ˌkɔːrtɪkɒsˈtɛrɔɪd/ : 皮质类固醇	gonadotropin /ɡoʊˌnædəˈtrɒpɪn/ : 促性腺激素
peak height velocity /piːk haɪt vɪˈləːsəti/ : 身高增长峰值速度	modulator therapy /ˈmɔːdʒəleɪtər ˈθerəpi/ : 调节剂治疗
height /haɪt/ : 身高	chronic disease /ˈkrɑːnɪk dɪˈziːz/ : 慢性疾病

A Comprehensive Review of Syndromic Forms of Obesity: Genetic Etiology, Clinical Features and Molecular Diagnosis.

综合征性肥胖的综合综述：遗传病因学、临床特征与分子诊断

期刊: Current obesity reports 日期: 2024

Abstract

Syndromic **obesity** refers to **obesity** occurring with additional clinical findings, such as **intellectual disability/developmental delay**, **dysmorphic features**, and congenital malformations. PURPOSE OF REVIEW: To present a narrative review regarding the **genetic etiology**, clinical description, and **molecular diagnosis** of syndromic **obesity**, which is a rare condition with high phenotypic variability and genetic heterogeneity. The following syndromes are presented in this review: Prader-Willi, Bardet-Biedl, Pseudohypoparathyroidism, Alstrom, Smith-Magenis, Cohen, Temple, 1p36 deletion, 16p11.2 microdeletion, Kleeftstra, SIM1-related, Borjeson-Forssman-Lehmann, WAGRO, Carpenter, MORM, and MYT1L-related syndromes. RECENT FINDINGS: There are three main groups of mechanisms for syndromic **obesity**: **imprinting**, **transcriptional activity regulation**, and cellular **cilia** function. For molecular diagnostic, methods of genome-wide investigation should be prioritized over sequencing of panels of syndromic **obesity** genes. In addition, we present novel syndromic conditions that need further delineation, but evidences suggest they have a higher frequency of **obesity**. The etiology of syndromic **obesity** tends to be linked to disrupted neurodevelopment (central) and is associated with a diversity of genes and biological pathways. In the genetic investigation of individuals with syndromic **obesity**, the possibility that the etiology of the syndromic condition is independent of **obesity** should be considered. The accurate genetic diagnosis impacts medical management, treatment, and prognosis, and allows proper **genetic counseling**.

中文翻译

综合征性肥胖是指肥胖伴随其他临床表现，如智力障碍/发育迟缓、畸形特征和先天性畸形。综述目的：本文对综合征性肥胖的遗传病因、临床描述和分子诊断进行叙述性综述，该疾病是一种罕见病，具有高度表型变异性和遗传异质性。本综述涵盖以下综合征：普拉德-威利综合征、巴德-比德尔综合征、假性甲状腺功能减退症、阿尔斯特伦综合征、史密斯-马吉利综合征、科恩综合征、坦普尔综合征、1p36缺失综合征、16p11.2微缺失综合征、克利夫斯特拉综合征、SIM1相关综合征、博耶森-福斯曼-莱曼综合征、WAGRO综合征、卡彭特综合征、MORM综合征和MYT1L相关综合征。最新发现：综合征性肥胖的机制主要分为三类：基因组印记、转录活性调控和细胞纤毛功能。在分子诊断方面，应优先采用全基因组检测方法而非综合征性肥胖基因组合测序。此外，本文提出了一些需要进一步界定的新型综合征，但已有证据表明这些综合征的肥胖发生率较高。综合征性肥胖的病因往往与神经发育障碍（中枢性）相关，并涉及多种基因和生物学通路。在对综合征性肥胖患者进行遗传学调查时，应考虑综合征的病因可能与肥胖无关。准确的遗传诊断会影响医疗管理、治疗和预后，并有助于进行适当的遗传咨询。

核心词汇

obesity /ou'bi:seɪti/ : 肥胖	syndrome /'sɪndroum/ : 综合征
intellectual disability /'ɪntə'lektʃuəl ,dɪsə'bi:leɪti/ : 智力障碍	developmental delay /dɪ'veləp'mentəl dɪ'leɪ/ : 发育迟缓
dysmorphic features /dɪs'mɔ:rfɪk 'fi:tʃəz/ : 畸形特征	congenital malformation /kən'dʒɛnɪtl ,mælfɔ:r'meɪʃən/ : 先天性畸形
genetic etiology /dʒə'netɪk ,i:ti'a:lədʒi/ : 遗传病因学	molecular diagnosis /mə'lekjʊlə ,daɪəg'nouɪsɪs/ : 分子诊断
imprinting /ɪm'prɪntɪŋ/ : 印记	transcriptional activity regulation /træn'skrɪpʃənəl æk'tɪvəti ,rɛgju'leɪʃən/ : 转录活性调控
cilia /'sɪliə/ : 纤毛	genetic counseling /dʒə'netɪk 'kaʊnsəlɪŋ/ : 遗传咨询

Advances in Parathyroid Hormone-based medicines.

基于甲状旁腺激素的药物的进展。

期刊: Journal of bone and mineral research : the official journal of the American Society for Bone and Mineral Research

日期: 2025

Abstract

parathyroid hormone (PTH), produced by the parathyroid glands, plays a critical role in the regulation of calcium and phosphate homeostasis, acting primarily on bone and kidney to maintain serum calcium levels within a narrow range. parathyroid hormone also plays important roles in bone remodeling by directly stimulating osteoblasts and osteocytes, integrating its calcemic response with stimulation of bone formation. Through the RANK/RANK-ligand system, these cells activate osteoclasts, promoting a balanced process of bone formation and resorption that maintains bone density and strength. Dysregulation of PTH, as seen in disorders such as hyper- and hypoparathyroidism, can lead to significant clinical complications. In recent years, major advancements have been made in the development of PTH analogs, aimed at leveraging PTH's physiological effects on bone to treat conditions such as osteoporosis and hypoparathyroidism. While PTH promotes both bone formation and bone resorption, the net outcome may be a gain or loss of bone mass, depending largely on the administration pattern of PTH or its analogs. When PTH is given intermittently (eg, as once-daily subcutaneous injection), bone formation is favored. Continuous administration of PTH or chronic elevation of blood PTH levels as seen in primary hyperparathyroidism tend to promote bone resorption. parathyroid hormone analogs, such as teriparatide (PTH(1-34)) and the PTHrP analog abaloparatide, administered once daily, have significant efficacy in stimulating bone formation, making them valuable options for the treatment of osteoporosis. Given this capacity to improve bone structure, these analogs hold broader therapeutic potential for other skeletal disorders, including fracture healing and oral bone repair, which expands the scope of PTH-based therapies beyond osteoporosis. Long-acting PTH analogs have applications in treating hypoparathyroidism, offering an alternative to conventional treatment with calcium and active vitamin D. This article reviews the molecular mechanisms of approved and emerging PTH-based medicines, their clinical applications, and recent advances in optimizing their therapeutic potential. We also discuss ongoing research aimed at developing next-generation PTH analogs with improved efficacy for skeletal and metabolic disorders.

中文翻译

甲状旁腺激素（PTH）由甲状旁腺分泌，在调节钙磷稳态中发挥关键作用，主要通过作用于骨骼和肾脏将血清钙水平维持在狭窄范围内。甲状旁腺激素还通过直接刺激成骨细胞和骨细胞，将钙调节反应与骨形成刺激相结合，在骨重塑中发挥重要作用。这些细胞通过RANK/RANK配体系统激活破骨细胞，促进骨形成与骨吸收的平衡过程，从而维持骨密度和强度。PTH调节异常（如甲状旁腺功能亢进或减退）可导致显著的临床并发症。近年来，基于PTH生理效应的类似物研发取得重大进展，旨在利用其对骨骼的作用治疗骨质疏松症和甲状旁腺功能减退等疾病。虽然PTH既能促进骨形成也能促进骨吸收，但净效应可能是骨量增加或减少，这主要取决于PTH或其类似物的给药方式。间歇性给药（如每日一次皮下注射）时，骨形成占优势；而持续性给药或原发性甲状旁腺功能亢进导致的慢性血PTH升高则倾向于促进骨吸收。每日一次给药的甲状旁腺激素类似物（如特立帕肽[PTH(1-34)]和PTHrP类似物阿巴洛肽）在刺激骨形成方面具有显著疗效，成为治疗骨质疏松症的重要选择。鉴于其改善骨结构的能力，这些类似物在骨折愈合和口腔骨修复等其他骨骼疾病中具有更广泛的治疗潜力，拓展了PTH类药物的应用范围。长效PTH类似物可用于治疗甲状旁腺功能减退，为传统钙剂和活性维生素D治疗提供替代方案。本文综述了已获批和新兴PTH类药物的分子机制、临床应用及优化其治疗潜力的最新进展，并讨论了旨在开发新一代PTH类似物以改善骨骼和代谢疾病疗效的持续研究。

核心词汇

parathyroid hormone /ˌpærəˈθaɪrɔɪd ˈhɔːrmoun/ : 甲状旁腺激素	calcium /ˈkælsiəm/ : 钙
phosphate /ˈfɒsfeɪt/ : 磷酸盐	bone remodeling /ˌbaʊn rɪˈmɒdəlɪŋ/ : 骨重塑
osteoblast /ˈɒstiəblæst/ : 成骨细胞	osteoclast /ˈɒstiəklæst/ : 破骨细胞
osteoporosis /ˌɒstiəpəˈrəʊsɪs/ : 骨质疏松症	hyperparathyroidism /ˌhaɪpərˌpærəˈθaɪrɔɪdɪzəm/ : 甲状旁腺功能亢进症
hypoparathyroidism /ˌhaɪpəʊˌpærəˈθaɪrɔɪdɪzəm/ : 甲状旁腺功能减退症	bone resorption /ˌbaʊn rɪˈsɔːrpsən/ : 骨吸收
bone formation /ˌbaʊn fɔːrˈmeɪʃən/ : 骨形成	analog /ˈænəlɒɡ/ : 类似物

Revised European Society of Endocrinology Clinical Practice Guideline: Treatment of Chronic Hypoparathyroidism in Adults.

修订版欧洲内分泌学会临床实践指南：成人慢性甲状旁腺功能减退症的治疗

期刊: European journal of endocrinology 日期: 2025

Abstract

In the present European Society of **endocrinology** (ESE) clinical **guideline**, we present recommendations for the **diagnosis**, **management**, and **monitoring** of chronic **hypoparathyroidism** (HypoPT) in adults. **management** of HypoPT has changed since the first ESE clinical **guideline** was published in 2015, as has the knowledge on patient burden of the disease, and the understanding of associated morbidities. In line with the ESE policy, the 2015 guidelines were updated based on up-to-date scientific evidence. As HypoPT is an **orphan disease**, strong evidence for most outcomes is scarce and recommendations were based on careful synthesis of the literature and expert opinion. **postsurgical** HypoPT should be defined as persisting more than 12 months following surgery; recovery could be expected even thereafter (clinical question I [Q I]). For Q II (optimal treatment of chronic HypoPT), relevant data regarding conventional treatment are lacking for clinically relevant endpoints and long-term effects. PTH **replacement therapy** reduces the pill burden of conventional therapy, improves various biochemical parameters, and potentially improves QoL. We cannot recommend a substantial role for **parathyroid allotransplantation** in the treatment of chronic HypoPT (Q III). In conclusion, we present recommendations for the **diagnosis**, **management**, and **monitoring** of chronic HypoPT in adults, to give health care providers practical clinical guidance on the **management** of this condition. The **guideline** can serve as a source for preparation of educational materials to empower patients and clinicians.

中文翻译

在当前欧洲内分泌学会（ESE）临床指南中，我们针对成人慢性甲状旁腺功能减退症（HypoPT）的诊断、管理和监测提出建议。自2015年首版ESE临床指南发布以来，HypoPT的管理方式已发生改变，同时人们对疾病给患者带来的负担及相关并发症的认识也有所深化。根据ESE政策，2015版指南基于最新科学证据进行了更新。由于HypoPT属于罕见病，多数结局指标缺乏强有力的证据支持，相关建议基于对文献的审慎整合及专家意见制定。术后HypoPT应定义为术后持续超过12个月；即使在此时间点后仍可能恢复（临床问题I [Q I]）。针对问题II（慢性HypoPT的最佳治疗），目前缺乏关于常规治疗对临床相关终点及长期影响的充分数据。甲状旁腺激素替代疗法可减少常规治疗的药片负担，改善多项生化指标，并可能提升生活质量。我们不建议将甲状旁腺同种异体移植作为慢性HypoPT的主要治疗手段（问题III [Q III]）。总之，我们提出成人慢性HypoPT诊断、管理和监测的建议，为医疗工作者提供该疾病管理的实用临床指导。本指南可作为编写患者及临床医生教育材料的参考依据。

核心词汇

hypoparathyroidism /ˌhaɪpəʊˈpærəˈθaɪrɔɪdɪzəm/ : 甲状旁腺功能减退症	parathyroid /ˌpærəˈθaɪrɔɪd/ : 甲状旁腺
endocrinology /ˌendəʊkrɪˈnɒlədʒi/ : 内分泌学	guideline /ˈɡaɪdlaɪn/ : 指南
diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断	management /ˈmænɪdʒmənt/ : 管理
monitoring /ˈmɒnɪtərɪŋ/ : 监测	orphan disease /ˈɔːrfən dɪˈziːz/ : 罕见病
postsurgical /ˈpəʊstˈsɜːrdʒɪkəl/ : 术后	replacement therapy /rɪˈpleɪsmənt ˈθerəpi/ : 替代治疗
allotransplantation /ˌæləʊˌtrænsplænˈteɪʃən/ : 同种异体移植	quality of life /ˈkwɒləti ɒv laɪf/ : 生活质量

Endocrine features of primary mitochondrial diseases.

原发性线粒体疾病的内分泌特征。

期刊: Current opinion in endocrinology, diabetes, and obesity 日期: 2024

Abstract

PURPOSE OF REVIEW: Primary mitochondrial diseases are one of the most prevalent groups of multisystem genetic disorders. Endocrinopathies associated with mitochondrial diseases may have clinical features that are distinct from the more common forms. We provide an overview of mitochondrial disorder genetics and phenotypes, focusing on recent studies regarding identification and treatment of associated endocrinopathies. RECENT FINDINGS: Known endocrine phenotypes of mitochondrial disorders continue to expand, and now include growth hormone deficiency, hypogonadism, precocious puberty, hypoparathyroidism, hypo- and hyperthyroidism, diabetes, and adrenal insufficiency. Recent studies suggest several genotype-phenotype correlations, including those related to nuclear variants. Diagnosis is important, as special considerations should be made in the management of endocrinopathies in mitochondrial patients. Finally, new mitochondrial replacement strategies may soon be available for women interested in preventing mitochondrial disease transmission to offspring. SUMMARY: Patients with multiple endocrinopathies or atypical endocrinopathies should be evaluated for primary mitochondrial disease, as a diagnosis may impact management of these individuals.

中文翻译

综述目的：原发性线粒体疾病是最常见的多系统遗传性疾病之一。线粒体疾病相关的内分泌病变可能具有与常见类型不同的临床特征。本文概述线粒体疾病的遗传学与表型，重点关注近期关于相关内分泌病变识别与治疗的研究。最新发现：线粒体疾病已知的内分泌表型持续扩展，目前包括生长激素缺乏、性腺功能减退、性早熟、甲状旁腺功能减退、甲状腺功能减退与亢进、糖尿病及肾上腺皮质功能不全。近期研究表明存在多种基因型-表型关联，包括与核基因变异相关的关联。诊断至关重要，因为线粒体疾病患者的内分泌病变管理需特殊考量。此外，针对希望预防线粒体疾病遗传给后代的女性，新的线粒体替代策略可能很快问世。总结：对于存在多种内分泌病变或非典型内分泌病变的患者，应评估其是否患有原发性线粒体疾病，因为诊断可能影响这些患者的治疗管理。

核心词汇

mitochondrial disease /ˌmaɪtəˈkɑːndriəl dɪˈziːz/ : 线粒体疾病	endocrinopathy /ˌendoukrɪˈnɒpəθi/ : 内分泌病
genetic disorder /dʒəˈnetɪk dɪsˈɔːdə/ : 遗传性疾病	phenotype /ˈfiːnətaɪp/ : 表型
growth hormone deficiency /ɡrouθ ˈhɔːrmoun dɪˈfɪjənsi/ : 生长激素缺乏症	hypogonadism /ˌhaɪpouˈɡounədɪzəm/ : 性腺功能减退症
precocious puberty /prɪˈkɒʃəs ˈpuːbərti/ : 性早熟	hypoparathyroidism /ˌhaɪpouˈpærəˈθaɪrɔɪdɪzəm/ : 甲状旁腺功能减退症
hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症	diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病
adrenal insufficiency /əˈdriːnəl ˌɪnsəˈfɪjənsi/ : 肾上腺功能不全	mitochondrial replacement /ˌmaɪtəˈkɑːndriəl rɪˈpleɪsmənt/ : 线粒体替代疗法

Hypoparathyroidism and assisted reproductive technology: considerations while undergoing ovarian hyperstimulation.

甲状旁腺功能减退与辅助生殖技术：卵巢过度刺激过程中的注意事项

期刊: Current opinion in endocrinology, diabetes, and obesity 日期: 2024

Abstract

PURPOSE OF REVIEW: hypoparathyroidism (hypoPTH) is a rare disease that requires diligent adherence to treatment regimens to prevent hypocalcemia but also treatment-induced hypercalcemia and hypercalciuria. The menstrual cycle, pregnancy, and lactation can all impact calcium homeostasis but there is little known regarding the impact of ovarian stimulation. Furthermore, the limited reports suggest no clear association between menstrual phase and calcium balance among those with hypoPTH. With increasing patient utilization of assisted reproductive technology (ART), there is a need for better understanding the care required for patients with hypoparathyroidism pursuing fertility technology. RECENT FINDINGS: There is currently no literature available on patients with hypoparathyroidism and the impact of controlled ovarian stimulation on calcium homeostasis. We present information regarding physiologic changes in pregnancy that impact calcium homeostasis and the first case presentation of a patient with hypoparathyroidism pursuing ART. SUMMARY: This article provides the first insights and guidance when providing fertility care for patients with hypoparathyroidism.

中文翻译

综述目的：甲状旁腺功能减退症（hypoPTH）是一种罕见疾病，需要患者严格遵循治疗方案以预防低钙血症，同时避免治疗引发的高钙血症和高钙尿症。月经周期、妊娠和哺乳均可能影响钙稳态，但关于卵巢刺激的影响知之甚少。此外，有限的研究报告表明，在甲状旁腺功能减退症患者中，月经周期与钙平衡之间无明显关联。随着患者对辅助生殖技术（ART）的使用增加，需更深入了解甲状旁腺功能减退症患者在接受生育技术时的护理需求。最新发现：目前尚无关于甲状旁腺功能减退症患者及控制性卵巢刺激对钙稳态影响的文献。本文介绍了妊娠期影响钙稳态的生理变化，并首次报告一例甲状旁腺功能减退症患者接受辅助生殖技术的病例。总结：本文为甲状旁腺功能减退症患者的生育护理提供了初步见解与指导。

核心词汇

hypoparathyroidism /ˌhaɪpəʊˈpærəˈθaɪrɔɪdɪzəm/：甲状旁腺功能减退症	hypocalcemia /ˌhaɪpəʊkælˈsiːmiə/：低钙血症
hypercalcemia /ˌhaɪpərˈkælˈsiːmiə/：高钙血症	hypercalciuria /ˌhaɪpərˈkælsiːˈjʊəriə/：高钙尿症
calcium homeostasis /ˈkælsiəmˌhoʊmiəˈsteɪsɪs/：钙稳态	menstrual cycle /ˈmenstruəlˈsaɪkl/：月经周期
pregnancy /ˈpregnənsi/：妊娠	lactation /lækˈteɪʃən/：哺乳
ovarian stimulation /oʊˈveriənˌstɪmjʊˈleɪʃən/：卵巢刺激	assisted reproductive technology /əˈsɪstɪdˌrɪːprəˈdʌktɪvˌtɛkˈnɒlədʒi/：辅助生殖技术
fertility /fərˈtɪlɪti/：生育力	parathyroid hormone /ˌpærəˈθaɪrɔɪdˈhɔːrmoun/：甲状旁腺激素

Epidemiologic changes in thyroid disease.

甲状腺疾病的流行病学变化。

期刊: Current opinion in endocrinology, diabetes, and obesity 日期: 2024

Abstract

PURPOSE OF REVIEW: To analyze the evolving epidemiologic trends in thyroid disease, focusing on risk factors, underlying drivers of these changes, and their implications on clinical practice and research priorities. RECENT FINDINGS: thyroid disease remains one of the most prevalent groups of disorders globally, and the shift in its frequency and distribution is multifactorial. The prevalence of hypothyroidism increases with age, although normal thyrotropin ranges appear to be age-dependent, raising concern for potentially inappropriate levothyroxine use. hyperthyroidism and Graves' disease continue to be predominant in reproductive-age women but exhibit a milder phenotype at diagnosis. Thyroid nodules are increasingly found in asymptomatic patients, likely from more widespread use of neck and chest imaging. thyroid cancer incidence has risen exponentially over the years, mostly driven by overdiagnosis of low-risk tumors; however, a small rise in incidence of higher risk tumors has been noted. obesity appears to be a risk factor for thyroid cancer occurrence and more aggressive forms of the disease. SUMMARY: Understanding epidemiologic trends in thyroid disease is crucial for guiding clinical practice and research efforts, aiming to optimize patient outcomes while preventing unnecessary and potentially harmful interventions.

中文翻译

综述目的：分析甲状腺疾病流行病学趋势的演变，重点关注风险因素、这些变化的潜在驱动因素及其对临床实践和研究重点的影响。最新发现：甲状腺疾病仍是全球最常见的疾病群之一，其频率和分布的变化受多因素影响。甲状腺功能减退症的患病率随年龄增长而增加，但正常促甲状腺激素范围似乎与年龄相关，这引发了对左甲状腺素可能不当使用的担忧。甲状腺功能亢进症和格雷夫斯病在育龄女性中仍占主导地位，但诊断时表型较轻。甲状腺结节在无症状患者中越来越常见，可能源于颈部和胸部影像学检查的广泛使用。甲状腺癌发病率多年来呈指数级增长，主要归因于低风险肿瘤的过度诊断；然而，高风险肿瘤的发病率略有上升。肥胖似乎是甲状腺癌发生及更具侵袭性疾病的危险因素。总结：理解甲状腺疾病的流行病学趋势对于指导临床实践和研究工作至关重要，旨在优化患者预后，同时避免不必要且可能有害的干预措施。

核心词汇

thyroid disease /ˈθaɪrɔɪd dɪˈziːz/：甲状腺疾病	hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/：甲状腺功能减退症
hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/：甲状腺功能亢进症	Graves disease /ɡreɪvz dɪˈziːz/：格雷夫斯病
thyroid nodule /ˈθaɪrɔɪd ˈnɒdjuːl/：甲状腺结节	thyroid cancer /ˈθaɪrɔɪd ˈkænsər/：甲状腺癌
levothyroxine /ˌliːvəʊˈθaɪˈrɒksiːn/：左甲状腺素	thyrotropin /θaɪˈrɒtrəpɪn/：促甲状腺激素
overdiagnosis /ˌoʊvərˌdaɪəɡˈnoʊsɪs/：过度诊断	risk factor /rɪsk ˈfæktər/：危险因素
obesity /oʊˈbiːsəti/：肥胖	phenotype /ˈfiːnətaɪp/：表型

Management Aspects of Medical Therapy in Graves Disease.

格雷夫斯病药物治疗的管理方面

期刊: Endocrine practice : official journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists

日期: 2025

Abstract

OBJECTIVE: Graves disease (GD) is the most common cause of hyperthyroidism. Treatment options include antithyroid drugs (ATDs), radioactive iodine, and surgery. In this review, we focus on the medical aspects of managing GD. METHODS: The authors conducted a literature review of PubMed to include studies and review articles on GD management, ATDs, long-term safety of antithyroid drugs, hyperthyroidism in pregnancy, Graves ophthalmopathy, and special circumstances related to hyperthyroidism. RESULTS: In adjunction to ATDs, medical management for GD also includes beta-blockers, glucocorticoids, and iodine containing agents. ATDs are currently the preferred option for initial management of GD, reflecting a shift in practice observed in the United States over the past 2 decades. ATDs in appropriate doses are well-tolerated and safe when used for longer duration, during pregnancy, and other circumstances discussed in this article. Routine thyroid function tests are important for monitoring. thyrotropin receptor antibody plays an essential role in determining duration of treatment and assessing the likelihood of recurrence. CONCLUSION: Medical management of GD with antithyroid drug is safe and effective. Long-term use beyond 24 months in patients with elevated thyrotropin receptor antibody is a reasonable alternative option to surgery and radioactive iodine due to higher rates of remission.

中文翻译

目的：格雷夫斯病（GD）是甲状腺功能亢进症最常见的病因。治疗方案包括抗甲状腺药物（ATDs）、放射性碘和手术。本综述重点关注GD的药物治疗管理。方法：作者对PubMed进行文献综述，纳入关于GD管理、ATDs、抗甲状腺药物长期安全性、妊娠期甲状腺功能亢进症、格雷夫斯眼病及甲状腺功能亢进症相关特殊情况的文献和综述文章。结果：除ATDs外，GD的药物治疗还包括β受体阻滞剂、糖皮质激素和含碘制剂。目前ATDs是GD初始治疗的首选方案，这反映了美国过去20年临床实践的转变。在适当剂量下，ATDs在长期使用、妊娠期及本文讨论的其他情况下具有良好的耐受性和安全性。常规甲状腺功能检测对监测至关重要。促甲状腺激素受体抗体在确定治疗时长和评估复发可能性中发挥关键作用。结论：使用抗甲状腺药物进行GD的药物治疗安全有效。对于促甲状腺激素受体抗体升高的患者，持续使用超过24个月的长期治疗因缓解率更高，是手术和放射性碘治疗的合理替代方案。

核心词汇

Graves disease /greɪvz dɪˈziːz/ : 格雷夫斯病	hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进
antithyroid drugs /ˌæntiˈθaɪrɔɪd drʌgz/ : 抗甲状腺药物	radioactive iodine /ˌreɪdiəʊˈæktɪv ˈaɪədaɪn/ : 放射性碘
surgery /ˈsɜːrdʒəri/ : 手术	glucocorticoids /ˌgluːkəʊˈkɔːrtɪkɔɪdz/ : 糖皮质激素
Graves ophthalmopathy /greɪvz ˌɒfθəˈlɒpəθi/ : 格雷夫斯眼病	thyroid function tests /ˈθaɪrɔɪd ˈfʌŋkʃən tests/ : 甲状腺功能检查
thyrotropin receptor antibody /ˌθaɪrəˈtrəʊpɪn ˌɪˈsɛptər ˈæntɪˌbɒdi/ : 促甲状腺激素受体抗体	remission /rɪˈmɪʃən/ : 缓解
pregnancy /ˈpregnənsi/ : 妊娠	

The genetics of Graves' disease.

格雷夫斯病的遗传学

期刊: Reviews in endocrine & metabolic disorders 日期: 2024

Abstract

Graves' disease (GD) is the commonest cause of hyperthyroidism and has a strong female preponderance. Everyday clinical practice suggests strong aggregation within families and twin studies demonstrate that genetic factors account for 60-80% of risk of developing GD. In this review, we collate numerous genetic studies and outline the discoveries over the years, starting with historic candidate gene studies and then exploring more recent genome-wide linkage and association studies, which have involved substantial cohorts of East Asian patients as well as those of European descent. Variants in genes including HLA, CTLA4, and PTPN22 have been shown to have substantial individual effects on disease susceptibility. In addition, we examine emerging evidence concerning the possibility that genetic variants may correlate with relevant clinical phenotypes including age of onset of GD, severity of thyrotoxicosis, goitre size and relapse of hyperthyroidism following antithyroid drug therapy, as well as thyroid eye disease. This review supports the inheritance of GD as a complex genetic trait, with a growing number of more than 80 susceptibility loci identified so far. Future implementation of more targeted clinical therapies requires larger studies investigating the influence of these genetic variants on the various phenotypes and different outcomes of conventional treatments.

中文翻译

格雷夫斯病（GD）是甲状腺功能亢进症最常见的病因，且具有显著的女性发病优势。日常临床实践表明该病存在明显的家族聚集性，双胞胎研究证实遗传因素占GD发病风险的60-80%。本综述系统梳理了众多遗传学研究，从早期的候选基因研究到近期涉及大量东亚裔及欧洲裔人群的全基因组连锁与关联研究，揭示了HLA、CTLA4、PTPN22等基因变异对疾病易感性的显著个体效应。此外，我们探讨了基因变异与相关临床表型（包括GD发病年龄、甲状腺毒症严重程度、甲状腺肿大、抗甲状腺药物治疗后甲亢复发及甲状腺眼病）之间可能关联的新证据。本综述支持GD作为复杂遗传性状的遗传模式，目前已发现超过80个易感位点。未来实施更具靶向性的临床治疗，需要更大规模的研究来探究这些遗传变异对各类表型及常规治疗不同结局的影响。

核心词汇

Graves' disease /greɪvz dɪ'zi:z/ : 格雷夫斯病	hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症
genetic /dʒəˈnetɪk/ : 遗传的	susceptibility /səˌseptɪˈbɪlɪti/ : 易感性
variant /ˈvəriənt/ : 变异	phenotype /ˈfiːnətaɪp/ : 表型
goitre /ˈɡɔɪtər/ : 甲状腺肿	relapse /rɪˈlæps/ : 复发
antithyroid drug /ˌæntiˈθaɪrɔɪd drʌɡ/ : 抗甲状腺药物	thyroid eye disease /ˈθaɪrɔɪd aɪ dɪ'zi:z/ : 甲状腺眼病
locus /ˈləʊkəs/ : 基因座	inheritance /ɪnˈheɪtəns/ : 遗传

Considerations in the Diagnosis and Management of Thyroid Dysfunction in Older Adults.

老年人甲状腺功能障碍诊断与管理的考量因素

期刊: Thyroid : official journal of the American Thyroid Association 日期: 2025

Abstract

Background: **thyroid dysfunction** is common in older adults and poses diagnostic and management challenges for clinicians. In this narrative review, we present published data focusing on special considerations in the diagnosis and management of **hypothyroidism** and **hyperthyroidism** in older adults. Methods: A comprehensive literature search of the PubMed and Ovid MEDLINE databases was conducted from January 2000 to December 2024 to identify pertinent articles in English for this narrative review. Results: Due to significant **cardiovascular risk** if untreated, both overt **hypothyroidism** and **hyperthyroidism** should be treated in older adults. Findings from observational studies do not support treating older adults with **subclinical hypothyroidism** with a **thyrotropin** (TSH) <7 mIU/L. However, observational data have demonstrated an increased risk of cardiovascular mortality and stroke in older adults with **subclinical hypothyroidism** with TSH 7.0-9.9 mIU/L and of coronary heart disease, cardiovascular mortality, and heart failure in those with TSH ≥10 mIU/L, suggesting **levothyroxine** treatment in these individuals should be considered. Data from clinical trials failed to show improvement with **levothyroxine** in **hypothyroidism** symptoms or fatigue in older adults with **subclinical hypothyroidism** compared with placebo. Over- and under-replacement with thyroid hormone is common and should be avoided, as population-based studies have shown associations with adverse cardiovascular and skeletal events. **subclinical hyperthyroidism** with a TSH <0.1 mIU/L should be treated in older individuals as it has been associated with increased **cardiovascular risk** and **bone density loss** based on observational data. Randomized controlled trials have shown that long-term low-dose **methimazole** is a viable alternative to **radioactive iodine** in older adults with **hyperthyroidism**. Conclusions: A personalized approach should be undertaken in the diagnosis and management of **thyroid dysfunction** in older adults. Multiple factors should be considered, including physiological age-related changes in thyroid function, comorbidities, and polypharmacy. Care should be taken to maintain **euthyroidism** in order to avoid adverse events.

中文翻译

背景：甲状腺功能障碍在老年人中较为常见，给临床医生的诊断和管理带来挑战。在这篇叙述性综述中，我们重点介绍了关于老年人甲状腺功能减退症和甲状腺功能亢进症诊断与管理的特殊考量因素的相关已发表数据。方法：对PubMed和Ovid MEDLINE数据库进行2000年1月至2024年12月的全面文献检索，以确定本叙述性综述相关的英文文章。结果：由于未经治疗会带来显著的心血管风险，老年人显性甲状腺功能减退症和甲状腺功能亢进症均应接受治疗。观察性研究结果不支持对促甲状腺激素（TSH）<7 mIU/L的亚临床甲状腺功能减退症老年患者进行治疗。然而，观察性数据显示，TSH为7.0-9.9 mIU/L的亚临床甲状腺功能减退症老年患者心血管死亡和卒中风险增加，而TSH≥10 mIU/L的患者冠心病、心血管死亡和心力衰竭风险增加，提示应考虑对这些个体进行左甲状腺素治疗。临床试验数据未能显示与安慰剂相比，左甲状腺素能改善亚临床甲状腺功能减退症老年患者的甲状腺功能减退症状或疲劳。甲状腺激素的过度或不足替代治疗较为常见，应予以避免，因为基于人群的研究显示其与不良心血管和骨骼事件相关。对于TSH<0.1 mIU/L的亚临床甲状腺功能亢进症老年患者应进行治疗，因为观察性数据显示其与心血管风险增加和骨密度下降相关。随机对照试验表明，在老年甲状腺功能亢进症患者中，长期低剂量甲硫咪唑是放射性碘的可行替代方案。结论：老年人甲状腺功能障碍的诊断和管理应采取个体化方法。需考虑多种因素，包括甲状腺功能随年龄的生理性变化、合并症及多重用药。应谨慎维持甲状腺功能正常状态，以避免不良事件发生。

核心词汇

thyroid dysfunction /ˈθaɪrɔɪd dɪsˈfʌŋkʃən/ : 甲状腺功能障碍	hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症
hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症	subclinical hypothyroidism /sʌbˈklɪnɪkəl ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 亚临床甲状腺功能减退症
subclinical hyperthyroidism /sʌbˈklɪnɪkəl ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 亚临床甲状腺功能亢进症	thyrotropin /θaɪˈrɒtrəpɪn/ : 促甲状腺激素
levothyroxine /ˌliːvəʊˈθaɪˈrɒksɪːn/ : 左甲状腺素	methimazole /meˈθɪməzəʊl/ : 甲硫咪唑
radioactive iodine /ˌreɪdiəʊˈæktɪv ˈaɪədaɪn/ : 放射性碘	cardiovascular risk /ˌkɑːrdɪəʊˈvæskjələr rɪsk/ : 心血管风险
bone density loss /bəʊn ˈdensɪti lɔːs/ : 骨密度丢失	euthyroidism /juːˈθaɪrɔɪdɪzəm/ : 甲状腺功能正常状态

Thyroid dysfunction and female infertility. A comprehensive review.

甲状腺功能障碍与女性不孕症：一项全面综述

期刊: Diabetes & metabolic syndrome 日期: 2023

Abstract

INTRODUCTION: Female **infertility** is defined as the inability to achieve pregnancy following one year of consistent, unprotected sexual intercourse. Among the various endocrine factors contributing to this complex issue, **thyroid dysfunction** assumes a pivotal and noteworthy role. METHODS: A narrative review, encompassing 134 articles up to 2023, was conducted utilizing the PubMed/Medline, EMBASE, and Scielo databases. The primary focus of this review was to investigate the effects of **thyroid dysfunction** on female **infertility**. RESULTS: Thyroid disorders exert a significant influence on **folliculogenesis**, **fertilization**, and **implantation** processes. Thyroid **autoimmunity**, although associated with diminished **ovarian reserve**, does not typically necessitate levothyroxine therapy. On the other hand, both subclinical and overt **hypothyroidism** often require levothyroxine treatment to enhance fertility and optimize obstetric outcomes. **hyperthyroidism** warrants prompt intervention due to its heightened risk of **miscarriage**. Furthermore, **thyroid dysfunction** exerts notable effects on assisted reproductive technologies, underscoring the importance of achieving **euthyroidism** prior to **ovarian stimulation**. CONCLUSION: Women presenting with **thyroid dysfunction** must undergo meticulous and individualized assessments since fertility outcomes, whether achieved through natural conception or assisted reproductive technologies, can be significantly influenced by thyroid-related factors.

中文翻译

引言：女性不孕症定义为在规律无保护性交一年后未能妊娠。在导致这一复杂问题的多种内分泌因素中，甲状腺功能障碍扮演着关键且值得关注的角色。方法：通过检索PubMed/Medline、EMBASE和Scielo数据库，对截至2023年的134篇文献进行叙述性综述。本综述主要探讨甲状腺功能障碍对女性不孕症的影响。结果：甲状腺疾病对卵泡发育、受精及胚胎着床过程具有显著影响。甲状腺自身免疫虽与卵巢储备功能下降相关，但通常无需左甲状腺素治疗。而亚临床和显性甲状腺功能减退症常需左甲状腺素治疗以改善生育能力及优化产科结局。甲状腺功能亢进症因流产风险升高需及时干预。此外，甲状腺功能障碍对辅助生殖技术具有显著影响，凸显了在卵巢刺激前实现甲状腺功能正常化的重要性。结论：甲状腺功能障碍女性必须接受细致且个体化的评估，因为无论通过自然受孕还是辅助生殖技术实现的生育结局，均可能受到甲状腺相关因素的显著影响。

核心词汇

infertility /ˌɪnfərˈtɪləti/ : 不孕症	thyroid dysfunction /ˈθaɪrɔɪd dɪsˈfʌŋkʃən/ : 甲状腺功能障碍
folliculogenesis /fəˌlɪkjʊləʊˈdʒɛnɪsɪs/ : 卵泡发生	fertilization /ˌfɜːrtɪlɪˈzeɪʃən/ : 受精
implantation /ˌɪmplænˈteɪʃən/ : 着床	autoimmunity /ˌɔːtəʊˈmjʊːnɪti/ : 自身免疫
ovarian reserve /oʊˈvɛəriən ɪˈzɜːrv/ : 卵巢储备	hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症
hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症	miscarriage /ˈmɪskəriɪdʒ/ : 流产
euthyroidism /juːˈθaɪrɔɪdɪzəm/ : 甲状腺功能正常状态	ovarian stimulation /oʊˈvɛəriən ˌstɪmjʊˈleɪʃən/ : 卵巢刺激

Hypothyroidism in Older Adults: A Narrative Review.

老年人甲状腺功能减退症：叙述性综述。

期刊: Endocrine, metabolic & immune disorders drug targets 日期: 2024

Abstract

The prevalence of **hypothyroidism** increases along with **aging**, resulting in one of the most common comorbidities among patients over 75 years. The leading causes of **hypothyroidism** in older adults are **iatrogenic**, **Hashimoto's thyroiditis**, and medications. The narrative review aimed to discuss the clinical characteristics of **hypothyroidism** in older adults and the impact of **hormonal replacement therapy** on survival rates. Thyroid function declines over time due to physiological changes in the **thyroid stimulating hormone** signaling, **iodine** absorption and metabolism, thyroid hormone metabolism, and activity at peripheral sites. A **serum TSH** value over the upper limit of the normal reference range is not necessarily attributable to **hypothyroidism**. However, an appropriate diagnostic work-up is required to rule out true **hypothyroidism** and discriminate the etiology (i.e., thyroid autoimmune diseases, **iodine** deficiency, drug-induced **hypothyroidism**). **levothyroxine** treatment should be considered in cases of overt **hypothyroidism**. A complete risk-to-benefit assessment, particularly considering the overall health status, life expectancy, **cognitive function**, mood, and **cardiovascular** and **neurological** background, should be considered before treating **subclinical hypothyroidism** with more potential benefits in patients under 75 years old. **levothyroxine** formulations facilitating hormone absorption and increasing compliance to long-term treatment should be preferred. TSH target should usually be set over 3 mIU/ml. Defining optimal diagnostic approaches and targeted therapeutic strategies should be considered in the personalized management of aged patients with **hypothyroidism**.

中文翻译

甲状腺功能减退症的患病率随年龄增长而升高，成为75岁以上患者最常见的合并症之一。老年人甲状腺功能减退的主要病因包括医源性因素、桥本甲状腺炎及药物影响。本综述旨在探讨老年甲减的临床特征及激素替代治疗对生存率的影响。由于促甲状腺激素信号传导、碘吸收与代谢、甲状腺激素代谢及外周组织活性等生理变化，甲状腺功能会随时间推移逐渐下降。血清TSH值超过正常参考范围上限不一定归因于甲减，但需通过适当诊断流程排除真性甲减并鉴别病因（如甲状腺自身免疫性疾病、碘缺乏、药物性甲减）。显性甲减应考虑左甲状腺素治疗。对于75岁以下患者，治疗亚临床甲减前需进行完整的风险-获益评估，尤其需综合考虑整体健康状况、预期寿命、认知功能、情绪状态及心血管与神经系统基础疾病。应优先选择促进激素吸收且能提高长期治疗依从性的左甲状腺素制剂。TSH目标值通常设定为高于3 mIU/ml。在老年甲减患者的个体化管理中，应明确最佳诊断及靶向治疗策略。

核心词汇

hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症	aging /ˈeɪdʒɪŋ/ : 衰老
comorbidity /ˌkɒmɔːrˈbɪdɪti/ : 合并症	iatrogenic /aɪˌætrəˈdʒɛnɪk/ : 医源性的
Hashimoto's thyroiditis /hɑːˈʃɪːmoʊz θaɪrɔɪˈdaɪtɪs/ : 桥本甲状腺炎	hormonal replacement therapy /hɔːrˈmounəl rɪˈpleɪsmənt ˈθerəpi/ : 激素替代疗法
thyroid stimulating hormone /ˈθaɪrɔɪd ˈstɪmjuleɪtɪŋ ˈhɔːrmoun/ : 促甲状腺激素	iodine /ˈaɪədaɪn/ : 碘
serum TSH /ˈsɪrəm tiː ɛs ɛɪtʃ/ : 血清促甲状腺激素	autoimmune disease /ˌɔːtəʊɪˈmjʊːn dɪˈziːz/ : 自身免疫性疾病
levothyroxine /ˌliːvoʊθaɪˈrɒksiːn/ : 左甲状腺素	subclinical hypothyroidism /sʌbˈklɪnɪkəl ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 亚临床甲状腺功能减退症
cognitive function /ˈkɒɡnɪtv ˈfʌŋkʃən/ : 认知功能	cardiovascular /ˌkɑːrdɪəvˈvæskjələr/ : 心血管的

Reimagining the Therapeutic Approach for Anaplastic Thyroid Cancer: The Roadmap to a Cure.

重新构想间变性甲状腺癌的治疗方法：治愈之路

期刊: Thyroid : official journal of the American Thyroid Association 日期: 2025

Abstract

Background: **anaplastic thyroid cancer** (ATC) is an aggressive **cancer** that leads to rapid death if left untreated. However, recent advances in ATC treatment have dramatically changed the **prognosis** in a select group of patients with BRAF(V600E) mutations. In these patients, BRAF/MEK inhibitors have been shown to dramatically and rapidly shrink tumors. Yet, these responses are short-lived unless additional treatment modalities are applied. In patients without a BRAF(V600E) **mutation**, the current available therapies are far less effective. Summary: In this article, we review the relevant literature and propose applying the "Total **therapy**" approach used since the 1960s for another deadly but curable disease, acute lymphocytic leukemia, to ATC. We have adapted the concepts of Induction, Consolidation, and Maintenance, applying them to ATC. This regimen integrates the treatments we have found to be successful in ATC: combination systemic **therapy** using **targeted therapy** plus **immunotherapy**, **surgery**, **radiation**, and continuation of the systemic **therapy** for several years, thereby attempting to eradicate all residual ATC cells. Conclusions: There has been a renewed interest in understanding the **genomics** of ATC and treating these patients with urgency rather than just providing **palliative care**. This shift has led to significant improvements in the **prognosis** of ATC. With the right tools and a clear roadmap to guide us, we now aim to take on the challenge of curing these patients.

中文翻译

背景：未分化甲状腺癌（ATC）是一种侵袭性癌症，若不治疗将迅速致命。然而，近年来ATC治疗的进展显著改变了部分携带BRAF(V600E)突变患者的预后。研究表明，BRAF/MEK抑制剂可快速且显著缩小这类患者的肿瘤。但若不联合其他治疗手段，这种疗效难以持久。对于无BRAF(V600E)突变的患者，现有疗法的效果则远不理想。摘要：本文通过文献回顾，提出将自20世纪60年代起用于治疗另一种致命但可治愈疾病——急性淋巴细胞白血病的"整体治疗"策略应用于ATC。我们借鉴了诱导、巩固和维持治疗的概念，将其整合为针对ATC的治疗方案：联合靶向治疗与免疫治疗的全身性疗法、手术、放疗，并持续数年维持全身治疗，旨在彻底清除所有残留的ATC细胞。结论：近年来，学界重新聚焦于ATC基因组学研究，并以紧迫态度积极治疗患者，而非仅提供姑息护理。这一转变显著改善了ATC的预后。凭借合适的工具与清晰的路线图，我们正致力于攻克治愈这些患者的挑战。

核心词汇

anaplastic thyroid cancer /ˌænəˈplæstɪk ˈθaɪrɔɪd ˈkænsər/ : 间变性甲状腺癌	cancer /ˈkænsər/ : 癌症
mutation /mjuːˈteɪʃən/ : 突变	tumor /ˈtuːmə/ : 肿瘤
therapy /ˈθerəpi/ : 治疗	surgery /ˈsɜːrdʒəri/ : 手术
radiation /ˌreɪdiˈeɪʃən/ : 放射治疗	immunotherapy /ˌɪmjʊnəʊˈθerəpi/ : 免疫治疗
targeted therapy /ˈtɑːrɡɪtɪd ˈθerəpi/ : 靶向治疗	prognosis /prɒɡˈnəʊsɪs/ : 预后
palliative care /ˈpæliətɪv keər/ : 姑息治疗	genomics /dʒɪˈnɒmɪks/ : 基因组学

Thyroid cancer and insulin resistance.

甲状腺癌与胰岛素抵抗。

期刊: Reviews in endocrine & metabolic disorders 日期: 2024

Abstract

thyroid cancer has shown a parallel increase with diabetes in the last few years. This narrative review aims to explain the association between these two entities, focusing on insulin resistance as the mediator and exploring the effects of antidiabetic agents on thyroid cancer incidence and progression. We searched Pubmed for English-written articles on insulin resistance, diabetes, antidiabetic treatments, and thyroid cancer reported from January 2019 to April 2023. Exclusion criteria were preclinical and clinical studies involving a population with thyroid dysfunction, benign nodular goiter, or those that only analyzed thyroid cancer's association with obesity. The results of the narrative literature review revealed 96 articles. Additionally, four studies from a manual search were retrieved. After the exclusion criteria were applied, we included 20 studies. Out of 8 studies on insulin-resistant or metabolic syndrome patients, all suggest a positive association with thyroid cancer. At the same time, for diabetes, four out of five publications support a link with thyroid cancer. The seven remaining studies on antidiabetics suggest that metformin might benefit thyroid cancer. In contrast, the evidence for an association between Glucagon-like peptide-1 receptor agonists (GLP-1 RA) and increased thyroid cancer findings is limited. In conclusion, the association between thyroid cancer and diabetes may be explained by insulin resistance, as shown in observational studies. However, the causal role is yet to be defined. Although the wide use of different antidiabetic agents has been related to thyroid cancer prevalence and progression, future research with drugs such as metformin or GLP-1 RA is still needed.

中文翻译

过去几年中，甲状腺癌与糖尿病的发病率同步上升。本叙述性综述旨在解释这两种疾病之间的关联，重点关注胰岛素抵抗作为中介因素，并探讨抗糖尿病药物对甲状腺癌发病率和进展的影响。我们检索了PubMed上2019年1月至2023年4月期间发表的关于胰岛素抵抗、糖尿病、抗糖尿病治疗与甲状腺癌的英文文献。排除标准包括涉及甲状腺功能障碍、良性结节性甲状腺肿人群的临床前和临床研究，以及仅分析甲状腺癌与肥胖关联的研究。叙述性文献综述结果显示共96篇文章，另通过手动检索获得4项研究。应用排除标准后，我们纳入20项研究。在8项针对胰岛素抵抗或代谢综合征患者的研究中，所有结果均提示与甲状腺癌呈正相关；而关于糖尿病，5篇文献中有4篇支持其与甲状腺癌的关联。其余7项关于抗糖尿病药物的研究表明，二甲双胍可能对甲状腺癌有益，而胰高血糖素样肽-1受体激动剂（GLP-1 RA）与甲状腺癌风险增加的关联证据有限。结论：观察性研究表明，甲状腺癌与糖尿病的关联可能通过胰岛素抵抗解释，但因果关系尚未明确。尽管不同抗糖尿病药物的广泛使用与甲状腺癌患病率和进展相关，但未来仍需开展针对二甲双胍或GLP-1 RA等药物的研究。

核心词汇

thyroid cancer /ˈθaɪrɔɪd ˈkænsər/ : 甲状腺癌	diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病
insulin resistance /ˈɪnsəlɪn rɪˈzɪstəns/ : 胰岛素抵抗	antidiabetic agents /ˌæntɪˈdaɪəˈbetɪk ˈeɪdʒənts/ : 抗糖尿病药物
metformin /ˈmetˈfɔːrmɪn/ : 二甲双胍	obesity /oʊˈbiːsəti/ : 肥胖
metabolic syndrome /ˌmetəˈbɒlɪk ˈsɪndroum/ : 代谢综合征	thyroid dysfunction /ˈθaɪrɔɪd dɪsˈfʌŋkʃən/ : 甲状腺功能障碍
benign nodular goiter /bɪˈnaɪn ˈnɒdʒulər ˈɡoɪtər/ : 良性结节性甲状腺肿	observational study /ˌɒbzərˈveɪʃənəl ˈstʌdi/ : 观察性研究
narrative review /ˈnærətɪv rɪˈvjuː/ : 叙述性综述	incidence /ˈɪnsɪdəns/ : 发病率
progression /prəˈɡreʃən/ : 进展	

Dermatologic manifestations of thyroid disease: a literature review.

甲状腺疾病的皮肤表现：文献综述。

期刊: Frontiers in endocrinology 日期: 2023

Abstract

INTRODUCTION: **thyroid hormone** is considered one of the key regulatory hormones for **skin homeostasis**. Multiple organs are affected by the release of peripheral thyroid hormones (T4 and T3) further regulating various functions at a cellular level. Specifically, skin is considered an important target organ in which the **thyroid hormone** has a significant impact. Multiple skin diseases are associated with **thyroid hormone dysregulation**. However, other striking **dermatologic** manifestations are seen in nails and **hair** as well. **hypothyroidism**, **hyperthyroidism**, and **thyroid cancer** can have an array of cutaneous manifestations, and we present the recent updates in this field. METHODS: A PubMed search was performed for updates in any new skin disease findings and treatments between 2010 and 2022. Research published in the past decade and previously known foundational skin findings associated with thyroid disease were presented in this **review**. CONCLUSION: Cutaneous manifestations of thyroid disease is one of the first notable signs of **thyroid hormone dysregulation**. This article reviews the recent updates on the thyroid and skin interplay, and it further discusses overt visible findings and various available treatment modalities.

中文翻译

引言：甲状腺激素被认为是皮肤稳态的关键调节激素之一。外周甲状腺激素（T4和T3）的释放会影响多个器官，并在细胞层面进一步调节多种功能。特别是，皮肤被视为甲状腺激素发挥重要作用的靶器官。多种皮肤疾病与甲状腺激素失调相关。此外，指甲和毛发也会出现显著的真皮表现。甲状腺功能减退、甲状腺功能亢进和甲状腺癌可引发一系列皮肤表现，本文介绍了该领域的最新进展。方法：通过PubMed检索2010年至2022年间关于新发现的皮肤疾病及其治疗的最新研究。本综述纳入了过去十年发表的研究以及既往已知的与甲状腺疾病相关的基础皮肤表现。结论：甲状腺疾病的皮肤表现是甲状腺激素失调最早出现的显著体征之一。本文回顾了甲状腺与皮肤相互作用的最新进展，并进一步讨论了明显的可见表现及多种现有治疗方式。

核心词汇

thyroid hormone /ˈθaɪrɔɪd ˈhɔːrmoun/ : 甲状腺激素	skin homeostasis /skɪn ˌhəʊmɪəˈsteɪsɪs/ : 皮肤稳态
hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症	hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症
thyroid cancer /ˈθaɪrɔɪd ˈkænsər/ : 甲状腺癌	cutaneous manifestation /kjuːˈteɪniəs ˌmænɪfɛˈsteɪʃən/ : 皮肤表现
dermatologic /ˌdɜːrmətəˈlɒdʒɪk/ : 皮肤科的	nail /neɪl/ : 指甲
hair /hɛr/ : 毛发	treatment modality /ˈtriːtmənt moʊˈdæləti/ : 治疗方式
review /rɪˈvjuː/ : 综述	dysregulation /dɪsˌrɛɡjʊˈleɪʃən/ : 失调

Progress in Thyroid Cancer Genomics: A 40-Year Journey.

甲状腺癌基因组学进展：40年历程

期刊: Thyroid : official journal of the American Thyroid Association 日期: 2023

Abstract

Background: Very little was known about the molecular pathogenesis of thyroid cancer until the late 1980s. As part of the Centennial celebration of the American Thyroid Association, we review the historical discoveries that contributed to our current understanding of the genetic underpinnings of thyroid cancer. Summary: The pace of discovery was heavily dependent on scientific breakthroughs in nucleic acid sequencing technology, cancer biology, thyroid development, thyroid cell signaling, and growth regulation. Accordingly, we attempt to link the primary observations on thyroid cancer molecular genetics with the methodological and scientific advances that made them possible. Conclusions: The major genetic drivers of the common forms of thyroid cancer are now quite well established and contribute to a significant extent to how we diagnose and treat the disease. However, many challenges remain. Future work will need to unravel the complexity of thyroid cancer ecosystems, which is likely to be a major determinant of their biological behavior and on how they respond to therapy.

中文翻译

背景：直到20世纪80年代末，人们对甲状腺癌的分子发病机制仍知之甚少。作为美国甲状腺协会百年庆典的一部分，我们回顾了那些有助于当前理解甲状腺癌遗传基础的历史性发现。摘要：这些发现的步伐在很大程度上依赖于核酸测序技术、癌症生物学、甲状腺发育、甲状腺细胞信号传导和生长调控等领域的科学突破。因此，我们试图将甲状腺癌分子遗传学的主要发现与促成这些发现的方法学和科学进步联系起来。结论：目前，常见类型甲状腺癌的主要遗传驱动因素已相当明确，并在很大程度上影响了我们对该疾病的诊断和治疗方式。然而，仍有许多挑战有待解决。未来的研究需要揭示甲状腺癌生态系统的复杂性，这可能是决定其生物学行为及治疗反应的关键因素。

核心词汇

thyroid cancer /ˈθaɪrɔɪd ˈkænsər/ : 甲状腺癌	pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制
molecular genetics /məˈleɪkjələr dʒəˈnetɪks/ : 分子遗传学	nucleic acid sequencing /njuːˈkliːk ˈæsɪd ˈsiːkwənsɪŋ/ : 核酸测序
cancer biology /ˈkænsər baɪˈɒlədʒi/ : 癌症生物学	thyroid development /ˈθaɪrɔɪd dɪˈvɛləpmənt/ : 甲状腺发育
cell signaling /sɛl ˈsɪɡnəlɪŋ/ : 细胞信号传导	growth regulation /ɡroʊθ ˌrɛɡjʊˈleɪʃən/ : 生长调控
genetic driver /dʒəˈnetɪk ˈdraɪvər/ : 遗传驱动因子	diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断
therapy /ˈθerəpi/ : 治疗	tumor ecosystem /ˈtuːmə ˈiːkəʊsɪstəm/ : 肿瘤生态系统
biological behavior /ˌbaɪəˈlɒdʒɪkəl bɪˈheɪvjər/ : 生物学行为	

Sirolimus for the treatment of Graves' orbitopathy.

西罗莫司治疗格雷夫斯眼病

期刊: Thyroid research 日期: 2025

Abstract

OBJECTIVES: A role of mTOR (mammalian target of rapamycin) in the pathogenesis of Graves' Orbitopathy (GO) has been proposed and rapamycin, better known as sirolimus, an mTOR inhibitor, was recently used in patients with GO. Here we review the available studies evaluating the role of mTOR in GO pathogenesis and the effects of sirolimus on GO. DESIGN: A comprehensive search of PubMed was conducted using the following keywords: "Graves' Orbitopathy", or "thyroid eye disease", or "Graves' ophthalmopathy", or "thyroid-associated ophthalmopathy"; and: "mTOR", or "sirolimus", or "rapamycin". INCLUSION CRITERIA: 1) original articles (preclinical and clinical studies); 2) English language. The articles that did not adequately explore the role of mTOR in GO pathogenesis and those in which the effects of sirolimus on GO were not investigated were excluded. At the end of the screening process, nine studies were included in this systematic review. RESULTS: mTOR signaling pathway was found to be upregulated in patients with GO. In addition, studies in a mouse model of GO, in orbital fibroblasts and peripheral blood mononuclear cells (PBMCs) derived from GO patients showed a significant reduction in inflammatory mediators, adipogenesis and fibrosis after treatment with sirolimus. In 2007 and 2019 two cases of patients with GO unresponsive to glucocorticoids and successfully treated with sirolimus were described. Consistently with these results, two retrospective investigations showed that treatment with sirolimus, used at low dosage for 12 weeks, was followed by a greater overall response of GO compared with methylprednisolone at 24 weeks. In addition, a good response in diplopia and ocular motility restriction after treatment with sirolimus was reported by two case series. All of these studies reported a good tolerability of sirolimus. Finally, GO response to treatment was shown to correlate with the serum levels of sirolimus at the end of treatment. CONCLUSIONS: Sirolimus may represent a cheap, effective, and safe alternative treatment for GO. In addition, serum levels of sirolimus may be used to predict the response to treatment. Randomized clinical trials are needed to confirm the efficacy of sirolimus on GO and establish the best possible treatment protocol.

中文翻译

目的: 已有研究提出mTOR (哺乳动物雷帕霉素靶蛋白) 在格雷夫斯眼病 (GO) 发病机制中的作用, 而雷帕霉素 (更常称为西罗莫司) 作为一种mTOR抑制剂, 近期已被用于GO患者。本文回顾了评估mTOR在GO发病机制中作用及西罗莫司对GO疗效的现有研究。设计: 通过PubMed数据库进行综合检索, 关键词包括: "Graves' orbitopathy"、"thyroid eye disease"、"Graves' ophthalmopathy"、"thyroid-associated ophthalmopathy"; 以及"mTOR"、"sirolimus"、"rapamycin"。纳入标准: 1) 原创性文章 (临床前及临床研究); 2) 英文文献。排除未充分探讨mTOR在GO发病机制中作用或未研究西罗莫司对GO疗效的文章。经筛选后, 共纳入9项研究进行系统综述。结果: 研究发现GO患者中mTOR信号通路表达上调。此外, 在GO小鼠模型及源自GO患者的眼眶成纤维细胞和外周血单核细胞 (PBMCs) 研究中, 西罗莫司治疗后炎症介质、脂肪生成及纤维化显著减少。2007年和2019年分别报道了两例对糖皮质激素无应答的GO患者经西罗莫司成功治疗的案例。与这些结果一致, 两项回顾性研究显示, 低剂量西罗莫司治疗12周后, GO总体应答率优于24周甲泼尼龙治疗。此外, 两项病例系列报告了西罗莫司治疗后复视和眼球运动受限的良好改善。所有研究均显示西罗莫司耐受性良好。最后, GO治疗应答与治疗结束时西罗莫司血清水平相关。结论: 西罗莫司可能成为GO经济、有效且安全的替代治疗方案。此外, 西罗莫司血清水平可用于预测治疗应答。需通过随机临床试验确认西罗莫司对GO的疗效并确定最佳治疗方案。

核心词汇

Graves' Orbitopathy /greɪvz ˌɔːrbiˈtɒpəθi/ : 格雷夫斯眼病	pathogenesis /ˌpæθəˈdʒenɪsɪs/ : 发病机制
inhibitor /ɪnˈhɪbɪtər/ : 抑制剂	fibroblast /ˈfaɪbrəˌblæst/ : 成纤维细胞
inflammation /ˌɪnfləˈmeɪʃən/ : 炎症	adipogenesis /ˌædɪpəʊˈdʒenɪsɪs/ : 脂肪生成
fibrosis /faɪˈbrəʊsɪs/ : 纤维化	glucocorticoid /ˌɡluːkəʊˈkɔːrtɪkɔɪd/ : 糖皮质激素
diplopia /dɪˈplʊpiə/ : 复视	ocular motility /ˈɒkjələr moʊˈtɪlɪti/ : 眼球运动
serum /ˈsɪrəm/ : 血清	randomized clinical trial /ˈrændəmaɪzd ˈklɪnɪkəl ˈtraɪəl/ : 随机临床试验

Advances in artificial intelligence in thyroid-associated ophthalmopathy.

人工智能在甲状腺相关眼病中的进展。

期刊: Frontiers in endocrinology 日期: 2024

Abstract

Thyroid-associated ophthalmopathy (TAO), also referred to as Graves' ophthalmopathy, is a medical condition wherein ocular complications arise due to autoimmune thyroid illness. The diagnosis of TAO, reliant on imaging, typical ocular symptoms, and abnormalities in thyroid function or thyroid-associated antibodies, is generally graded and staged. In recent years, Artificial intelligence(AI), particularly deep learning(DL) technology, has gained widespread use in the diagnosis and treatment of ophthalmic diseases. This paper presents a discussion on specific studies involving AI, specifically DL, in the context of TAO, highlighting their applications in TAO diagnosis, staging, grading, and treatment decisions. Additionally, it addresses certain limitations in AI research on TAO and potential future directions for the field.

中文翻译

甲状腺相关眼病（TAO），又称格雷夫斯眼病，是一种因自身免疫性甲状腺疾病导致眼部并发症的疾病。TAO的诊断依赖于影像学检查、典型眼部症状以及甲状腺功能或甲状腺相关抗体异常，通常需要进行分级和分期。近年来，人工智能（AI），特别是深度学习（DL）技术，在眼科疾病的诊断和治疗中得到了广泛应用。本文讨论了AI（尤其是DL）在TAO中的具体研究，重点介绍了其在TAO诊断、分期、分级及治疗决策中的应用。此外，还探讨了AI在TAO研究中存在的某些局限性以及该领域未来的潜在发展方向。

核心词汇

Graves' ophthalmopathy /greɪvz ˌafθəɪl'mapəθi/ : 格雷夫斯眼病	autoimmune thyroid illness /ɔːtuː'mjuːn 'θaɪrɔɪd 'ɪlnəs/ : 自身免疫性甲状腺疾病
ocular complications /'akjələr ˌkɑmplɪ'keɪʃənz/ : 眼部并发症	imaging /'ɪmɪdʒɪŋ/ : 影像学检查
ocular symptoms /'akjələr 'sɪmptəmz/ : 眼部症状	thyroid function /'θaɪrɔɪd 'fʌŋkʃən/ : 甲状腺功能
antibodies /'æntɪˌbɒdiz/ : 抗体	diagnosis /ˌdaɪəg'nəʊsɪs/ : 诊断
staging /'steɪdʒɪŋ/ : 分期	grading /'greɪdɪŋ/ : 分级
deep learning /diːp 'lɜːrnɪŋ/ : 深度学习	ophthalmic diseases /af'θælmɪk dɪ'ziːzɪz/ : 眼科疾病
treatment decisions /'triːtmənt dɪ'sɪʒənz/ : 治疗决策	

Role of genetics and epigenetics in Graves' orbitopathy.

遗传学与表观遗传学在格雷夫斯眼病中的作用。

期刊: European thyroid journal 日期: 2024

Abstract

OBJECTIVES: The pathogenesis of Graves' orbitopathy (GO) remains to be fully elucidated. Here, we reviewed the role of genetics and epigenetics. DESIGN: We conducted a PubMed search with the following keywords: GO, thyroid eye disease; or Graves' ophthalmopathy; or thyroid-associated ophthalmopathy; and: genetic, or epigenetic, or gene expression, or gene mutation, or gene variant, or gene polymorphism, or DNA methylation, or DNA acetylation. Articles in which whole DNA and/or RNA sequencing, proteome, and methylome analyses were performed were chosen. RESULTS: The different prevalence of GO in the two sexes, as well as racial differences, suggest that genetics play a role in GO pathogenesis. In addition, the long-lasting phenotype of GO and patient-derived orbital fibroblasts suggests a genetic or epigenetic mechanism. Although no genes have been found to confer a specific risk for GO, differential gene expression has been reported in orbital fibroblasts from GO patients vs control fibroblasts, suggesting that an epigenetic mechanism may be involved. In this regard, a different degree of DNA methylation, which affects gene expression, has been found between GO and control fibroblasts, which was confirmed by whole methylome analysis. histone acetylation and deacetylation, which also affect gene expression, remain to be investigated. CONCLUSIONS: Although no pathogenic gene variants have been reported, epigenetic mechanisms elicited by an initial autoimmune insult seem to be needed for differential gene expression to occur and, thus, for GO to develop and persist over time.

中文翻译

目的：格雷夫斯眼病（GO）的发病机制尚未完全阐明。本文综述了遗传学与表观遗传学在其中的作用。设计：我们在PubMed中检索了以下关键词：GO、甲状腺眼病；或格雷夫斯眼病；或甲状腺相关眼病；以及：遗传、表观遗传、基因表达、基因突变、基因变异、基因多态性、DNA甲基化、DNA乙酰化。筛选出进行全DNA和/或RNA测序、蛋白质组学及甲基化组学分析的文章。结果：GO在性别及种族间的患病率差异提示遗传因素在其发病机制中发挥作用。此外，GO的持续性表型及患者来源的眼眶成纤维细胞表明存在遗传或表观遗传机制。尽管尚未发现特异性增加GO风险的基因，但GO患者与对照组眼眶成纤维细胞的差异基因表达已被报道，提示表观遗传机制可能参与其中。在此方面，GO与对照组成纤维细胞中影响基因表达的DNA甲基化程度存在差异，这一发现已通过全甲基化组分析得到证实。同样影响基因表达的组蛋白乙酰化与去乙酰化作用仍有待研究。结论：尽管尚未报道致病性基因变异，但初始自身免疫损伤引发的表观遗传机制似乎是差异基因表达发生、进而导致GO发生并持续存在的必要条件。

核心词汇

Graves' orbitopathy /greɪvz ˌɔːrbi ˈtɒpəθi/ : 格雷夫斯眼病	pathogenesis /ˌpæθəˈdʒenɪsɪs/ : 发病机制
genetics /dʒiˈnetɪks/ : 遗传学	epigenetics /ˌepɪdʒiˈnetɪks/ : 表观遗传学
gene expression /dʒiːn ɪkˈspreʃən/ : 基因表达	DNA methylation /ˌdiː.ɛnˈeɪ məθiˈleɪʃən/ : DNA甲基化
histone acetylation /ˈhɪstəʊn əˌsiːtɪˈleɪʃən/ : 组蛋白乙酰化	autoimmune /ˌɔːtəʊˈmjuːn/ : 自身免疫的
fibroblast /ˈfaɪbrəˌblæst/ : 成纤维细胞	phenotype /ˈfiːnəˌtaɪp/ : 表型
methylome /ˈmeθɪləʊm/ : 甲基化组	gene variant /dʒiːn ˈveriənt/ : 基因变异

2023 European Thyroid Association Clinical Practice Guidelines for thyroid nodule management.

2023年欧洲甲状腺协会甲状腺结节管理临床实践指南

期刊: European thyroid journal 日期: 2023

Abstract

With the widespread use of sensitive **imaging** techniques, which include neck visualization, a conspicuous number of thyroid nodules emerge and demand attention. Most lesions are **benign**, **asymptomatic**, and do not warrant treatment. In the case of cancer **diagnosis**, most are small, intrathyroidal and indolent neoplasms that can safely be managed conservatively. There is a pronounced need for more cost-effective, risk-adapted approaches to the **management** of this highly prevalent condition, taking the wishes of the patient into consideration. Thus, the present guidelines aim at providing a clinical practice guide for the initial **workup** and the subsequent **management** of adult individuals harboring thyroid nodules. Importantly, these guidelines are not intended to cover the **management** of thyroid **malignancy**. The manuscript and the specific recommendations were developed by reconciling the best available research evidence with the knowledge and clinical experience of the panelists and updating aspects of a number of previous European Thyroid Association guidelines.

中文翻译

随着颈部可视化等敏感成像技术的广泛应用，大量甲状腺结节被发现并需要关注。大多数病变为良性、无症状且无需治疗。在确诊为癌症的病例中，多数为微小、局限于甲状腺内的惰性肿瘤，可通过保守治疗安全处理。针对这种高发疾病，亟需更具成本效益、风险适应性的管理方案，同时需考虑患者意愿。因此，本指南旨在为成人甲状腺结节的初始评估及后续管理提供临床实践指导。需特别说明，本指南不涵盖甲状腺恶性肿瘤的管理。本文件及具体建议通过整合现有最佳研究证据与专家组成员的临床经验及知识制定，并更新了欧洲甲状腺协会既往多项指南的相关内容。

核心词汇

thyroid nodule /ˈθaɪrɔɪd ˈnɒdjuːl/ : 甲状腺结节	imaging /ˈɪmɪdʒɪŋ/ : 影像学检查
neoplasm /ˈniːəʊplæzəm/ : 肿瘤	benign /bɪˈneɪn/ : 良性
malignancy /məˈlɪɡnənsi/ : 恶性肿瘤	asymptomatic /eɪˌsɪmptəˈmættɪk/ : 无症状的
diagnosis /ˌdaɪəɡˈnəʊsɪs/ : 诊断	guideline /ˈgaɪdlaɪn/ : 指南
workup /ˈwɜːkʌp/ : 检查评估	management /ˈmænɪdʒmənt/ : 管理
biopsy /ˈbaɪɒpsi/ : 活检	ultrasound /ˈʌltrəsaʊnd/ : 超声

Thyroid nodules: Global, economic, and personal burdens.

甲状腺结节：全球、经济和个人负担。

期刊: Frontiers in endocrinology 日期: 2023

Abstract

Thyroid nodules have garnered attention due to changes in population surveillance systems and rising concerns about the associated financial burden on healthcare systems, payers, and patients. In this review, we find that prevalence rates vary widely based on method of detection and may particularly pronounced in asymptomatic patients undergoing routine screening. incidence rates may be particularly rising in lower-income and middle-income countries and may be declining in higher-income countries. Despite high incidence rates, survival rates continue to be as high as 97% for papillary thyroid cancer. Over the last few decades, thyroid nodule workup and management has grown more sophisticated with the advent of fine-needle aspiration biopsy, specialized biomarkers, and molecular testing. However, gaps remain in risk stratification that can lead to substantial costs of care. Certain molecular tests, such as the Afirma Gene Sequencing Classifier can lead to a cost per diagnosis of \$17,873 while achieving only mild decreases in diagnostic lobectomies for patients (11.6% to 9.7% in one study). Out-of-pocket costs associated with thyroid nodule management continue to drive significant financial toxicity for patients, especially for individuals with thyroid cancer. financial toxicity has been defined as a term that describes how direct and indirect medical costs of cancer care strain patients and households via decreased income, assets, and spending on basic necessities. Recent studies suggest that such toxicity can lead to adverse financial outcomes, such as foreclosure and bankruptcy. Additional cost-effectiveness analyses are needed to improve existing thyroid nodule management systems and new clinical tools are needed to avoid unnecessary workup and management.

中文翻译

甲状腺结节因人群监测系统的变化以及对其给医疗系统、支付方和患者带来的经济负担日益增加的担忧而受到关注。本综述发现，基于检测方法的不同，患病率差异显著，尤其在接受常规筛查的无症状患者中更为突出。中低收入国家的发病率可能呈上升趋势，而高收入国家则可能有所下降。尽管发病率较高，但甲状腺乳头状癌的生存率仍高达97%。过去几十年间，随着细针穿刺活检、特异性生物标志物及分子检测的出现，甲状腺结节的评估与管理已日趋精细。然而，风险分层仍存在不足，可能导致高昂的医疗成本。某些分子检测（如Afirma基因测序分类器）的单次诊断费用可达17,873美元，而仅使患者诊断性甲状腺切除术的比例略有下降（一项研究中从11.6%降至9.7%）。与甲状腺结节管理相关的自付费用持续给患者带来显著的经济毒性，尤其对甲状腺癌患者而言。经济毒性被定义为描述癌症诊疗的直接和间接医疗成本如何通过收入减少、资产缩水及基本生活开支压缩而给患者及家庭造成压力的术语。近期研究表明，此类毒性可能导致不良经济后果，如房屋止赎和破产。未来需开展更多成本效益分析以完善现有甲状腺结节管理体系，并开发新型临床工具以避免不必要的评估与管理。

核心词汇

thyroid nodule /ˈθaɪrɔɪd ˈnɒdjuːl/ : 甲状腺结节	papillary thyroid cancer /pəˈpɪləri ˈθaɪrɔɪd ˈkænsər/ : 甲状腺乳头状癌
biomarker /ˈbaɪəʊˌmɑːrkər/ : 生物标志物	molecular testing /məˈlɛkjʊlər ˈtɛstɪŋ/ : 分子检测
risk stratification /rɪsk ˌstrætɪfɪˈkeɪʃən/ : 风险分层	diagnostic lobectomy /ˌdaɪəɡˈnɒstɪk ləʊˈbɛktəmi/ : 诊断性甲状腺叶切除术
financial toxicity /faɪˈnænfəl tɒkˈsɪsɪti/ : 经济毒性	prevalence /ˈpreɪvələns/ : 患病率
incidence /ˈɪnsɪdəns/ : 发病率	survival rate /sərˈvaɪvəl reɪt/ : 生存率

Artificial Intelligence in Thyroidology: A Narrative Review of the Current Applications, Associated Challenges, and Future Directions.

甲状腺学中的人工智能：当前应用、相关挑战及未来方向的叙述性综述

期刊: Thyroid : official journal of the American Thyroid Association 日期: 2023

Abstract

Background: The use of artificial intelligence (AI) in health care has grown exponentially with the promise of facilitating biomedical research and enhancing **diagnosis**, **treatment**, monitoring, disease prevention, and health care delivery. We aim to examine the current state, limitations, and future directions of AI in thyroidology. Summary: AI has been explored in thyroidology since the 1990s, and currently, there is an increasing interest in applying AI to improve the care of patients with thyroid nodules (TNODs), **thyroid cancer**, and functional or **autoimmune thyroid disease**. These applications aim to automate processes, improve the accuracy and consistency of **diagnosis**, personalize **treatment**, decrease the burden for health care professionals, improve access to specialized care in areas lacking expertise, deepen the understanding of subtle pathophysiologic patterns, and accelerate the learning curve of less experienced clinicians. There are promising results for many of these applications. Yet, most are in the **validation** or early clinical evaluation stages. Only a few are currently adopted for **risk stratification** of TNODs by **ultrasound** and determination of the **malignant** nature of indeterminate TNODs by **molecular testing**. Challenges of the currently available AI applications include the lack of prospective and multicenter validations and utility studies, small and low diversity of training data sets, differences in data sources, lack of explainability, unclear clinical impact, inadequate stakeholder engagement, and inability to use outside of the research setting, which might limit the value of their future adoption. Conclusions: AI has the potential to improve many aspects of thyroidology; however, addressing the limitations affecting the suitability of AI interventions in thyroidology is a prerequisite to ensure that AI provides added value for patients with thyroid disease.

中文翻译

背景：人工智能在医疗领域的应用呈指数级增长，有望促进生物医学研究，并提升诊断、治疗、监测、疾病预防及医疗服务水平。本文旨在探讨人工智能在甲状腺学领域的应用现状、局限性及未来发展方向。摘要：自20世纪90年代起，人工智能已开始应用于甲状腺学领域。当前，利用人工智能改善甲状腺结节、甲状腺癌、功能性或自身免疫性甲状腺疾病患者诊疗的研究兴趣日益浓厚。这些应用旨在实现流程自动化、提高诊断准确性与一致性、制定个性化治疗方案、减轻医疗专业人员负担、改善缺乏专业知识的地区的专科医疗服务可及性、深化对细微病理生理模式的理解，并加速经验不足临床医生的学习曲线。多项应用已展现出令人鼓舞的成果，但多数仍处于验证或早期临床评估阶段。目前仅有少数应用被实际采纳，包括通过超声进行甲状腺结节风险分层，以及通过分子检测确定不确定性甲状腺结节的恶性特征。现有AI应用面临的挑战包括：缺乏前瞻性多中心验证及效用研究、训练数据集规模小且多样性不足、数据来源差异、可解释性缺失、临床影响不明确、利益相关方参与不足，以及无法在科研环境外使用等问题，这些因素可能限制其未来应用价值。结论：人工智能具有改善甲状腺学多领域实践的潜力，但解决影响AI干预措施在甲状腺学中适用性的局限性，是确保AI为甲状腺疾病患者创造附加价值的先决条件。

核心词汇

thyroid nodule /ˈθaɪrɔɪd ˈnɒdjuːl/ : 甲状腺结节	thyroid cancer /ˈθaɪrɔɪd ˈkænsər/ : 甲状腺癌
autoimmune thyroid disease /ˌɔːtəʊɪˈmjʊːn ˈθaɪrɔɪd dɪˈziːz/ : 自身免疫性甲状腺疾病	ultrasound /ˈʌltrəsaʊnd/ : 超声
risk stratification /rɪsk ˌstrætɪfɪˈkeɪʃən/ : 风险分层	malignant /məˈlɪɡnənt/ : 恶性的
molecular testing /məˈlekjʊlər ˈtestɪŋ/ : 分子检测	diagnosis /ˌdaɪəɡˈnəʊsɪs/ : 诊断
treatment /ˈtriːtmənt/ : 治疗	pathophysiologic pattern /ˌpæθəʊˌfɪziəˈlɒdʒɪk ˈpætən/ : 病理生理模式
validation /ˌvælɪˈdeɪʃən/ : 验证	prospective study /prəˈspektɪv ˈstʌdi/ : 前瞻性研究

Endogenous hyperinsulinemic hypoglycemia: case series and literature review.

内源性高胰岛素血症性低血糖：病例系列与文献综述。

期刊: Endocrine 日期: 2023

Abstract

PURPOSE: Endogenous hyperinsulinemic **hypoglycemia** (EHH) is an uncommon disease characterized by inappropriately high plasma insulin levels despite low plasma glucose levels. Some rare etiologies can lead to EHH. Correct diagnosis is a prerequisite for treatment. Hence, although challenging, it is crucial for patients with EHH to identify the different causes. **METHODS:** We describe a case series of three patients, all of whom had obvious hypoglycemic symptoms and extraordinary **hyperinsulinemia**. Their plasma glucose, insulin, and C-peptide levels were tested simultaneously when **hypoglycemia** occurred. Moreover, other biochemical indices and relevant **antibody** levels were measured and imaging examinations were conducted. **RESULTS:** According to their medical history, physical examination, laboratory results, and imaging findings, the three patients were diagnosed with **insulinoma**, type B **insulin resistance** syndrome, and **insulin autoimmune syndrome**. After precise treatments, **hypoglycemia** was ultimately eliminated. **CONCLUSION:** Although these diseases have similar symptoms and biochemical abnormalities, the treatment and **prognosis** are different. The case series presented here highlights the challenges in the **differential diagnosis** of EHH. An accurate diagnosis is necessary for **hypoglycemia** treatment.

中文翻译

目的：内源性高胰岛素血症性低血糖（EHH）是一种罕见疾病，其特征是尽管血糖水平较低，但血浆胰岛素水平却异常升高。某些罕见病因可导致EHH。正确诊断是治疗的前提。因此，尽管具有挑战性，但明确EHH患者的不同病因至关重要。**方法：**我们报告了三例病例系列，所有患者均有明显低血糖症状和异常高胰岛素血症。在低血糖发作时同步检测其血糖、胰岛素和C肽水平。此外，还测量了其他生化指标和相关抗体水平，并进行了影像学检查。**结果：**根据病史、体格检查、实验室结果和影像学发现，三名患者分别被诊断为胰岛素瘤、B型胰岛素抵抗综合征和胰岛素自身免疫综合征。经过精准治疗后，低血糖最终消除。**结论：**尽管这些疾病具有相似的症状和生化异常，但治疗和预后不同。本病例系列强调了EHH鉴别诊断的挑战性。准确诊断是低血糖治疗的必要条件。

核心词汇

hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖症

insulinoma /ˌɪnsjʊlɪˈnoʊmə/ : 胰岛素瘤

insulin resistance /ˌɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗

biochemical abnormality /ˌbaɪəʊˈkemɪkəl ˌæbnɔːrˈmələti/ : 生化异常

antibody /ˈæntɪˌbɒdi/ : 抗体

etiology /ˌiːtiˈɒlədʒi/ : 病因学

hyperinsulinemia /ˌhaɪpərɪnsjʊlɪˈniːmiə/ : 高胰岛素血症

insulin autoimmune syndrome /ˌɪnsjʊlɪn ˌɔːtəɪˈmjuːn ˈsɪndroʊm/ : 胰岛素自身免疫综合征

differential diagnosis /ˌdɪfəˈrenʃəl ˌdaɪəɡˈnoʊsɪs/ : 鉴别诊断

imaging examination /ˈɪmɪdʒɪŋ ɪɡˌzæmɪˈneɪʃən/ : 影像学检查

prognosis /ˈprɒɡˈnoʊsɪs/ : 预后

Glucagonoma and Glucagonoma Syndrome: An Updated Review.

胰高血糖素瘤与胰高血糖素瘤综合征：最新综述。

期刊: Clinical endocrinology 日期: 2025

Abstract

BACKGROUND: **glucagonoma** is a rare well-differentiated slowly proliferating **pancreatic neuroendocrine tumour**, characterized by several manifestations including **necrolytic migratory erythema**, **weight loss**, **diabetes** and **anaemia**. AIM: The purpose of the current review was to acknowledge literature about this rare **tumour** discerning the clinical features, diagnosis, treatment and prognosis of **glucagonoma** by comparing three different periods. RESULTS: A study published by Song reviewed 216 cases from studies published between 1999 and 2016, making a comparison with 407 cases reported before 1998 by Soga and Yakuwa. The current review consisted of 86 cases reported in studies published in and after 2017: 40 males and 46 females, with an average age of 48.2 years. The male-to-female ratio was 0.87. The rate of typical clinical findings was as follows: NME, 93.2% (69/74); DM, 70.3% (50/74); **weight loss**, 62.2% (46/74); **anaemia**, 44.6% (33/74); **glossitis** or **stomatitis** or **cheilitis**, 31.1% (23/74). A total of 85 cases reported the location of the **tumour** as the **pancreas** and 38.8% of these cases involved the tail of the **pancreas**. The average **tumour** size was 4.6 cm in patients between 2017 and 2024. **metastasis** was detected in 52.3% of patients (45/86). The comparison with previous series highlighted an earlier age of diagnosis of **glucagonoma** and a higher rate of NME, consistent with a higher diagnostic accuracy. CONCLUSIONS: **glucagonoma** is a rare pathology, with peculiar characteristics that need to be acknowledged to achieve a timely diagnosis and finally improve patient prognosis.

中文翻译

背景：胰高血糖素瘤是一种罕见的、分化良好且缓慢增殖的胰腺神经内分泌肿瘤，其特征性表现包括坏死松解性游走性红斑、体重减轻、糖尿病和贫血。目的：本综述旨在通过比较三个不同时期的相关文献，阐明这种罕见肿瘤的临床特征、诊断、治疗及预后。结果：Song发表的一项研究回顾了1999年至2016年间发表的216例病例，并与Soga和Yakuwa在1998年前报告的407例病例进行了比较。本综述纳入了2017年及之后发表的86例病例：其中男性40例，女性46例，平均年龄48.2岁，男女比例为0.87。典型临床表现的发生率如下：坏死松解性游走性红斑93.2%（69/74）；糖尿病70.3%（50/74）；体重减轻62.2%（46/74）；贫血44.6%（33/74）；舌炎、口腔炎或唇炎31.1%（23/74）。共85例病例报告肿瘤位于胰腺，其中38.8%的病例累及胰尾部。2017年至2024年间患者的平均肿瘤大小为4.6厘米。52.3%的患者（45/86）发现转移。与既往系列研究相比，本综述显示胰高血糖素瘤的诊断年龄更早，坏死松解性游走性红斑的发生率更高，这与更高的诊断准确性一致。结论：胰高血糖素瘤是一种罕见疾病，需认识其独特特征以实现及时诊断，最终改善患者预后。

核心词汇

glucagonoma /gluːˈkæɡənoʊmə/ : 胰高血糖素瘤	pancreatic neuroendocrine tumour /ˌpæŋkriˈætiːk ˌnjuːəreɪ ˈendəʊkraɪn ˈtjuːmə/ : 胰腺神经内分泌肿瘤
necrolytic migratory erythema /ˌnekroʊˈlɪtɪk ˈmaɪɡrətəri ˌeri ˈθiːmə/ : 坏死性游走性红斑	diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病
anaemia /əˈniːmiə/ : 贫血	weight loss /weɪt lɒs/ : 体重减轻
glossitis /ɡloʊˈsɑɪtɪs/ : 舌炎	stomatitis /ˌsteʊməˈtaɪtɪs/ : 口腔炎
cheilitis /keɪˈlaɪtɪs/ : 唇炎	metastasis /meˈtæstəsɪs/ : 转移
pancreas /ˈpæŋkriəs/ : 胰腺	tumour /ˈtjuːmə/ : 肿瘤

Practical therapeutic approach in the management of diabetes mellitus secondary to Cushing's syndrome, acromegaly and neuroendocrine tumours.

库欣综合征、肢端肥大症及神经内分泌肿瘤继发糖尿病的实用治疗策略

期刊: Frontiers in endocrinology 日期: 2023

Abstract

Cushing's syndrome, acromegaly and neuroendocrine disorders are characterized by an excess of counterregulatory hormones, able to induce insulin resistance and glucose metabolism disorders at variable degrees and requiring immediate treatment, until patients are ready to undergo surgery. This review focuses on the management of diabetes mellitus in endocrine disorders related to an excess of counterregulatory hormones. Currently, the landscape of approved agents for treatment of diabetes is dynamic and is mainly patient-centred and not glycaemia-centred. In addition, personalized medicine is more and more required to provide a precise approach to the patient's disease. For this reason, we aimed to define a practical therapeutic algorithm for management of diabetes mellitus in patients with glucagonoma, pheochromocytoma, Cushing's syndrome and acromegaly, based on our practical experience and on the physiopathology of the specific endocrine disease taken into account. This document is addressed to all specialists who approach patients with diabetes mellitus secondary to endocrine disorders characterized by an excess of counterregulatory hormones, in order to take better care of these patients. Care and control of diabetes mellitus should be one of the primary goals in patients with an excess of counterregulatory hormones requiring immediate and aggressive treatment.

中文翻译

库欣综合征、肢端肥大症及神经内分泌疾病的特征为反调节激素过量，这些激素可在不同程度上诱发胰岛素抵抗和糖代谢紊乱，需在患者接受手术前进行紧急治疗。本综述聚焦于与反调节激素过量相关的内分泌疾病中糖尿病的治疗管理。当前获批的糖尿病治疗药物格局动态变化，主要遵循以患者为中心而非以血糖为中心的原则。此外，精准医疗日益要求针对患者疾病提供个体化方案。基于此，我们结合临床实践经验及特定内分泌疾病的病理生理机制，旨在为胰高血糖素瘤、嗜铬细胞瘤、库欣综合征及肢端肥大症患者的糖尿病管理制定实用治疗路径。本文面向所有接诊继发于反调节激素过量内分泌疾病所致糖尿病患者的专科医师，以期优化此类患者的诊疗。对于需要紧急强化治疗的反调节激素过量患者，糖尿病的管控应作为首要目标之一。

核心词汇

Cushing's syndrome /'kʊʃɪŋz 'sɪndroum/ : 库欣综合征

neuroendocrine disorders /,njʊəʊ'endekrɪn dɪs'ɔ:rdəz/ : 神经内分泌疾病

insulin resistance /'ɪnsəlɪn rɪ'zɪstəns/ : 胰岛素抵抗

diabetes mellitus /,daɪə'bi:tɪz 'mɛlɪtəs/ : 糖尿病

glucagonoma /,glu:kə'gounoumə/ : 胰高血糖素瘤

personalized medicine /'pɜ:rsənəlaɪzd 'mɛdɪsɪn/ : 个体化医学

acromegaly /,ækre'mɛgəli/ : 肢端肥大症

counterregulatory hormones /,kaʊntər'regjelətɔ:ri 'hɔ:rmounz/ : 反调节激素

glucose metabolism disorders /,glu:kous mə'tæbəlaɪzəm dɪs'ɔ:rdəz/ : 糖代谢紊乱

endocrine disorders /'endekrɪn dɪs'ɔ:rdəz/ : 内分泌疾病

pheochromocytoma /,fi:ou'kroumousaɪ'toumə/ : 嗜铬细胞瘤

therapeutic algorithm /,θɛrə'pjʊ:tɪk 'ælɡərɪðəm/ : 治疗算法

Nuclear medicine and perspective thoughts in the diagnosis and treatment of pheochromocytoma and paraganglioma.

嗜铬细胞瘤和副神经节瘤诊断与治疗中的核医学及前瞻性思考。

期刊: Journal of neuroendocrinology 日期: 2025

Abstract

pheochromocytoma and **paraganglioma** (PPGL) are rare and life-threatening tumors of the **adrenal medulla** and extra-adrenal paraganglia, respectively. Management of PPGL depends on accurate diagnosis and treatment. This review provides a very comprehensive overview of **nuclear medicine** in PPGL with some of our perspective views on diagnostic challenges and pitfalls, treatment, and new phase studies. It describes **nuclear medicine** techniques including (64)Cu/(68)Ga-DOTA-SSA, (18)F-FDOPA, (18)F-FDG, (123)I-MIBG, and some emerging **molecular imaging** agents and PRRT therapies, such as (203)Pb VMT-alpha-NET and (212)Pb VMT-alpha-NET **targeted alpha therapy** (TAT). It also provides insight into the use of **proliferating cell nuclear antigen** (PCNA) inhibitors in combination with therapeutics in aggressive/**metastatic** PPGL. Through a comprehensive review of the latest developments and clinical practice, this review aims to guide healthcare professionals in improving the **diagnostic accuracy** and **therapeutic efficacy** of PPGL.

中文翻译

嗜铬细胞瘤和副神经节瘤（PPGL）分别是发生于肾上腺髓质和肾上腺外副神经节的罕见且危及生命的肿瘤。PPGL的诊疗依赖于精准诊断与规范治疗。本综述全面阐述了核医学在PPGL中的应用，并结合我们的观点探讨了诊断挑战与陷阱、治疗策略及新药临床研究。文中系统介绍了包括(64)Cu/(68)Ga-DOTA-SSA、(18)F-FDOPA、(18)F-FDG、(123)I-MIBG等核医学技术，以及(203)Pb VMT-alpha-NET和(212)Pb VMT-alpha-NET靶向α疗法（TAT）等新兴分子影像剂与PRRT疗法。同时深入探讨了增殖细胞核抗原（PCNA）抑制剂联合治疗在侵袭性/转移性PPGL中的应用价值。通过系统梳理最新研究进展与临床实践，本综述旨在指导医疗专业人员提升PPGL的诊断准确性与治疗效能。

核心词汇

pheochromocytoma /fi:oukroumousai'toumə/ : 嗜铬细胞瘤	paraganglioma /,pæɹəgæŋgli'oumə/ : 副神经节瘤
adrenal medulla /ə'dri:nəl mɪ'dʌlə/ : 肾上腺髓质	nuclear medicine /'nu:kliə 'mɛdɪsɪn/ : 核医学
molecular imaging /mə'lekju:lər 'ɪmɪdʒɪŋ/ : 分子影像	targeted alpha therapy /'tɑ:rgɪtɪd 'ælfə 'θɛrəpi/ : 靶向α治疗
proliferating cell nuclear antigen /prə'ɪfɪəreɪtɪŋ sɛl 'nu:kliə 'æntɪdʒən/ : 增殖细胞核抗原	inhibitor /ɪn'hɪbɪtə/ : 抑制剂
metastatic /,mɛtə'stæɪtɪk/ : 转移性的	diagnostic accuracy /,daɪəg'nɒstɪk 'ækjəreɪsi/ : 诊断准确性
therapeutic efficacy /θɛrə'pju:tɪk 'ɛfɪkəsi/ : 治疗有效性	

A review of the safety of sodium-glucose co-transporter-2 inhibitors.

钠-葡萄糖共转运蛋白2抑制剂的安全性综述。

期刊: Diabetes, obesity & metabolism 日期: 2025

Abstract

The development of sodium-glucose co-transporter-2 (SGLT2) inhibitors represents a major turning point in the effort to preserve **kidney function** and prevent **cardiovascular events** and **heart failure** hospitalisations in those at high risk. These agents have now transcended their original glucose-lowering indication and provide a range of clinical benefits in people both with and without **diabetes**, and at varying levels of **kidney function**. Despite this, SGLT2 inhibitors remain underutilized by the medical community. One potential barrier to improved uptake may be concern about **adverse effects**. The following review summarizes the wealth of information garnered from **clinical trials** and real-world data in recent years to examine the **safety** of SGLT2 inhibitors and to provide practical advice to promote safer use of these important **therapies**.

中文翻译

钠-葡萄糖协同转运蛋白2 (SGLT2) 抑制剂的发展, 标志着在保护高危人群肾功能、预防心血管事件及心力衰竭住院方面的重要转折点。这类药物已超越其最初的降糖适应症, 为伴或不伴糖尿病、不同肾功能水平的患者提供了广泛的临床获益。尽管如此, SGLT2抑制剂在医学界的应用仍不充分。阻碍其推广的潜在因素之一可能是对不良反应的担忧。本综述总结了近年来从临床试验和真实世界数据中获取的丰富信息, 旨在评估SGLT2抑制剂的安全性, 并提供实用建议以促进这些重要疗法的安全应用。

核心词汇

kidney function /'kɪdni 'fʌŋkʃən/ : 肾功能	cardiovascular events /kɑ:rdiəʊ'væskjələr ɪ'vents/ : 心血管事件
heart failure /hɑ:rt 'feɪljər/ : 心力衰竭	diabetes /ˌdaɪə'bi:ti:z/ : 糖尿病
adverse effects /æd'vɜ:rs ɪ'fekts/ : 不良反应	clinical trials /'klɪnɪkəl 'traɪəlz/ : 临床试验
safety /'seɪfti/ : 安全性	therapies /'θerəpi:z/ : 治疗

Treatment recommendations for glycogen storage disease type IB-associated neutropenia and neutrophil dysfunction with empagliflozin: Consensus from an international workshop.

恩格列净治疗糖原贮积症IB型相关中性粒细胞减少症和中性粒细胞功能障碍的建议：国际研讨会共识。

期刊: Molecular genetics and metabolism 日期: 2024

Abstract

glycogen storage disease type Ib (GSD Ib, biallelic variants in SLC37A4) is a rare disorder of glycogen metabolism complicated by neutropenia/neutrophil dysfunction. Since 2019, the SGLT2-inhibitor empagliflozin has provided a mechanism-based treatment option for the symptoms caused by neutropenia/neutrophil dysfunction (e.g. mucosal lesions, inflammatory bowel disease). Because of the rarity of GSD Ib, the published evidence on safety and efficacy of empagliflozin is still limited and does not allow to develop evidence-based guidelines. Here, an international group of experts provides 14 best practice consensus treatment recommendations based on expert practice and review of the published evidence. We recommend to start empagliflozin in all GSD Ib individuals with clinical or laboratory signs related to neutropenia/neutrophil dysfunction with a dose of 0.3-0.4 mg/kg/d given as a single dose in the morning. Treatment can be started in an outpatient setting. The dose should be adapted to the weight and in case of inadequate clinical treatment response or side effects. We strongly recommend to pause empagliflozin immediately in case of threatening dehydration and before planned longer surgeries. Discontinuation of G-CSF therapy should be attempted in all individuals. If available, 1,5-AG should be monitored. Individuals who have previously not tolerated starches should be encouraged to make a new attempt to introduce starch in their diet after initiation of empagliflozin treatment. We advise to monitor certain safety and efficacy parameters and recommend continuous, alternatively frequent glucose measurements during the introduction of empagliflozin. We provide specific recommendations for special circumstances like pregnancy and liver transplantation.

中文翻译

糖原贮积症Ib型（GSD Ib，由SLC37A4基因双等位基因变异引起）是一种罕见的糖原代谢疾病，常伴有中性粒细胞减少/功能障碍。自2019年起，SGLT2抑制剂恩格列净为中性粒细胞减少/功能障碍引起的症状（如黏膜病变、炎症性肠病）提供了基于病理机制的治疗选择。由于GSD Ib的罕见性，关于恩格列净安全性和有效性的已发表证据仍然有限，尚不足以制定循证指南。为此，一个国际专家组基于临床实践经验和已发表证据的回顾，提出了14项最佳实践共识治疗建议。我们建议对所有出现中性粒细胞减少/功能障碍相关临床或实验室指征的GSD Ib患者启用恩格列净，初始剂量为0.3-0.4 mg/kg/d，早晨单次给药。治疗可在门诊启动。剂量应根据体重调整，若临床疗效不足或出现副作用时需调整剂量。我们强烈建议在面临脱水风险时及计划进行长时间手术前立即暂停恩格列净。所有患者均应尝试停用G-CSF治疗。如条件允许，应监测1,5-脱水葡萄糖醇水平。对于既往无法耐受淀粉类食物的患者，建议在启动恩格列净治疗后重新尝试在饮食中引入淀粉。我们建议在启用恩格列净期间监测特定安全性和有效性参数，并推荐持续或频繁进行血糖监测。针对妊娠和肝移植等特殊情况，我们提供了具体建议。

核心词汇

glycogen storage disease /ˈglʌkədʒən ˈstɔːrɪdʒ dɪˈziːz/ : 糖原贮积症	neutropenia /ˌnjuːtrəˈpiːniə/ : 中性粒细胞减少症
neutrophil dysfunction /ˌnjuːtrəˈfiːl dɪsˈfʌŋkʃən/ : 中性粒细胞功能障碍	mucosal lesions /mjuːˈkəʊsəl ˈliːʒənz/ : 黏膜病变
inflammatory bowel disease /ɪnˈflæmətɔːri ˈbaʊəl dɪˈziːz/ : 炎症性肠病	consensus treatment recommendations /kənˈsensəs ˈtriːtmənt ˌrekəmenˈdeɪʃənz/ : 共识治疗建议
dehydration /diːˈhaɪˈdreɪʃən/ : 脱水	starch /stɑːrtʃ/ : 淀粉
glucose measurements /ˈgluːkəʊs ˈmeʒərmənts/ : 血糖测量	liver transplantation /ˈlɪvər ˌtrænsplænˈteɪʃən/ : 肝移植
pregnancy /ˈpregnənsi/ : 妊娠	

Rewriting Diabetes Therapy: How Incretin Modulation is Transforming Cardiovascular and Renal Outcomes.

改写糖尿病治疗：肠促胰岛素调节如何改变心血管和肾脏结局

期刊: Diabetes therapy : research, treatment and education of diabetes and related disorders

日期: 2026

Abstract

INTRODUCTION: In patients with type 2 diabetes mellitus (T2DM), cardiovascular (CV) disease and chronic kidney disease (CKD) drive excess morbidity and mortality. Beyond glucose-lowering, incretin-based therapies may provide organ protection across the cardiorenal axis. **METHODS:** Narrative review of mechanistic pathways and randomized trials of GLP-1 receptor agonists (GLP-1RA), DPP-4 inhibitors, and newer dual/triple agonists, with targeted updates from recent pivotal programs (SELECT, FLOW, SOUL, SURPASS-CVOT) and emerging oral small-molecule GLP-1R agonists. **RESULTS:** Long-acting GLP-1RA reduces major adverse CV events (MACE), all-cause and CV death, heart-failure hospitalization, and kidney composites across CV outcome trials and meta-analyses. A 2019 pooled analysis and a 2025 update confirm consistent reductions in MACE and hard kidney outcomes independent of baseline HbA1c. In obesity without diabetes, semaglutide 2.4 mg lowered MACE in SELECT, expanding prevention beyond glycemia. In CKD with T2DM, FLOW showed that semaglutide reduced major kidney disease events and death from CV/kidney causes. In T2DM with ASCVD and/or CKD, the SOUL cardiovascular outcome trial (CVOT) demonstrated that oral semaglutide reduced three-point MACE versus placebo. In head-to-head CVOT, tirzepatide was non-inferior to dulaglutide on MACE while achieving greater weight and HbA1c reductions. Mechanistically, GLP-1R signaling spans Gs-cAMP/PKA, beta-arrestin-dependent pathways, and additional routes (including Gq contexts), aligning with anti-inflammatory, natriuretic, and antifibrotic effects observed preclinically and clinically. Oral non-peptide GLP-1R agonists (e.g., orforglipron) show phase 2 efficacy but lack long-term CV/renal outcome data. **CONCLUSIONS:** incretin-based therapy has shifted care from glucose-centric targets to cardiorenal risk reduction. GLP-1RA are guideline-endorsed for patients with T2DM and high CV/renal risk irrespective of HbA1c; dual agonists and oral small-molecule agents may broaden indications pending definitive outcome evidence.

中文翻译

引言：在2型糖尿病（T2DM）患者中，心血管疾病和慢性肾脏病是导致超额发病率和死亡率的主要原因。除降糖作用外，基于肠促胰岛素的疗法可能通过心肾轴提供器官保护。方法：对GLP-1受体激动剂、DPP-4抑制剂及新型双/三重激动剂的机制通路和随机试验进行叙述性综述，并针对近期关键研究项目（SELECT、FLOW、SOUL、SURPASS-CVOT）及新兴口服小分子GLP-1R激动剂进行靶向更新。结果：在心血管结局试验和荟萃分析中，长效GLP-1RA可降低主要不良心血管事件、全因及心血管死亡、心力衰竭住院和肾脏复合终点。2019年汇总分析及2025年更新证实，无论基线HbA1c水平如何，MACE和硬性肾脏结局均持续降低。在无糖尿病的肥胖患者中，司美格鲁肽2.4mg在SELECT试验中降低MACE，将预防范围扩展至血糖控制之外。在合并T2DM的CKD患者中，FLOW试验显示司美格鲁肽可减少主要肾脏疾病事件及心血管/肾脏原因死亡。在合并ASCVD和/或CKD的T2DM患者中，SOUL心血管结局试验证实口服司美格鲁肽较安慰剂降低三点MACE。头对头CVOT显示，替尔泊肽在MACE方面非劣效于度拉糖肽，同时实现更显著的体重和HbA1c降低。机制上，GLP-1R信号涵盖Gs-cAMP/PKA、β-arrestin依赖性通路及其他途径（包括Gq背景），与临床前和临床观察到的抗炎、利钠和抗纤维化效应一致。口服非肽类GLP-1R激动剂（如orforglipron）显示2期疗效，但缺乏长期心血管/肾脏结局数据。结论：基于肠促胰岛素的疗法已将治疗重点从以血糖为中心转向降低心肾风险。GLP-1RA被指南推荐用于合并高心血管/肾脏风险的T2DM患者，无论HbA1c水平如何；双激动剂和口服小分子药物可能扩大适应症，但需确证性结局证据。

核心词汇

cardiovascular disease /ˌkɑːrdiəʊˈvæskjələr dɪˈziːz/ : 心血管疾病	chronic kidney disease /ˈkrɒnɪk ˈkɪdni dɪˈziːz/ : 慢性肾脏病
incretin /ɪnˈkriːtɪn/ : 肠促胰岛素	heart failure /hɑːt ˈfeɪljər/ : 心力衰竭
obesity /ouˈbiːsəti/ : 肥胖	atherosclerotic cardiovascular disease /ˌæθərəʊskləˈrɒtɪk ˌkɑːrdiəʊˈvæskjələr dɪˈziːz/ : 动脉粥样硬化性心血管疾病
natriuretic /ˌnɛtriəˈreɪtɪk/ : 利尿钠	antifibrotic /ˌænti faɪˈbrɒtɪk/ : 抗纤维化
cardiorenal /ˌkɑːrdiəʊˈriːnəl/ : 心肾的	

Semaglutide and obesity: beyond the nutritional and lifestyle intervention?

司美格鲁肽与肥胖：超越营养与生活方式干预？

期刊: Minerva endocrinology 日期: 2024

Abstract

Semaglutide is the second marketed glucagon-like peptide 1 receptor agonist that can be used safely and efficiently in non-diabetic people with excess weight, providing a new milestone in the pharmacological treatment of obesity. This narrative review aims to describe the clinical actions of this new drug in weight management in non-diabetic patients along with possible side-effects and dropout reasons. To accomplish this, the PubMed database was searched to retrieve the most relevant clinical studies published to date on this topic, using the following keywords "semaglutide and obesity". Currently, semaglutide is on the market in two formulations, the once-weekly subcutaneous (s.c.) semaglutide and once-daily oral semaglutide. Data in the literature on the anti-obesity action of semaglutide are available for both routes of administration of the drug, with a prevalence of studies using the s.c. one. However, given its dosage, oral semaglutide may provide greater attractiveness and better treatment adherence, but further research is needed in this field.

中文翻译

司美格鲁肽是第二种上市的胰高血糖素样肽-1受体激动剂，可安全有效地用于非糖尿病超重人群，为肥胖症的药物治疗树立了新里程碑。本叙述性综述旨在阐述该新药在非糖尿病患者体重管理中的临床作用，以及可能的副作用和停药原因。为此，我们检索了PubMed数据库中迄今发表的相关临床研究，使用关键词"semaglutide and obesity"。目前，司美格鲁肽有两种剂型上市：每周一次皮下注射剂和每日一次口服剂。文献中关于司美格鲁肽抗肥胖作用的数据涵盖两种给药途径，其中以皮下注射剂研究为主。然而，考虑到剂量因素，口服司美格鲁肽可能具有更高的吸引力和更好的治疗依从性，但该领域仍需进一步研究。

核心词汇

obesity /ou'bi:seɪti/ : 肥胖	weight management /weɪt 'mæɪnɪdʒmənt/ : 体重管理
subcutaneous /ˌsʌbkju:'teɪniəs/ : 皮下的	oral /'ɔ:rel/ : 口服的
formulation /ˌfɔ:rmju'leɪʃən/ : 制剂	dosage /'dɔʊsɪdʒ/ : 剂量
adherence /əd'hɪərəns/ : 依从性	clinical study /'klɪnɪkəl 'stʌdi/ : 临床研究
pharmacological treatment /ˌfɑ:rməke'lɒdʒɪkəl 'tri:tment/ : 药物治疗	dropout /'drɒpaʊt/ : 退出

Strategies for minimizing muscle loss during use of incretin-mimetic drugs for treatment of obesity.

使用肠促胰岛素类似物治疗肥胖期间减少肌肉流失的策略

期刊: Obesity reviews : an official journal of the International Association for the Study of Obesity 日期: 2025

Abstract

The rapid and widespread clinical adoption of highly effective incretin-mimetic drugs (IMDs), particularly semaglutide and tirzepatide, for the treatment of **obesity** has outpaced the updating of clinical practice guidelines. Consequently, many patients may be at risk for adverse effects and uncertain long-term outcomes related to the use of these drugs. Of emerging concern is the loss of **skeletal muscle** mass and function that can accompany rapid substantial **weight reduction**; such losses can lead to reduced functional and metabolic health, weight cycling, compromised quality of life, and other adverse outcomes. Available evidence suggests that **clinical trial** participants receiving IMDs for the treatment of **obesity** lost 10% or more of their **muscle mass** during the 68- to 72-week interventions, approximately equivalent to 20 years of age-related muscle loss. The ability to maintain **muscle mass** during **caloric restriction**-induced **weight reduction** is influenced by two key factors: nutrition and physical exercise. **nutrition therapy** should ensure adequate intake and absorption of high-quality **protein** and micronutrients, which may require the use of oral nutritional supplements. Additionally, concurrent physical activity, especially **resistance training**, has been shown to effectively minimize loss of **muscle mass** and function during **weight reduction** therapy. All patients receiving IMDs for **obesity** should participate in comprehensive treatment programs emphasizing adequate **protein** and **micronutrient** intakes, as well as **resistance training**, to preserve **muscle mass** and function, maximize the benefit of IMD therapy, and minimize potential risks.

中文翻译

高效模拟肠促胰岛素药物 (IMDs)，特别是司美格鲁肽和替尔泊肽，在肥胖治疗中的快速广泛临床应用已超越了临床实践指南的更新速度。因此，许多患者可能面临药物相关不良反应及不确定长期结局的风险。新近引发关注的是，快速大幅减重可能伴随骨骼肌质量与功能的丧失，这种流失会导致功能代谢健康下降、体重循环、生活质量受损及其他不良后果。现有证据表明，接受IMDs治疗肥胖的临床试验参与者在68至72周的干预期间，肌肉质量减少了10%以上，约相当于20年年龄相关的肌肉流失量。在热量限制诱导的减重过程中维持肌肉质量的能力受两大关键因素影响：营养与体育锻炼。营养治疗应确保优质蛋白质和微量营养素的充分摄入与吸收，必要时需使用口服营养补充剂。此外，同步进行体育锻炼（尤其是抗阻训练）已被证实能有效减少减重治疗期间的肌肉质量与功能流失。所有接受IMDs治疗肥胖的患者都应参与综合治疗方案，强调充足蛋白质和微量营养素摄入及抗阻训练，以保护肌肉质量与功能，最大化IMDs治疗获益，并最小化潜在风险。

核心词汇

obesity /ou'bi:səti/ : 肥胖	skeletal muscle /'skelɪtl 'mʌsl/ : 骨骼肌
muscle mass /'mʌsl mæs/ : 肌肉质量	weight reduction /weɪt rɪ'dʌkʃən/ : 减重
caloric restriction /kə'lo:rɪk rɪ'strɪkʃən/ : 热量限制	nutrition therapy /nu:'trɪʃən 'θerəpi/ : 营养治疗
protein /'prəuti:n/ : 蛋白质	micronutrient /,maɪkrou'nu:trɪənt/ : 微量营养素
resistance training /rɪ'zɪstəns 'treɪnɪŋ/ : 抗阻训练	adverse effect /æd'vɜ:rs ɪ'fekt/ : 不良反应
clinical trial /'klɪnɪkl 'traɪəl/ : 临床试验	oral nutritional supplement /'ɔ:rəl nu:'trɪʃənəl 'sʌplɪmənt/ : 口服营养补充剂

Incretin-Based Weight Loss Pharmacotherapy: Can Resistance Exercise Optimize Changes in Body Composition?

基于肠促胰岛素的减重药物治疗：抗阻运动能否优化身体成分变化？

期刊: Diabetes care 日期: 2024

Abstract

This narrative review highlights the degree to which new **antiobesity medications** based on gut-derived nutrient-stimulated hormones (**incretins**) cause loss of **lean mass**, and the importance of **resistance exercise** to preserve **muscle**. Glucagon-like peptide 1 receptor agonists (GLP-1RA) induce substantial weight loss in randomized trials, effects that may be enhanced in combination with glucose-dependent insulinotropic polypeptide (GIP) receptor agonists. Liraglutide and semaglutide (GLP-1RA), tirzepatide (GLP-1 and GIP receptor dual agonist), and retatrutide (GLP-1, GIP, and glucagon receptor triple agonist) are peptides with incretin agonist activity that induce approximately 15-24% weight loss in adults with **overweight** and **obesity**, alongside beneficial impacts on **blood pressure**, **cholesterol**, **blood glucose**, and **insulin**. However, these agents also cause rapid and significant loss of **lean mass** (approximately 10% or approximately 6 kg), comparable to a decade or more of aging. Maintaining **muscle** mass and function as humans age is crucial to avoiding **sarcopenia** and **frailty**, which are strongly linked to morbidity and mortality. Studies indicate that supervised **resistance exercise** training interventions with a duration >10 weeks can elicit large increases in **lean mass** (approximately 3 kg) and strength (approximately 25%) in men and women. After a low-calorie diet, combining **aerobic exercise** with liraglutide improved weight loss maintenance compared with either alone. Retaining **lean mass** during incretin therapy could blunt body weight (and fat) regain on cessation of weight loss pharmacotherapy. We propose that tailored **resistance exercise** training be recommended as an adjunct to incretin therapy to optimize changes in **body composition** by preserving **lean mass** while achieving fat loss.

中文翻译

本综述强调，基于肠道源性营养刺激激素（肠促胰岛素）的新型抗肥胖药物在多大程度上导致瘦体重流失，以及抗阻运动对维持肌肉的重要性。随机试验表明，胰高血糖素样肽-1受体激动剂（GLP-1RA）可诱导显著体重下降，与葡萄糖依赖性促胰岛素多肽（GIP）受体激动剂联用时效果可能增强。利拉鲁肽和司美格鲁肽（GLP-1RA）、替尔泊肽（GLP-1与GIP受体双重激动剂）以及瑞他鲁肽（GLP-1、GIP与胰高血糖素受体三重激动剂）均具有肠促胰岛素激动活性，可使超重和肥胖成人减重约15-24%，同时对血压、胆固醇、血糖和胰岛素产生有益影响。然而，这些药物也会导致瘦体重快速显著流失（约10%或约6公斤），相当于十年以上的衰老过程。随着年龄增长，维持肌肉质量和功能对避免肌少症和衰弱至关重要，而这两者与发病率和死亡率密切相关。研究表明，持续超过10周的有监督抗阻运动训练干预可使男性和女性的瘦体重增加约3公斤，力量提升约25%。在低热量饮食后，将有氧运动与利拉鲁肽联合使用比单独使用任一方法更能改善体重维持效果。在肠促胰岛素治疗期间保留瘦体重，可减缓停药后体重（及脂肪）的反弹。我们建议将定制化抗阻运动训练作为肠促胰岛素治疗的辅助手段，通过保留瘦体重同时实现减脂，优化身体成分变化。

核心词汇

antiobesity medications /ˌænti ɒʊˈbiːsəti ˌmɛdɪˈkeɪʃənz/ : 抗肥胖药物	incretins /ˈɪnkrɪtɪnz/ : 肠促胰岛素
lean mass /liːn mæs/ : 瘦体重	resistance exercise /rɪˈzɪstəns ˈeksəsaɪz/ : 抗阻运动
muscle /ˈmʌsəl/ : 肌肉	obesity /ɒʊˈbiːsəti/ : 肥胖
overweight /ˈoʊvərweɪt/ : 超重	blood pressure /blʌd ˈpreʃər/ : 血压
cholesterol /kəˈleʃtəroːl/ : 胆固醇	blood glucose /blʌd ˈgluːkəʊs/ : 血糖
insulin /ˈɪnsəlɪn/ : 胰岛素	sarcopenia /ˌsɑːrkeˈpiːniə/ : 肌少症
frailty /ˈfreɪlti/ : 衰弱	aerobic exercise /ɛˈrəʊbɪk ˈeksəsaɪz/ : 有氧运动

Emerging perspectives on once-weekly insulins in type 1 and type 2 diabetes: a mini-review.

关于每周一次胰岛素在1型和2型糖尿病中的新兴观点：小型综述。

期刊: Frontiers in endocrinology 日期: 2025

Abstract

The development of once-weekly **basal insulin** analogues, such as **insulin** icodec and efsitora alfa, represents a promising strategy to reduce injection burden and improve adherence in **diabetes** management. This mini-review summarizes the recent findings from clinical trials evaluating once-weekly **insulin** therapies in both type 1 and type 2 **diabetes**. In type 1 **diabetes**, available data remain limited; however, the ONWARDS 6 and QWINT-5 trials demonstrated that once-weekly icodec and efsitora, respectively, achieved comparable reductions in HbA1c to once-daily **insulin** degludec, when used in combination with **prandial insulin**. In type 2 **diabetes**, accumulating evidences from randomized clinical trials supports the **efficacy** of once-weekly icodec and efsitora, showing non-inferiority-and in some cases, superiority-compared to once-daily **basal insulin**, both in **insulin**-naive individuals and in those previously treated with **insulin**. **safety** profiles of once-weekly insulins in type 2 **diabetes** are reassuring, with similar rates of clinically significant and severe **hypoglycemia** compared to once-daily regimens. In contrast, trials in type 1 **diabetes** reported higher **hypoglycemia** rates with once-weekly insulins. Recent findings from the COMBINE program demonstrated that the fixed-ratio combination of icodec and semaglutide (IcoSema) produced superior HbA1c reductions compared to either agent alone, though not superior to a basal-bolus regimen with glargine and aspart **insulin**. However, several important questions remain to be addressed regarding once-weekly insulins, including their long-term **efficacy** on **cardiovascular** outcomes and overall long-term **safety**.

中文翻译

每周一次的基础胰岛素类似物（如胰岛素icodec和efsitora alfa）的开发，是减轻注射负担、提高糖尿病治疗依从性的重要策略。本综述总结了近期评估每周一次胰岛素疗法在1型和2型糖尿病中应用的临床试验结果。在1型糖尿病中，现有数据仍有限；但ONWARDS 6和QWINT-5试验表明，联合餐时胰岛素使用时，每周一次的icodec和efsitora分别实现了与每日一次德谷胰岛素相当的HbA1c降幅。在2型糖尿病中，随机临床试验积累的证据支持每周一次icodec和efsitora的疗效，显示其在未使用过胰岛素及既往接受胰岛素治疗的患者中，疗效非劣效于每日一次基础胰岛素，部分情况下甚至更优。每周一次胰岛素在2型糖尿病中的安全性令人放心，临床显著和严重低血糖发生率与每日一次方案相似。相比之下，1型糖尿病试验报告每周一次胰岛素组低血糖发生率更高。COMBINE项目最新结果显示，icodec与司美格鲁肽的固定比例复方制剂（IcoSema）在降低HbA1c方面优于单药治疗，但未优于甘精胰岛素联合门冬胰岛素的基础-餐时方案。然而，关于每周一次胰岛素仍存在若干重要问题待解答，包括其对心血管结局的长期疗效及整体长期安全性。

核心词汇

insulin /'ɪnsjʊlɪn/ : 胰岛素	diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病
hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖	basal insulin /ˈbeɪsəl ˈɪnsjʊlɪn/ : 基础胰岛素
prandial insulin /ˈprændiəl ˈɪnsjʊlɪn/ : 餐时胰岛素	clinical trial /ˈklɪnɪkəl ˈtraɪəl/ : 临床试验
efficacy /ˈefɪkəsi/ : 疗效	safety /ˈseɪfti/ : 安全性
cardiovascular /ˌkɑːrdiəʊˈvæskjʊlə/ : 心血管的	combination therapy /ˌkɒmbɪˈneɪʃən ˈθerəpi/ : 联合治疗
insulin analogue /ˈɪnsjʊlɪn ˈænələɡ/ : 胰岛素类似物	

Once-weekly Insulin Icodec Versus Once-daily Long-acting Insulin for Type II Diabetes: A Meta-analysis of Randomized Controlled Trials.

每周一次胰岛素Icodec对比每日一次长效胰岛素治疗2型糖尿病：随机对照试验的荟萃分析。

期刊: Journal of the Endocrine Society 日期: 2024

Abstract

BACKGROUND: **insulin** icodec is a novel basal **insulin** analog with once-weekly subcutaneous administration. We aim to estimate the efficacy and safety of **insulin** icodec vs long-acting **insulin** (**insulin** glargine and degludec) in type II diabetic patients. METHODS: We conducted a **systematic review** and meta-analysis synthesizing randomized controlled trials (RCTs), which were retrieved by systematically searching PubMed, Web of Science, SCOPUS, and Cochrane through May 29, 2023. We used RevMan V. 5.4 to pool dichotomous data using **risk ratio** (RR) and continuous data using **mean difference** (MD) with a 95% **confidence interval** (CI). Our primary outcome was **glycated hemoglobin** (HbA1C) change. RESULTS: We included 7 RCTs with a total of 3183 patients. **insulin** icodec was associated with significantly decreased HbA1C (MD: -0.15 with 95% CI [-0.24, -0.06], P = .002) and increased percentage of time with glucose in range (TIR) (MD: 4.06 with 95% CI [2.06, 6.06], P = .0001). However, **insulin** icodec was associated with increased **body weight** (MD: 0.57 with 95% CI [0.45, 0.70], P = .00001). Also, there was no difference regarding any serious adverse events (AEs) (RR: 0.96 with 95% CI [0.76, 1.20], P = .7) or AEs leading to withdrawal (RR: 1.54 with 95% CI [0.84, 2.82], P = .16). However, **insulin** icodec was associated with increased any AEs incidence (RR: 1.06 with 95% CI [1.01, 1.12], P = .02). CONCLUSION: **insulin** icodec was associated with decreased HbA1C, increased TIR, with similar hypoglycemic and serious AEs. However, it was also associated with increased **body weight** and the incidence of any AEs.

中文翻译

背景：胰岛素icodec是一种新型基础胰岛素类似物，每周皮下注射一次。本研究旨在评估胰岛素icodec与长效胰岛素（甘精胰岛素和德谷胰岛素）在2型糖尿病患者中的疗效和安全性。方法：我们进行了一项系统综述和荟萃分析，综合了截至2023年5月29日通过系统检索PubMed、Web of Science、SCOPUS和Cochrane数据库获得的随机对照试验（RCT）。使用RevMan V.5.4软件，以风险比（RR）合并二分类数据，以均值差（MD）合并连续数据，并计算95%置信区间（CI）。主要结局指标为糖化血红蛋白（HbA1C）变化。结果：共纳入7项RCT，总计3183例患者。胰岛素icodec与HbA1C显著降低相关（MD: -0.15, 95% CI [-0.24, -0.06], P = .002），且血糖在目标范围内时间（TIR）百分比增加（MD: 4.06, 95% CI [2.06, 6.06], P = .0001）。然而，胰岛素icodec与体重增加相关（MD: 0.57, 95% CI [0.45, 0.70], P = .00001）。此外，两组在严重不良事件（RR: 0.96, 95% CI [0.76, 1.20], P = .7）或导致退出研究的不良事件（RR: 1.54, 95% CI [0.84, 2.82], P = .16）方面无显著差异。但胰岛素icodec与任何不良事件发生率增加相关（RR: 1.06, 95% CI [1.01, 1.12], P = .02）。结论：胰岛素icodec可降低HbA1C、增加TIR，且低血糖和严重不良事件风险与对照组相似，但同时也与体重增加及任何不良事件发生率升高相关。

核心词汇

insulin /ˈɪnsjʊlɪn/ : 胰岛素	diabetes mellitus /ˌdaɪəˈbiːtiːz ˈmɛlɪtəs/ : 糖尿病
glycated hemoglobin /ˈglʌɪkeɪtɪd ˌhiːməˈɡloʊbɪn/ : 糖化血红蛋白	systematic review /ˌsɪstəˈmætɪk rɪˈvjuː/ : 系统评价
randomized controlled trial /ˈrændəmaɪzd kənˈtrɔʊld ˈtraɪəl/ : 随机对照试验	hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖
adverse event /ˈædvɜːrs ɪˈvɛnt/ : 不良事件	body weight /ˈbɒdi weɪt/ : 体重
confidence interval /ˈkɒnfɪdəns ˈɪntəvəl/ : 置信区间	risk ratio /rɪsk ˈreɪʃiʊ/ : 风险比
mean difference /miːn ˈdɪfərəns/ : 均数差	

Advances in Cell Replacement Therapies for Diabetes.

糖尿病细胞替代疗法的最新进展

期刊: Diabetes 日期: 2025

Abstract

islet cell replacement therapies have evolved as a viable treatment option for type 1 diabetes complicated by problematic **hypoglycemia** and glycemic lability. Refinements of islet manufacturing, islet **transplantation** procedures, peritransplant recipient management, and immunosuppressive protocols allowed most recipients to achieve favorable outcomes. Subsequent phase 3 trials of **transplantation** of deceased donor islets documented the effectiveness of transplanted islets in restoring near-normoglycemia, glycemic stability, and protection from severe **hypoglycemia**, with an acceptable safety profile for the enrolled high-risk population. Health authorities in several countries have approved deceased donor islet **transplantation** for treating patients with type 1 diabetes and recurrent severe **hypoglycemia**. These achievements amplified academic and industry efforts to generate **pluripotent stem cell**-derived beta-cells through directed differentiation for beta-cell replacement. Preliminary results of ongoing clinical trials suggest that the **transplantation** of stem cell-derived beta-cells can consistently restore **insulin independence** in immunosuppressed recipients with type 1 diabetes, thus signaling the profound progress made in generating an unlimited and a uniform supply of cells for transplant. Avoiding the risks of chronic **immunosuppression** represents the next frontier. Several strategies have entered or are approaching clinical investigation, including immune-isolating islets, engineering immune-privileged islet implantation sites, rendering islets immune evasive, and inducing **immune tolerance** in transplanted islets. Capitalizing on high-dimensional, **multiomic** technologies for deep profiling of **graft**-directed immunity and the fate of the **graft** will provide new insights that promise to translate into sustaining functional **graft** survival long-term. Leveraging these parallel progression paths will facilitate the wider clinical adoption of cell replacement therapies in diabetes care. ARTICLE HIGHLIGHTS: **transplantation** of deceased donor-derived primary human islets has restored near-normoglycemia and protection from severe **hypoglycemia** in immunosuppressed recipients with type 1 diabetes. **transplantation** of embryonic stem cell-derived beta-cells has restored **insulin independence** in immunosuppressed recipients with type 1 diabetes. Clinical trials are underway and planned to evaluate the safety and efficacy of **transplantation** of mature stem cell-derived beta-cells with transient, local, minimal, and/or no-maintenance **immunosuppression** in recipients with type 1 diabetes. The high-dimensional, **multiomic** monitoring of immunity to transplanted islets and of the fate of the islet **graft** will facilitate the identification of determinants of sustained islet **graft** function and of patients most likely to benefit from cell replacement therapies.

中文翻译

胰岛细胞替代疗法已成为治疗伴有问题性低血糖和血糖不稳定的1型糖尿病的可行选择。胰岛制备、胰岛移植流程、围移植期受者管理及免疫抑制方案的改进，使大多数受者获得了良好预后。后续的死亡供体胰岛移植三期临床试验证实，移植胰岛能有效恢复接近正常的血糖水平、稳定血糖并预防严重低血糖，且对纳入的高风险人群具有可接受的安全性。多个国家的卫生机构已批准使用死亡供体胰岛移植治疗伴有复发性严重低血糖的1型糖尿病患者。这些成就推动了学术界和工业界通过定向分化生成多能干细胞来源的β细胞以替代β细胞的努力。正在进行的临床试验初步结果表明，移植干细胞来源的β细胞可稳定恢复免疫抑制的1型糖尿病受者的胰岛素独立性，这标志着在生成无限且均一的移植细胞供应方面取得了重大进展。避免慢性免疫抑制的风险是下一个前沿领域。多项策略已进入或即将进入临床研究阶段，包括免疫隔离胰岛、构建免疫豁免的胰岛植入位点、使胰岛具备免疫逃逸能力，以及诱导移植胰岛的免疫耐受。利用高维度多组学技术深度分析针对移植物的免疫反应及移植物命运，将提供新的见解，有望转化为长期维持功能性移植物存活策略。利用这些并行的发展路径，将推动细胞替代疗法在糖尿病治疗中的更广泛应用。文章亮点：- 移植死亡供体来源的原代人胰岛，可使免疫抑制的1型糖尿病受者恢复接近正常的血糖水平并预防严重低血糖。- 移植胚胎干细胞来源的β细胞，可使免疫抑制的1型糖尿病受者恢复胰岛素独立性。- 正在进行的及计划中的临床试验，旨在评估在1型糖尿病受者中移植成熟干细胞来源的β细胞，并采用短暂、局部、最小化或无需维持免疫抑制方案的安全性和有效性。- 对移植胰岛的免疫反应及胰岛移植命运的高维度多组学监测，将有助于识别维持胰岛移植功能的决定因素，以及最可能从细胞替代疗法中获益的患者群体。

核心词汇

islet cell /'aɪlt sel/ : 胰岛细胞

transplantation /ˌtrænsplæn'teɪʃən/ : 移植

pluripotent stem cell /pluː'rɪpə'tənt stem sel/ : 多能干细胞

immune tolerance /ɪ'mju:n 'tɒlərəns/ : 免疫耐受

hypoglycemia /ˌhaɪpəʊglə'siːmiə/ : 低血糖

immunosuppression /ɪmjuːnə'sʌpreʃən/ : 免疫抑制

insulin independence /ɪnsəlɪn ɪndɪ'pendəns/ : 胰岛素独立

graft /græft/ : 移植

multiomic /ˌmʌltiˈoʊmɪk/ : 多组学

cell replacement therapy /sel ɪˈpleɪsmənt ˈθerəpi/ : 细胞替代疗法

云梦山-侵权必究

Clinical Presentation and Diagnostic Approach to Hypoglycemia in Adults Without Diabetes Mellitus.

非糖尿病成人低血糖的临床表现与诊断方法

期刊: Endocrine practice : official journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists

日期: 2023

Abstract

OBJECTIVE: To review the clinical presentation, causes, and diagnostic approach to spontaneous hypoglycemia in adults without diabetes mellitus. **METHODS:** A literature review was performed using the PubMed and Google Scholar databases. **RESULTS:** hypoglycemia is uncommon in people who are not on glucose-lowering medications. Under normal physiologic conditions, multiple neural and hormonal counterregulatory mechanisms prevent the development of abnormally low levels of plasma glucose. If spontaneous hypoglycemia is suspected, the Whipple triad should be used to confirm hypoglycemia before pursuing further diagnostic workup. The Whipple criteria include the following: (1) low levels of plasma glucose, (2) signs or symptoms that would be expected with low levels of plasma glucose, and (3) improvement in those signs or symptoms when the level of plasma glucose increases. Spontaneous hypoglycemia can be caused by conditions that cause endogenous hyperinsulinism, including insulinoma, postbariatric hypoglycemia, and noninsulinoma pancreatogenous hypoglycemia. Spontaneous hypoglycemia can also be seen with critical illness, hepatic or renal dysfunction, hormonal deficiency, non-diabetes-related medications, and non-islet cell tumors. The initial diagnostic approach should begin by obtaining a detailed history of the nature and timing of the patient's symptoms, medications, underlying comorbid conditions, and any acute illness. A laboratory evaluation should be conducted at the time of the spontaneous symptomatic episode. Supervised tests such as a 72-hour fast or mixed-meal test may be needed to recreate the situation under which the patient is likely to experience symptoms. **CONCLUSION:** We provide an overview of the physiology of counterregulatory response to hypoglycemia, its causes, and diagnostic approaches to spontaneous hypoglycemia in adults.

中文翻译

目的: 回顾无糖尿病的成人自发性低血糖的临床表现、病因及诊断方法。方法: 通过PubMed和Google Scholar数据库进行文献综述。结果: 低血糖在未服用降糖药物的人群中并不常见。在正常生理条件下, 多种神经和激素反调节机制可防止血浆葡萄糖水平异常降低。若怀疑自发性低血糖, 应在进一步诊断检查前使用Whipple三联征确认低血糖。Whipple标准包括: (1) 血浆葡萄糖水平低, (2) 出现与低血糖水平相符的体征或症状, (3) 当血浆葡萄糖水平升高时, 这些体征或症状得到改善。自发性低血糖可由导致内源性高胰岛素血症的疾病引起, 包括胰岛素瘤、减重术后低血糖及非胰岛素瘤性胰源性低血糖。此外, 危重症、肝肾功能障碍、激素缺乏、非糖尿病相关药物及非胰岛细胞肿瘤也可导致自发性低血糖。初始诊断方法应从详细询问患者症状的性质与发生时间、用药史、基础合并症及任何急性疾病开始。应在自发性症状发作时进行实验室评估。可能需要进行72小时禁食或混合餐试验等监督下的测试, 以重现患者可能出现症状的情境。结论: 我们概述了低血糖反调节反应的生理机制、病因及成人自发性低血糖的诊断方法。

核心词汇

hypoglycemia /ˌhaɪpəʊglaiˈsiːmiə/: 低血糖症

insulinoma /ˌɪnsəlɪˈnoʊmə/: 胰岛素瘤

counterregulatory /ˌkaʊntərˈrɛɡjələtɔːri/: 反向调节的

endogenous /ɛnˈdɒdʒənəs/: 内源性的

noninsulinoma pancreatogenous hypoglycemia /nɒnˈɪnsəlɪnoʊməˌpæŋkriˈeɪtədʒənəsˌhaɪpəʊglaiˈsiːmiə/: 非胰岛素瘤性胰源性低血糖

renal dysfunction /ˈriːnəl dɪsˈfʌŋkʃən/: 肾功能不全

islet cell tumor /ˈaɪlɪt sɛlˈtuːmə/: 胰岛细胞瘤

Whipple triad /ˈwɪpəlˈtraɪæd/: 惠普尔三联征

hyperinsulinism /ˌhaɪpərˈɪnsəlɪnɪzəm/: 高胰岛素血症

plasma glucose /ˈplæzməˈɡluːkəʊs/: 血浆葡萄糖

postbariatric hypoglycemia /pəʊstˌbɛrɪˈætrɪkˌhaɪpəʊglaiˈsiːmiə/: 减重术后低血糖

hepatic dysfunction /hɪˈpætrɪk dɪsˈfʌŋkʃən/: 肝功能不全

hormonal deficiency /ˈhɔːrˈmounəl dɪˈfɪʃənsi/: 激素缺乏

A Review on Diabetic Retinopathy.

糖尿病视网膜病变综述

期刊: Current diabetes reviews 日期: 2024

Abstract

diabetic retinopathy is a well-recognised microvascular complication of diabetes and is among the leading cause of blindness all over the world. Over the last decade, there have been advances in the diagnosis of diabetic retinopathy and diabetic macular edema. At the same time, newer therapies for the management of diabetic retinopathy have evolved. As a result of these advances, a decline in severe vision loss due to diabetes has been witnessed in some developing countries. However, there is a steady increase in the number of people affected with diabetes, and is expected to rise further in the coming years. Therefore, it is prudent to identify diabetic retinopathy, and timely intervention is needed to decrease the burden of severe vision loss. An effort has been made to review all the existing knowledge regarding diabetic retinopathy in this article and summarize the present treatment options for diabetic retinopathy.

中文翻译

糖尿病视网膜病变是糖尿病公认的微血管并发症，也是全球导致失明的主要原因之一。过去十年间，糖尿病视网膜病变及糖尿病黄斑水肿的诊断技术取得了进展。与此同时，针对糖尿病视网膜病变的新型治疗方法也不断涌现。得益于这些进步，部分发展中国家因糖尿病导致的严重视力丧失病例有所减少。然而，糖尿病患者人数仍在持续增长，预计未来几年将进一步上升。因此，及早识别糖尿病视网膜病变并进行及时干预，对于减轻严重视力丧失的负担至关重要。本文旨在系统回顾糖尿病视网膜病变的现有知识，并总结当前的治疗方案。

核心词汇

diabetic retinopathy /ˌdaɪəˈbetɪk ˌretɪˈnɒpəθi/ : 糖尿病视网膜病变	microvascular complication /ˌmaɪkrəʊˈvæskjələr ˌkʌmplɪˈkeɪʃən/ : 微血管并发症
diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病	blindness /ˈblaɪndnəs/ : 失明
diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断	diabetic macular edema /ˌdaɪəˈbetɪk ˈmækjʊlər ɪˈdiːmə/ : 糖尿病性黄斑水肿
therapy /ˈθerəpi/ : 治疗	vision loss /ˈvɪʒən lɔːs/ : 视力丧失
intervention /ˌɪntərˈvenʃən/ : 干预	retina /ˈretɪnə/ : 视网膜
edema /ɪˈdiːmə/ : 水肿	complication /ˌkʌmplɪˈkeɪʃən/ : 并发症

Inflammation, diabetic foot and related treatments.

炎症、糖尿病足及相关治疗。

期刊: Frontiers in endocrinology 日期: 2025

Abstract

diabetic foot, one of the most severe chronic complications of diabetes mellitus, arises from complex interactions among **neuropathy**, vascular **ischemia**, and inflammatory dysregulation. This review systematically explores the inflammatory regulatory network centered on NF-kappaB and the interleukin (IL) family in the **diabetic foot** microenvironment and examines their roles in promoting tissue damage. Under hyperglycemic conditions, the AGE-RAGE axis potently activates the NF-kappaB signaling pathway, leading to upregulation of pro-inflammatory factors and suppression of anti-inflammatory mediators, thereby forming a vicious cycle of "inflammation-oxidative stress-tissue damage." **Charcot foot**, a distinct subtype characterized by neurogenic **osteoarthropathy**, is strongly linked to abnormal IL-6/RANKL pathway activation and impaired anti-inflammatory neuropeptide signaling. We further assess the diagnostic utility of inflammatory biomarkers, including **procalcitonin** (PCT), high-sensitivity C-reactive protein (hs-CRP), and the neutrophil-to-lymphocyte ratio (NLR), and summarize current therapeutic challenges along with emerging targeted strategies, such as NF-kappaB inhibition via salicylates. This review underscores the importance of spatiotemporal heterogeneity and dynamic equilibrium within inflammatory networks for clinical stratification and proposes that integrating multi-omics data with artificial intelligence models holds promise for developing personalized interventions. This approach offers novel theoretical insights to overcome therapeutic bottlenecks in **diabetic foot** management.

中文翻译

糖尿病足是糖尿病最严重的慢性并发症之一，其发病机制涉及神经病变、血管缺血及炎症调控失调的复杂相互作用。本综述系统探讨了糖尿病足微环境中以NF-κB和白细胞介素（IL）家族为核心的炎症调控网络，并阐明其在促进组织损伤中的作用。在高血糖条件下，AGE-RAGE轴强烈激活NF-κB信号通路，导致促炎因子上调及抗炎介质抑制，形成"炎症-氧化应激-组织损伤"的恶性循环。夏科足作为以神经源性骨关节病为特征的独特亚型，与IL-6/RANKL通路异常激活及抗炎神经肽信号受损密切相关。我们进一步评估了降钙素原（PCT）、超敏C反应蛋白（hs-CRP）及中性粒细胞-淋巴细胞比值（NLR）等炎症生物标志物的诊断价值，并总结了当前治疗挑战及新兴靶向策略（如水杨酸盐抑制NF-κB）。本综述强调炎症网络的时空异质性与动态平衡对临床分层的重要性，并提出整合多组学数据与人工智能模型有望开发个性化干预方案，为突破糖尿病足治疗瓶颈提供全新理论视角。

核心词汇

diabetic foot /ˌdaɪəˈbɛtrɪk fʊt/ : 糖尿病足	neuropathy /ˌnjuːˈrɒpəθi/ : 神经病变
ischemia /ˈɪskiːmiə/ : 缺血	inflammation /ˌɪnfləˈmeɪʃən/ : 炎症
hyperglycemia /ˌhaɪpəˈɡliːsiːmiə/ : 高血糖	oxidative stress /ˈɒksɪdɪtɪv stres/ : 氧化应激
Charcot foot /ˈʃɑːrkou fʊt/ : 夏科足	osteoarthropathy /ˌɒstiouɑːrˈθrɒpəθi/ : 骨关节病
biomarker /ˈbaɪəʊˌmaːrkər/ : 生物标志物	procalcitonin /ˌprɒvʌkælˈsætənɪn/ : 降钙素原
salicylate /səˈlɪsɪleɪt/ : 水杨酸盐	

Current status and progress in research on dressing management for diabetic foot ulcer.

糖尿病足溃疡敷料管理的研究现状与进展

期刊: Frontiers in endocrinology 日期: 2023

Abstract

diabetic foot ulcer (DFU) is a major complication of diabetes and is associated with a high risk of lower limb **amputation** and mortality. During their lifetime, 19%-34% of patients with diabetes can develop DFU. It is estimated that 61% of DFU become infected and 15% of those with DFU require **amputation**. Furthermore, developing a DFU increases the risk of mortality by 50%-68% at 5 years, higher than some cancers. Current standard management of DFU includes surgical **debridement**, the use of topical dressings and wound decompression, vascular assessment, and glycemic control. Among these methods, local treatment with dressings builds a protective physical barrier, maintains a moist environment, and drains the **exudate** from DFU wounds. This review summarizes the development, **pathophysiology**, and healing mechanisms of DFU. The latest research progress and the main application of dressings in laboratory and clinical stage are also summarized. The dressings discussed in this review include traditional dressings (gauze, oil yarn, traditional Chinese medicine, and others), basic dressings (**hydrogel**, hydrocolloid, sponge, foam, film agents, and others), bacteriostatic dressings, composite dressings (**collagen**, nanomaterials, chitosan dressings, and others), bioactive dressings (scaffold dressings with stem cells, decellularized wound matrix, autologous platelet enrichment plasma, and others), and dressings that use modern technology (3D **bioprinting**, photothermal effects, bioelectric dressings, microneedle dressings, smart bandages, orthopedic prosthetics and **regenerative medicine**). The **dressing** management challenges and limitations are also summarized. The purpose of this review is to help readers understand the pathogenesis and healing mechanism of DFU, help physicians select dressings correctly, provide an updated overview of the potential of biomaterials and devices and their application in DFU management, and provide ideas for further exploration and development of dressings. Proper use of dressings can promote DFU healing, reduce the cost of treating DFU, and reduce patient pain.

中文翻译

糖尿病足溃疡 (DFU) 是糖尿病的主要并发症, 与下肢截肢和死亡的高风险相关。19%-34%的糖尿病患者一生中可能发生DFU。据估计, 61%的DFU会感染, 其中15%需要截肢。此外, 发生DFU会使5年死亡风险增加50%-68%, 高于某些癌症。目前DFU的标准治疗包括手术清创、局部敷料使用、伤口减压、血管评估和血糖控制。其中, 敷料局部治疗可构建保护性物理屏障、维持湿润环境并引流DFU伤口渗出液。本综述总结了DFU的发展、病理生理学和愈合机制, 同时概述了敷料在实验室和临床阶段的最新研究进展及主要应用。文中讨论的敷料包括传统敷料 (纱布、油纱、中药等)、基础敷料 (水凝胶、水胶体、海绵、泡沫、薄膜剂等)、抑菌敷料、复合敷料 (胶原蛋白、纳米材料、壳聚糖敷料等)、生物活性敷料 (含干细胞支架敷料、脱细胞伤口基质、自体富血小板血浆等) 以及应用现代技术的敷料 (3D生物打印、光热效应、生物电敷料、微针敷料、智能绷带、骨科假体与再生医学)。同时总结了敷料管理的挑战与局限性。本综述旨在帮助读者理解DFU的发病机制与愈合机制, 指导医生正确选择敷料, 提供生物材料与器械在DFU管理中的潜力及应用的更新概述, 并为敷料的进一步探索与开发提供思路。合理使用敷料可促进DFU愈合、降低治疗成本并减轻患者痛苦。

核心词汇

diabetic foot ulcer /ˌdaɪəˈbɛtɪk fʊt ˈʌlsər/ : 糖尿病足溃疡

debridement /dɪˈbriːdmənt/ : 清创术

exudate /ˈɛksjʊdeɪt/ : 渗出液

hydrogel /ˈhaɪdrədʒəl/ : 水凝胶

stem cell /stɛm sɛl/ : 干细胞

regenerative medicine /rɪˈdʒɛnəreɪtɪv ˈmɛdɪsɪn/ : 再生医学

amputation /ˌæmpjuˈteɪʃən/ : 截肢

dressing /ˈdresɪŋ/ : 敷料

pathophysiology /ˌpæθəʊˌfɪziˈɒlədʒi/ : 病理生理学

collagen /ˈkɒlədʒən/ : 胶原蛋白

bioprinting /ˈbaɪəʊˌprɪntɪŋ/ : 生物打印

biomaterial /ˈbaɪəʊməˌtɪriəl/ : 生物材料

Mechanisms of diabetic foot ulceration: A review.

糖尿病足溃疡的机制：综述。

期刊: Journal of diabetes 日期: 2023

Abstract

Diabetic foot ulcers (DFUs) are associated with complex pathogenic factors and are considered a serious **complication** of diabetes. The potential mechanisms underlying DFUs have been increasingly investigated. Previous studies have focused on the three aspects of diabetic **peripheral vascular disease**, **neuropathy**, and wound infections. With advances in technology, researchers have been gradually conducting studies using immune cells, endothelial cells, keratinocytes, and fibroblasts, as they are involved in **wound healing**. It has been reported that the upregulation or downregulation of molecular signaling pathways is essential for the healing of DFUs. With a recent increase in the awareness of **epigenetics**, its regulatory role in **wound healing** has become a much sought-after trend in the treatment of DFUs. This review focuses on four aspects involved in the **pathogenesis** of DFUs: physiological and pathological mechanisms, cellular mechanisms, **molecular signaling pathway** mechanisms, and **epigenetics**. Given the challenge in the treatment of DFUs, we are hopeful that our review will provide new ideas for peers.

中文翻译

糖尿病足溃疡（DFUs）的致病因素复杂，被视为糖尿病的严重并发症。近年来，关于DFUs潜在机制的研究日益深入。既往研究主要聚焦于糖尿病周围血管病变、神经病变及创面感染三个方面。随着技术进步，研究者逐步开展涉及免疫细胞、内皮细胞、角质形成细胞和成纤维细胞在创面愈合中作用的研究。已有报道指出，分子信号通路的上调或下调对DFUs的愈合至关重要。随着表观遗传学认知的深化，其在创面愈合中的调控作用已成为DFUs治疗领域备受关注的新趋势。本综述从DFUs发病机制涉及的四个维度展开：生理病理机制、细胞机制、分子信号通路机制及表观遗传学机制。鉴于DFUs治疗的挑战性，我们期待本综述能为同行提供新的研究思路。

核心词汇

diabetic foot ulcer /ˌdaɪəˈbetɪk fʊt ˈʌlsər/ : 糖尿病足溃疡	complication /ˌkɑːmplɪˈkeɪʃən/ : 并发症
peripheral vascular disease /pəˈrɪfərəl ˈvæskjələr dɪˈziːz/ : 外周血管疾病	neuropathy /nuːˈrɑːpəθi/ : 神经病变
wound infection /wuːnd ɪnˈfekʃən/ : 伤口感染	immune cell /ɪˈmjuːn sel/ : 免疫细胞
endothelial cell /ˌendəʊˈθiːliəl sel/ : 内皮细胞	keratinocyte /kəˈræʈɪnəsaɪt/ : 角质形成细胞
fibroblast /ˈfaɪbrəblæst/ : 成纤维细胞	wound healing /wuːnd ˈhiːlɪŋ/ : 伤口愈合
molecular signaling pathway /məˈlekjələr ˈsɪɡnəlɪŋ ˈpæθweɪ/ : 分子信号通路	epigenetics /ˌepɪdʒəˈnetɪks/ : 表观遗传学
pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制	

Developing a Protocol for Management of Euglycemic Diabetic Ketoacidosis.

制定正常血糖糖尿病酮症酸中毒管理方案

期刊: Current diabetes reports 日期: 2025

Abstract

PURPOSE OF REVIEW: **euglycemic diabetic ketoacidosis** (euDKA) has been described since the 1970s, however the incidence appears to be increasing in association with the increased use of sodium-glucose cotransporter 2 inhibitor (SGLT2i) medications. Traditional hospital-based DKA protocols in which an **insulin infusion** is adjusted based on glucose levels are not effective in euDKA due to the presence of **euglycemia** which limits the capacity for insulin administration. This review was completed to review the data on euDKA and introduce a **protocol** for targeted management of this condition. RECENT FINDINGS: Data comparing euDKA outcomes to traditional **hyperglycemia** DKA demonstrate longer hospital length of stay and mean time to **anion gap** closure in euDKA based on current DKA management standards. Furthermore, the increase in prescribing SGLT2i medications thereby increases the risk of euDKA. At present, there are no reported protocols specific for euDKA and it is not directly addressed in the most recent guidelines issued by **endocrinology** specialty societies. We created a **protocol** within our hospital **intensive care unit** to standardize treatment of euDKA using fixed **insulin infusion** and **titration** of **dextrose**-containing fluids. The **protocol** has been approved by our hospital regulatory committees and is currently being utilized in intensive care units. Future studies should review ongoing safety and efficacy of **protocol** use in various hospital settings.

中文翻译

综述目的：正常血糖性糖尿病酮症酸中毒（euDKA）自20世纪70年代起已有记载，但随着钠-葡萄糖协同转运蛋白2抑制剂（SGLT2i）药物使用的增加，其发病率似乎呈上升趋势。传统基于医院的DKA治疗方案（根据血糖水平调整胰岛素输注）对euDKA无效，因为正常血糖限制了胰岛素给药能力。本综述旨在总结euDKA相关数据，并引入针对该病症的靶向管理方案。最新发现：比较euDKA与传统高血糖DKA结局的数据显示，基于当前DKA管理标准，euDKA患者的住院时间更长，阴离子间隙闭合平均时间也更长。此外，SGLT2i药物处方的增加进一步提高了euDKA风险。目前尚无针对euDKA的特定治疗方案报告，且内分泌专业学会最新发布的指南也未直接涉及该问题。我们在医院重症监护室制定了一项方案，通过固定胰岛素输注和含糖液体滴定来标准化euDKA治疗。该方案已获医院监管委员会批准，目前正在重症监护室应用。未来研究应持续评估该方案在不同医院环境中的安全性和有效性。

核心词汇

euglycemic diabetic ketoacidosis /juːˈɡlaɪˈsiːmɪk ˌdaɪəˈbiːtɪk ˌkiːtəʊˈæsiːˈdɔʊzɪs/ : 正常血糖性糖尿病酮症酸中毒	diabetic ketoacidosis /ˌdaɪəˈbiːtɪk ˌkiːtəʊˈæsiːˈdɔʊzɪs/ : 糖尿病酮症酸中毒
euglycemia /juːˈɡlaɪˈsiːmiə/ : 正常血糖	hyperglycemia /ˌhaɪpərˈɡlaɪˈsiːmiə/ : 高血糖
insulin infusion /ˈɪnsəlɪn ɪnˈfjuːʒən/ : 胰岛素输注	anion gap /ˈænaɪən ɡæp/ : 阴离子间隙
dextrose /ˈdekstroʊs/ : 葡萄糖	intensive care unit /ɪnˈtensɪv keər ˈjuːnɪt/ : 重症监护病房
protocol /ˈprəʊtɒkɒl/ : 方案	endocrinology /ˌendəʊkrɪˈnɒlədʒi/ : 内分泌学
ketoacidosis /ˌkiːtəʊˈæsiːˈdɔʊzɪs/ : 酮症酸中毒	titration /taɪˈtreɪʃən/ : 滴定

Euglycemic diabetic ketoacidosis in the era of SGLT-2 inhibitors.

SGLT-2抑制剂时代的正常血糖性糖尿病酮症酸中毒。

期刊: BMJ open diabetes research & care 日期: 2023

Abstract

euglycemic diabetic ketoacidosis (EDKA) is an emerging complication of diabetes associated with an increasing use of sodium-glucose transporter type 2 (SGLT-2) inhibitor drugs. This review highlights the growing incidence of EDKA and its diagnostic challenges due to the absence of hallmark **hyperglycemia** seen in **diabetic ketoacidosis** (DKA). The paper presents a classification system for the severity of EDKA, categorizing it into mild, moderate, and severe based on serum pH and **bicarbonate** levels. Another classification system is proposed to define stages of EDKA based on **anion gap** and **ketones** at the time of **diagnosis** and during the treatment period. A treatment algorithm is proposed to guide clinicians in managing EDKA. This treatment algorithm includes monitoring **anion gap** and **ketones** to guide **insulin** and fluid management, and slower transition to **subcutaneous insulin** to prevent a **relapse**. Increased awareness of EDKA is essential for a timely **diagnosis** because an early **diagnosis** and treatment can improve clinical outcomes.

中文翻译

正常血糖性糖尿病酮症酸中毒（EDKA）是糖尿病的一种新兴并发症，与2型钠-葡萄糖转运体（SGLT-2）抑制剂药物的使用增加相关。本综述强调了EDKA发病率的上升及其诊断挑战，因为该病缺乏糖尿病酮症酸中毒（DKA）典型的高血糖特征。本文提出了一种基于血清pH值和碳酸氢盐水平的EDKA严重程度分类系统，将其分为轻度、中度和重度。另一分类系统则根据诊断时及治疗期间的阴离子间隙和酮体水平，定义了EDKA的不同阶段。此外，本文提出了一种治疗流程以指导临床医生管理EDKA，该流程包括监测阴离子间隙和酮体以指导胰岛素和液体管理，并建议更缓慢地过渡至皮下胰岛素以防止复发。提高对EDKA的认识对于及时诊断至关重要，因为早期诊断和治疗可改善临床结局。

核心词汇

diabetic ketoacidosis /ˌdaɪəˈbetɪk ˌkiːtəʊˈæsiˈdɔʊsis/ : 糖尿病酮症酸中毒	euglycemic diabetic ketoacidosis /juːˈɡlaɪˈsiːmɪk ˌdaɪəˈbetɪk ˌkiːtəʊˈæsiˈdɔʊsis/ : 正常血糖性糖尿病酮症酸中毒
hyperglycemia /ˌhaɪpəˈɡlaɪˈsiːmiə/ : 高血糖	ketones /ˈkiːtəʊnz/ : 酮体
anion gap /ˈænaɪən ɡæp/ : 阴离子间隙	bicarbonate /baɪˈkɑːrbənət/ : 碳酸氢盐
insulin /ˈɪnsəlɪn/ : 胰岛素	subcutaneous insulin /ˌsʌbkjuːˈteɪniəs ˈɪnsəlɪn/ : 皮下胰岛素
acidosis /ˌæsiˈdɔʊsis/ : 酸中毒	relapse /rɪˈlæps/ : 复发
diagnosis /ˌdaɪəˈɡnoʊsis/ : 诊断	

Diabetic ketoacidosis and hyperosmolar hyperglycemic state in patients with end-stage renal disease on dialysis: a review.

终末期肾病透析患者中的糖尿病酮症酸中毒和高渗性高血糖状态：综述。

期刊: Endocrine research 日期: 2026

Abstract

PURPOSE: This review aims to discuss the prevalence, risk factors, and management of **diabetic ketoacidosis** (DKA) and **hyperosmolar hyperglycemic state** (HHS) in patients with end-stage renal disease (ESRD) on **hemodialysis**. **METHODS:** A comprehensive literature review of retrospective studies, case reports, and clinical guidelines was conducted. **RESULTS:** DKA and HHS incidence is lower in ESRD patients due to delayed **insulin clearance** and regular **dialysis**. Management requires cautious fluid administration, reduced **insulin** infusion rates, conservative **potassium replacement**, and gradual correction of **hyperosmolarity**. Emergency **hemodialysis** may be necessary for severe **acidosis** or **hyperkalemia**. **CONCLUSION:** Treatment must be individualized. Emerging strategies, including extended **dialysis** sessions and closed-loop **insulin** delivery, require further validation.

中文翻译

目的：本综述旨在探讨终末期肾病（ESRD）血液透析患者中糖尿病酮症酸中毒（DKA）和高渗性高血糖状态（HHS）的患病率、危险因素及管理策略。方法：对回顾性研究、病例报告及临床指南进行系统性文献综述。结果：由于胰岛素清除延迟及规律透析，ESRD患者中DKA和HHS的发生率较低。管理需谨慎补液、降低胰岛素输注速率、保守补钾并逐步纠正高渗状态。严重酸中毒或高钾血症时可能需要紧急血液透析。结论：治疗需个体化。延长透析疗程及闭环胰岛素输注等新兴策略仍需进一步验证。

核心词汇

diabetic ketoacidosis /ˌdaɪəˈbetɪk ˌkiːtəʊˈæsiˈdɔʊsis/ : 糖尿病酮症酸中毒	hyperosmolar hyperglycemic state /ˌhaɪpəʊzˈmɒlələr ˌhaɪpəɡlaɪˈsiːmɪk steɪt/ : 高渗性高血糖状态
hemodialysis /ˌhiːmɒdaɪˈælsɪs/ : 血液透析	insulin /ˈɪnsəlɪn/ : 胰岛素
hyperkalemia /ˌhaɪpərkeˈliːmiə/ : 高钾血症	acidosis /ˌæsiˈdɔʊsis/ : 酸中毒
hyperosmolarity /ˌhaɪpəʊzˈmɒləriti/ : 高渗性	dialysis /daɪˈælsɪs/ : 透析
insulin clearance /ˈɪnsəlɪn ˈkliːrəns/ : 胰岛素清除率	potassium replacement /pəˈtæsiəm rɪˈpleɪsmənt/ : 钾补充

The role of automated insulin delivery technology in diabetes.

自动化胰岛素输送技术在糖尿病中的作用

期刊: Diabetologia 日期: 2024

Abstract

The role of **automated insulin delivery** systems in **diabetes** is expanding. Hybrid closed-loop systems are being used in routine clinical practice for treating people with type 1 **diabetes**. Encouragingly, real-world data reflects the performance and usability observed in clinical trials. We review the commercially available hybrid closed-loop systems, their distinctive features and the associated real-world data. We also consider emerging indications for closed-loop systems, including the treatment of type 2 **diabetes** where variability of day-to-day **insulin** requirements is high, and other challenging applications for this technology. We discuss issues around access and implementation of closed-loop technology, and consider the limitations of present closed-loop systems, as well as innovative approaches that are being evaluated to improve their performance.

中文翻译

自动化胰岛素输注系统在糖尿病治疗中的作用正在不断扩大。混合闭环系统已广泛应用于1型糖尿病的常规临床治疗。令人振奋的是，真实世界数据反映了临床试验中观察到的性能和可用性。我们回顾了市售的混合闭环系统、其独特特征及相关真实世界数据。同时探讨了闭环系统的新兴适应症，包括治疗每日胰岛素需求波动较大的2型糖尿病，以及该技术在其他具有挑战性场景中的应用。我们讨论了闭环技术的可及性与实施问题，分析了当前闭环系统的局限性，以及正在评估的旨在提升其性能的创新方法。

核心词汇

insulin /ˈɪnsəlɪn/ : 胰岛素	diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病
automated insulin delivery /ˈɔːtəmeɪtɪd ˈɪnsəlɪn dɪˈlɪvəri/ : 自动化胰岛素输注	clinical trial /ˈklɪnɪkəl ˈtraɪəl/ : 临床试验
insulin pump /ˈɪnsəlɪn pʌmp/ : 胰岛素泵	continuous glucose monitoring /kənˈtɪnjuəs ˈɡluːkəʊs ˈmɒnɪtərɪŋ/ : 持续葡萄糖监测
glycemic control /ˈɡlaɪˈsiːmɪk kənˈtrəʊl/ : 血糖控制	hypoglycemia /ˌhaɪpəʊˈɡlaɪˈsiːmiə/ : 低血糖

Is It Time to Move Beyond TIR to TITR? Real-World Data from Over 20,000 Users of Continuous Glucose Monitoring in Patients with Type 1 and Type 2 Diabetes.

是时候从TIR转向TITR了吗？来自超过20,000名1型和2型糖尿病患者持续血糖监测的真实世界数据。

期刊: Diabetes technology & therapeutics 日期: 2024

Abstract

The growing use of **continuous glucose monitoring** (CGM) has been supported by expert consensus and clinical guidelines on **glycemic management** in **diabetes** with **time in range** (TIR 70-180 mg/dL) representing a key CGM-derived glucose metric. **time in tight range** (TITR) has also been proposed for clinical use, spanning largely normal glucose levels of 70-140 mg/dL. However, keeping such narrow glucose ranges can be challenging, and understanding the factors modulating TITR can help achieve these tight glycemic targets. Our real-life study aimed to evaluate the relationship between **average glucose** (AG) and TIR/TITR in a large cohort (n = 22,006) of CGM users, divided into four groups: self-identified as having type 1 **diabetes** (T1D) treated with **insulin** using **multiple daily injections** (MDI) or pumps; type 2 **diabetes** (T2D) on MDI or **insulin** pumps; T2D on **basal insulin** only; and T2D not on **insulin** treatment. The T2D groups, regardless of treatment type, displayed the highest TIR and TITR values, associated with lowest **glycemic variability** measured as glucose **coefficient of variation** (CV; 23-30%). The T1D group showed the lowest TIR and TITR, associated with the highest CVs (36-38%). Overall, higher CV was associated with lower TIR and TITR for AG values below 180 and 140 mg/dL, respectively, with the reverse holding true for AG values above these thresholds. The discordance between AG and TIR/TITR was less pronounced in T2D compared with T1D, attributed to lower CV in the former group. It was also observed that TITR has advantages over TIR for assessing glycemia status and progress toward more stringent A1C, particularly when approaching normal glucose levels. The data detail how CV affects the AG relationship with TIR/TITR, which has implications for CGM interpretation. In many instances TITR, rather than TIR, may be preferable to employ once AG falls below 140 mg/dL and near-normal glucose levels are required clinically.

中文翻译

持续葡萄糖监测（CGM）的广泛应用已获得糖尿病血糖管理专家共识和临床指南的支持，其中目标范围内时间（TIR 70-180 mg/dL）是CGM衍生的关键血糖指标。严格目标范围内时间（TITR）也被提出用于临床实践，其范围主要覆盖正常血糖水平70-140 mg/dL。然而，维持如此狭窄的血糖范围具有挑战性，理解影响TITR的因素有助于实现这些严格的血糖目标。本真实世界研究旨在评估大规模CGM用户队列（n=22,006）中平均血糖（AG）与TIR/TITR的关系。研究对象分为四组：自我报告为1型糖尿病（T1D）且接受每日多次胰岛素注射（MDI）或胰岛素泵治疗者；2型糖尿病（T2D）接受MDI或胰岛素泵治疗者；仅使用基础胰岛素的T2D患者；以及未接受胰岛素治疗的T2D患者。无论采用何种治疗方式，T2D组的TIR和TITR值均最高，且血糖变异系数（CV；23-30%）最低。T1D组的TIR和TITR值最低，CV值最高（36-38%）。总体而言，当AG低于180 mg/dL和140 mg/dL时，较高的CV与较低的TIR和TITR相关；当AG高于这些阈值时则相反。与T1D相比，T2D组中AG与TIR/TITR的不一致性更小，这归因于前者的CV较低。研究还发现，在评估血糖状态及向更严格糖化血红蛋白（A1C）目标进展时，TITR比TIR更具优势，尤其在接近正常血糖水平时。这些数据详细说明了CV如何影响AG与TIR/TITR的关系，这对CGM解读具有重要意义。当AG低于140 mg/dL且临床需要接近正常血糖水平时，多数情况下优先使用TITR而非TIR更为合适。

核心词汇

continuous glucose monitoring /kənˈtɪnjuəs ˈɡluːkəʊs ˈmɒnɪtərɪŋ/ : 持续葡萄糖监测	diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病
glycemic management /ɡlaɪˈsiːmɪk ˈmænɪdʒmənt/ : 血糖管理	time in range /taɪm ɪn reɪndʒ/ : 目标范围内时间
time in tight range /taɪm ɪn taɪt reɪndʒ/ : 严格目标范围内时间	average glucose /ˈævərɪdʒ ˈɡluːkəʊs/ : 平均血糖
glycemic variability /ɡlaɪˈsiːmɪk ˌveəriəˈbɪləti/ : 血糖变异性	coefficient of variation /ˌkəʊɪˈfɪʃənt əv ˌveəriˈeɪʃən/ : 变异系数
insulin /ˈɪnsəlɪn/ : 胰岛素	multiple daily injections /ˈmʌltɪpl ˈdeɪli ɪnˈdʒɛkʃənz/ : 每日多次注射
insulin pump /ˈɪnsəlɪn pʌmp/ : 胰岛素泵	basal insulin /ˈbeɪsəl ˈɪnsəlɪn/ : 基础胰岛素

Once-weekly insulin efsitora alfa: Design and rationale for the QWINT phase 3 clinical development programme.

每周一次胰岛素efsitora alfa：QWINT 3期临床开发项目的设计与原理

期刊: Diabetes, obesity & metabolism 日期: 2024

Abstract

AIMS: **insulin** efsitora alfa (efsitora) is a once-weekly **basal insulin**. This review describes the study design and rationale of the efsitora phase 3 Once Weekly (QW) **insulin** Therapy (QWINT) clinical development programme, including the five trials, QWINT-1 through QWINT-5. MATERIALS AND METHODS: The five trials included **insulin**-naïve adults (QWINT-1 and -2) with type 2 **diabetes** (T2D), adults with T2D previously treated with **basal insulin** (QWINT-3 and -4), and QWINT-5 in adults with type 1 **diabetes**. All five trials were designed as multicentre, randomized, controlled, open-label, treat-to-target studies to investigate the **efficacy** and **safety** of efsitora versus active once-daily **basal insulin** comparators (**insulin** glargine U100 or **insulin** degludec U100). The primary objective of each trial is to compare the change in HbA1c from baseline to week 26 or 52 between efsitora and the active comparator. The key secondary objectives include change in **fasting glucose**, **insulin dose** and **continuous glucose monitoring** variables, and patient-reported outcome questionnaires. CONCLUSIONS: The QWINT development programme includes a racially and geographically diverse population to provide important information regarding the **efficacy** and **safety** of efsitora and its clinical management of people with **diabetes**.

中文翻译

目的：胰岛素依非托拉α（依非托拉）是一种每周一次的基础胰岛素。本综述描述了依非托拉3期每周一次胰岛素治疗（QWINT）临床开发项目的研究设计和原理，包括五项试验（QWINT-1至QWINT-5）。材料与方法：这五项试验包括未使用过胰岛素的2型糖尿病（T2D）成人患者（QWINT-1和-2）、既往接受基础胰岛素治疗的T2D成人患者（QWINT-3和-4），以及1型糖尿病成人患者（QWINT-5）。所有五项试验均设计为多中心、随机、对照、开放标签、达标治疗研究，旨在评估依非托拉与每日一次活性基础胰岛素对照药（甘精胰岛素U100或德谷胰岛素U100）的有效性和安全性。每项试验的主要目标是比较依非托拉与活性对照药从基线至第26周或第52周HbA1c的变化。关键次要目标包括空腹血糖变化、胰岛素剂量、持续血糖监测变量以及患者报告结局问卷。结论：QWINT开发项目纳入了种族和地理背景多样的人群，为依非托拉的有效性、安全性及其在糖尿病患者中的临床管理提供了重要信息。

核心词汇

insulin /ˈɪnsjʊlɪn/ : 胰岛素	diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病
basal insulin /ˈbeɪsəl ˈɪnsjʊlɪn/ : 基础胰岛素	fasting glucose /ˈfɑːstɪŋ ˈɡluːkəʊs/ : 空腹血糖
insulin dose /ˈɪnsjʊlɪn dəʊs/ : 胰岛素剂量	continuous glucose monitoring /kənˈtɪnjuəs ˈɡluːkəʊs ˈmɒnɪtərɪŋ/ : 持续葡萄糖监测
efficacy /ˈefɪkəsi/ : 疗效	safety /ˈseɪfti/ : 安全性

Safety and glycaemic control during exercise with the MiniMed 780G: A narrative review of evidence from lab to real world.

MiniMed 780G运动期间的安全性与血糖控制：从实验室到现实世界证据的叙述性综述

期刊: Diabetes, obesity & metabolism 日期: 2026

Abstract

Physical activity plays a critical role in reducing **cardiovascular risk** among individuals with type 1 diabetes (PwT1D); however, more than two-thirds of PwT1D remain insufficiently active, primarily due to the unpredictability of **glucose** fluctuations and the fear of **hypoglycaemia** during **exercise**. Automated **insulin** delivery systems, such as the MiniMed 780G (MM780G), offer potential solutions by automating **insulin** dosing, supporting stable **glucose** control, and allowing for a higher temporary **glucose** target. This narrative review summarizes the current evidence on the safety and efficacy of the MM780G system during **exercise** in PwT1D, with a focus on the **temporary target** (TT) feature and real-world usability. The MM780G consistently supports glycaemic safety during **exercise**, especially moderate-intensity aerobic activity, with time in range frequently meeting or exceeding consensus guidelines. Activation of the TT feature 60- to 120-min pre-**exercise**, particularly when paired with reduced meal boluses, significantly lowers the risk of **hypoglycaemia**. Real-world studies confirm its safety across varied activities, including prolonged **aerobic exercise** and contact sports. While evidence for resistance and high-intensity **exercise** remains limited, initial findings are favourable. User education is crucial to optimize use, particularly in competitive and prolonged settings. The MM780G is a safe and effective tool for glycaemic management during **exercise** in PwT1D. Best outcomes are achieved through strategic use of the TT feature and **meal bolus** reduction, individualized carbohydrate strategies and proper education. Applying easy-to-follow pre-exercising recommendations may encourage better engagement with regular exercising. Further research is needed on long-term adaptations to training and resistance-based **exercise** protocols.

中文翻译

体育锻炼在降低1型糖尿病患者心血管风险方面具有关键作用，但超过三分之二的患者仍缺乏足够运动，主要原因是血糖波动不可预测以及对运动期间低血糖的担忧。以美敦力MiniMed 780G (MM780G) 为代表的自动胰岛素输注系统，通过自动化胰岛素输注、维持血糖稳定并允许设置更高的临时血糖目标，提供了潜在解决方案。本综述总结了MM780G系统在1型糖尿病患者运动期间安全性与有效性的现有证据，重点关注临时目标功能及真实世界应用体验。MM780G在运动期间（尤其是中等强度有氧运动）持续保障血糖安全，血糖达标时间常达到或超过共识指南标准。运动前60-120分钟激活临时目标功能，特别是配合减少餐前胰岛素剂量时，可显著降低低血糖风险。真实世界研究证实其在多种运动场景（包括长时间有氧运动和接触性运动）中的安全性。尽管抗阻训练和高强度运动的证据仍有限，但初步结果令人鼓舞。用户教育对优化使用至关重要，尤其在竞技运动和长时间运动场景中。MM780G是1型糖尿病患者运动期间血糖管理的安全有效工具。通过策略性使用临时目标功能、减少餐前胰岛素剂量、个体化碳水化合物策略及充分教育，可获得最佳效果。采用易于遵循的运动前建议可能有助于促进患者坚持规律运动。未来需进一步研究长期训练适应及抗阻运动方案。

核心词汇

hypoglycaemia /ˌhaɪpəʊglaiˈsiːmiə/ : 低血糖	cardiovascular risk /ˌkɑːrdiəvˈvæskjələr rɪsk/ : 心血管风险
insulin /ˈɪnsjʊlɪn/ : 胰岛素	exercise /ˈeksəsaɪz/ : 运动
glucose /ˈgluːkəʊs/ : 葡萄糖	automated insulin delivery system /ˈɔːtəmeɪtɪd ˈɪnsjʊlɪn dɪˈlɪvəri ˈsɪstəm/ : 自动胰岛素输注系统
glycaemic control /ˈglaiˈsiːmɪk kənˈtrəʊl/ : 血糖控制	temporary target /ˈtempərəri ˈtɑːɡɪt/ : 临时目标
aerobic exercise /eəˈrəʊbɪk ˈeksəsaɪz/ : 有氧运动	hypoglycaemia risk /ˌhaɪpəʊglaiˈsiːmiə rɪsk/ : 低血糖风险
meal bolus /miːl ˈbəʊləs/ : 餐前大剂量	

Diabetes Technology in People with Type 2 Diabetes: Novel Indications.

2型糖尿病患者中的糖尿病技术：新适应症。

期刊: Current diabetes reports 日期: 2024

Abstract

PURPOSE OF REVIEW: **diabetes** technology has been continuously evolving. Current versions of continuous **glucose** monitors (CGM) use minimally invasive designs, monitor **glucose** values with high accuracy, and can be used to guide **insulin** dosing. Extensive evidence supports the use of **diabetes** technology for monitoring and **insulin** administration in people with type 1 **diabetes**. However, there is emerging evidence for people with type 2 **diabetes**. In this review, we present the different technological devices used to monitor **glucose** and deliver **insulin** and the evidence supporting their use in people with type 2 **diabetes**. RECENT FINDINGS: The use of CGMs in people with type 2 **diabetes** treated with **insulin** or non-**insulin** therapies has been associated with improvements in **glycemic control** and time spent in **hypoglycemia**. Smart **insulin** pens and smart connected devices are options to track compliance and guide **insulin** delivery in people who do not require **insulin pump** therapy. Mechanical patch pumps can be used to reduce the burden of multiple daily **insulin** injections. **automated insulin delivery** algorithms improve **glycemic control** without an increase in **hypoglycemia**. The use of technology in the management of type 2 **diabetes** generates glycemic data previously inaccessible, reduces barriers for **insulin** initiation, improves **glycemic control**, tracks adherence to therapy, and improves user satisfaction.

中文翻译

综述目的：糖尿病技术持续发展。当前版本的持续葡萄糖监测系统采用微创设计，能以高精度监测血糖值，并可用于指导胰岛素剂量调整。大量证据支持糖尿病技术在1型糖尿病患者监测与胰岛素给药中的应用，但针对2型糖尿病患者的证据正在逐步积累。本综述将介绍用于血糖监测与胰岛素给药的不同技术设备，以及支持其在2型糖尿病患者中应用的循证依据。最新发现：在接受胰岛素或非胰岛素治疗的2型糖尿病患者中使用持续葡萄糖监测系统，可改善血糖控制并减少低血糖时间。智能胰岛素笔与智能连接设备为无需胰岛素泵治疗的患者提供了依从性追踪与胰岛素给药指导方案。机械贴片泵可减轻每日多次胰岛素注射的负担。自动化胰岛素输注算法能在不增加低血糖风险的前提下改善血糖控制。糖尿病管理技术的应用可获取既往无法获得的血糖数据，降低胰岛素启动障碍，改善血糖控制，追踪治疗依从性，并提升用户满意度。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病	glucose /ˈɡluːkʊs/ : 葡萄糖
insulin /ˈɪnsəlɪn/ : 胰岛素	hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖
glycemic control /ɡlaɪˈsiːmɪk kənˈtrɒl/ : 血糖控制	continuous glucose monitor /kənˈtɪnjuəs ˈɡluːkʊs ˈmɒnɪtər/ : 持续葡萄糖监测仪
insulin pump /ˈɪnsəlɪn pʌmp/ : 胰岛素泵	insulin pen /ˈɪnsəlɪn pen/ : 胰岛素笔
patch pump /pætʃ pʌmp/ : 贴片泵	automated insulin delivery /ˈɔːtəmeɪtɪd ˈɪnsəlɪn dɪˈlɪvəri/ : 自动化胰岛素输注

Obesity, Metabolic Syndrome, and Osteoarthritis-An Updated Review.

肥胖、代谢综合征与骨关节炎——最新综述

期刊: Current obesity reports 日期: 2023

Abstract

PURPOSE OF REVIEW: **metabolic syndrome** (MetS), also called the 'deadly quartet' comprising **obesity**, **diabetes**, **dyslipidemia**, and **hypertension**, has been ascertained to have a causal role in the **pathogenesis** of **osteoarthritis** (OA). This review is aimed at discussing the current knowledge on the contribution of **metabolic syndrome** and its various components to OA **pathogenesis** and progression. **RECENT FINDINGS:** Lately, an increased association identified between the various components of **metabolic syndrome** (**obesity**, **diabetes**, **dyslipidemia**, and **hypertension**) with OA has led to the identification of the 'metabolic **phenotype**' of OA. These metabolic perturbations alongside low-grade systemic **inflammation** have been identified to inflict detrimental effects upon multiple tissues of the joint including **cartilage**, bone, and **synovium** leading to complete joint failure in OA. Recent epidemiological and clinical findings affirm that adipokines significantly contribute to **inflammation**, tissue degradation, and OA **pathogenesis** mediated through multiple signaling pathways. OA is no longer perceived as just a 'wear and tear' disease and the involvement of the metabolic components in OA **pathogenesis** adds up to the complexity of the disease. Given the global surge in **obesity** and its allied metabolic perturbations, this review aims to throw light on the current knowledge on the pathophysiology of MetS-associated OA and the need to address MetS in the context of metabolic OA management. Better regulation of the constituent factors of MetS could be profitable in preventing MetS-associated OA. The identification of key roles for several metabolic regulators in OA **pathogenesis** has also opened up newer avenues in the recognition and development of novel therapeutic agents.

中文翻译

综述目的：代谢综合征（MetS），又称“致命四重奏”，包括肥胖、糖尿病、血脂异常和高血压，已被证实对骨关节炎（OA）的发病机制具有因果作用。本综述旨在探讨当前关于代谢综合征及其各组分对OA发病和进展影响的认识。最新发现：近期，代谢综合征各组分（肥胖、糖尿病、血脂异常和高血压）与OA之间关联的增强，促使了OA“代谢表型”的识别。这些代谢紊乱伴随低度全身性炎症，已被发现对关节的多种组织（包括软骨、骨骼和滑膜）造成有害影响，最终导致OA中的关节完全衰竭。最新的流行病学和临床研究证实，脂肪因子通过多种信号通路显著促进炎症、组织降解及OA发病机制。OA不再被视为单纯的“磨损”性疾病，代谢组分在OA发病机制中的参与增加了疾病的复杂性。鉴于全球肥胖及其相关代谢紊乱的激增，本综述旨在阐明当前关于MetS相关OA病理生理学的认识，以及在代谢性OA管理中应对MetS的必要性。更好地调控MetS的组成因素可能有助于预防MetS相关OA。此外，多种代谢调节因子在OA发病机制中关键作用的发现，也为识别和开发新型治疗药物开辟了新途径。

核心词汇

metabolic syndrome /ˌmetəˈbɒlɪk ˈsɪndroum/ : 代谢综合征

diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病

hypertension /ˌhaɪpərˈtɛnʃən/ : 高血压

inflammation /ˌɪnfləˈmeɪʃən/ : 炎症

synovium /sɪˈnɒvɪəm/ : 滑膜

pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制

obesity /ouˈbiːsɪti/ : 肥胖

dyslipidemia /dɪsˌlɪpɪˈdiːmiə/ : 血脂异常

osteoarthritis /ˌɒstiəʊɑːrˈθraɪtɪs/ : 骨关节炎

cartilage /ˈkɑːrtɪlɪdʒ/ : 软骨

adipokine /ˈædɪpoukəɪn/ : 脂肪因子

phenotype /ˈfiːnətaɪp/ : 表型

Obesity Paradox: Fact or Fiction?

肥胖悖论：事实还是虚构？

期刊: Current obesity reports 日期: 2023

Abstract

PURPOSE OF REVIEW: **obesity** is related to several comorbidities such as type 2 diabetes mellitus, **cardiovascular disease**, **heart failure**, and various types of cancers. While the detrimental effect of **obesity** in both **mortality** and morbidity has been well established, the concept of the **obesity** paradox in specific chronic diseases remains a topic of continuous interest. In the present review, we examine the controversial issues around the **obesity** paradox in certain conditions such as **cardiovascular disease**, several types of cancer and **chronic obstructive pulmonary disease**, and the factors that may confound the relation between **obesity** and **mortality**. RECENT FINDINGS: We refer to the **obesity** paradox when particular chronic diseases exhibit an interesting "paradoxical" protective association between the **body mass index** (BMI) and clinical outcomes. This association, however, may be driven by multiple factors among which the limitations of the BMI itself; the unintended weight loss precipitated by chronic illness; the various phenotypes of **obesity**, i.e., **sarcopenic obesity** or the athlete's **obesity phenotype**; and the **cardiorespiratory fitness** levels of the included patients. Recent evidence highlighted that previous cardioprotective medications, **obesity** duration, and smoking status seem to play a role in the **obesity** paradox. The **obesity** paradox has been described in a plethora of chronic diseases. It cannot be emphasized enough that the incomplete information received from a single BMI measurement may interfere with outcomes of studies arguing in favor of the **obesity** paradox. Thus, the development of carefully designed studies, unhampered by confounding factors, is of great importance.

中文翻译

综述目的：肥胖与多种合并症相关，如2型糖尿病、心血管疾病、心力衰竭及各类癌症。尽管肥胖对死亡率和发病率的负面影响已得到充分证实，但特定慢性疾病中“肥胖悖论”的概念仍是持续关注的焦点。本综述探讨了心血管疾病、多种癌症及慢性阻塞性肺疾病等特定疾病中围绕肥胖悖论的争议性问题，以及可能混淆肥胖与死亡率关系的因素。最新发现：当特定慢性疾病在体重指数（BMI）与临床结局之间呈现有趣的“矛盾性”保护关联时，我们称之为肥胖悖论。然而，这种关联可能受多重因素驱动，包括BMI本身的局限性、慢性疾病引发的非自愿性体重下降、肥胖的不同表型（如肌少性肥胖或运动员型肥胖表型），以及纳入患者的心肺适应能力水平。最新证据强调，既往心脏保护药物使用、肥胖持续时间及吸烟状态似乎也在肥胖悖论中发挥作用。肥胖悖论已在多种慢性疾病中被描述。必须反复强调，单次BMI测量所获的不完整信息可能干扰支持肥胖悖论的研究结论。因此，开展不受混杂因素影响的精心设计研究至关重要。

核心词汇

obesity /ou'bi:seɪ/ : 肥胖	comorbidity /,koumɔ:r'bi:deɪ/ : 合并症
cardiovascular disease /,ka:rdiəv'væskjələr dɪ'zi:z/ : 心血管疾病	heart failure /ha:rt 'feɪljər/ : 心力衰竭
chronic obstructive pulmonary disease /'krɒnɪk əb'strʌktɪv 'pʌlmə,nəri dɪ'zi:z/ : 慢性阻塞性肺疾病	mortality /mɔ:r'tæləti/ : 死亡率
body mass index /'bɒdi mæs 'ɪndeks/ : 体重指数	phenotype /'fi:nətaɪp/ : 表型
sarcopenic obesity /,sa:rke'pi:nɪk ou'bi:seɪ/ : 肌少性肥胖	cardiorespiratory fitness /,ka:rdiəv'respəreɪtɔ:ri 'fɪtnəs/ : 心肺适应能力
confounding factor /kən'faʊndɪŋ 'fæktər/ : 混杂因素	

Resmetirom: The First Disease-Specific Treatment for MASH.

瑞司美替罗：首个针对MASH的疾病特异性治疗药物

期刊: International journal of endocrinology 日期: 2025

Abstract

In 2023, the medical terminology for **Nonalcoholic Fatty Liver Disease** (NAFLD) and NA **Steatohepatitis** (NASH) was updated to **Metabolic Dysfunction**-Associated SteatoticLiver Disease (MASLD) and MA **Steatohepatitis** (MASH). This review highlights the critical epidemiological, pathophysiological, and therapeutic aspects of MASH, focusing on the novel treatment option, resmetirom. Resmetirom, a **Thyroid Hormone Receptor**-beta (THR-beta) agonist, specifically targets liver function to simulate localized **Hyperthyroidism**, effectively reducing **Lipid Accumulation** and **Liver Fibrosis** without the systemic effects commonly associated with thyroid hormone therapy. Clinical trials, particularly the pivotal MAESTRO-NASH trial, have demonstrated significant improvements in liver health metrics, achieving primary endpoints by resolving MASH without worsening fibrosis and showing a favorable **Safety Profile**. This paper reviews the mechanism of action, efficacy, and safety of resmetirom, providing insight into its potential to change the therapeutic landscape for patients suffering from MASH.

中文翻译

2023年，非酒精性脂肪性肝病（NAFLD）和非酒精性脂肪性肝炎（NASH）的医学术语更新为代谢功能障碍相关脂肪性肝病（MASLD）和代谢功能障碍相关脂肪性肝炎（MASH）。本综述重点阐述了MASH的关键流行病学、病理生理学及治疗学要点，聚焦于新型治疗药物瑞美替罗。瑞美替罗作为一种甲状腺激素受体β（THR-beta）激动剂，通过特异性靶向肝脏功能模拟局部甲状腺功能亢进状态，在有效减少脂质蓄积和肝纤维化的同时，避免了甲状腺激素治疗常见的全身性副作用。关键性MAESTRO-NASH临床试验证实，该药物在改善肝脏健康指标方面取得显著成效，在未加重纤维化的情况下实现MASH消退的主要终点，并展现出良好的安全性特征。本文系统回顾了瑞美替罗的作用机制、疗效及安全性数据，揭示了其改变MASH患者治疗格局的潜在价值。

核心词汇

Nonalcoholic Fatty Liver Disease /ˌnɒn.ælkəˈhɒlɪk ˈfæti ˈlɪvər dɪˈziːz/ : 非酒精性脂肪性肝病	Steatohepatitis /ˌstiːəʊ.ˈheɪpəˈtaɪtɪs/ : 脂肪性肝炎
Metabolic Dysfunction /ˌmetəˈbɒlɪk dɪsˈfʌŋkʃən/ : 代谢功能障碍	Liver Fibrosis /ˈlɪvər faɪˈbrəʊsɪs/ : 肝纤维化
Thyroid Hormone Receptor /ˈθaɪrɔɪd ˈhɔːmɒn rɪˈseptər/ : 甲状腺激素受体	Lipid Accumulation /ˈlɪpɪd əˌkjuːmjʊˈleɪʃən/ : 脂质积累
Clinical Trial /ˈklɪnɪkəl ˈtraɪəl/ : 临床试验	Primary Endpoint /ˈpraɪməri ˈendpɔɪnt/ : 主要终点
Safety Profile /ˈseɪfti ˈprəʊfaɪl/ : 安全性特征	Hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进
Epidemiology /ˌɛpɪˌdiːmiˈɒlədʒi/ : 流行病学	Pathophysiology /ˌpæθəʊˌfɪziˈɒlədʒi/ : 病理生理学

Metabolic dysfunction-associated steatotic liver disease and its link to cancer.

代谢功能障碍相关脂肪性肝病及其与癌症的关联。

期刊: Metabolism: clinical and experimental 日期: 2024

Abstract

Metabolic-dysfunction associated **steatotic liver disease** (MASLD), formerly known as nonalcoholic fatty liver disease (NAFLD), is a growing global health concern with significant implications for **oncogenesis**. This review synthesizes current evidence on the association between MASLD and cancer risk, highlighting its role as a risk factor for both **intrahepatic** and **extrahepatic** malignancies. MASLD is increasingly recognized as a major cause of **hepatocellular carcinoma** (HCC), with its incidence rising in parallel with the prevalence of **metabolic dysfunction**. Furthermore, MASLD is associated with an elevated risk of various gastrointestinal cancers, including colorectal, esophageal, stomach, and pancreatic cancers. Beyond the digestive tract, evidence suggests that MASLD may also contribute to an increased risk of other cancers such as breast, prostate, thyroid, gynecological, renal and lung cancers. Understanding the mechanisms underlying these associations and the impact of MASLD on cancer risk is crucial for developing targeted **screening** and prevention strategies.

中文翻译

代谢功能障碍相关脂肪性肝病（MASLD），原名非酒精性脂肪性肝病（NAFLD），是一个日益严重的全球健康问题，对肿瘤发生具有重要影响。本综述综合了当前关于MASLD与癌症风险关联的证据，强调其作为肝内和肝外恶性肿瘤危险因素的作用。MASLD日益被认为是肝细胞癌（HCC）的主要原因，其发病率随代谢功能障碍的流行而上升。此外，MASLD与多种胃肠道癌症风险升高相关，包括结直肠癌、食管癌、胃癌和胰腺癌。在消化道以外，证据表明MASLD可能还增加其他癌症的风险，如乳腺癌、前列腺癌、甲状腺癌、妇科癌症、肾癌和肺癌。理解这些关联背后的机制以及MASLD对癌症风险的影响，对于制定针对性的筛查和预防策略至关重要。

核心词汇

steatotic liver disease /sti:ə'tɒtɪk 'lɪvər dɪ'zi:z/ : 脂肪性肝病	metabolic dysfunction /ˌmetə'bɒlɪk dɪs'fʌŋkʃən/ : 代谢功能障碍
hepatocellular carcinoma /ˌhepə'tɒʊ'seljʊlə 'kɑ:rsɪ'nɒmə/ : 肝细胞癌	malignancy /mə'ɪgnənsi/ : 恶性肿瘤
intrahepatic /ˌɪntərə'pæɪtɪk/ : 肝内的	extrahepatic /ˌɛkstrə'pæɪtɪk/ : 肝外的
gastrointestinal cancer /ˌgæstrəʊɪn'testɪnəl 'kænsər/ : 胃肠道癌	colorectal cancer /ˌkɒlə'rektəl 'kænsər/ : 结直肠癌
esophageal cancer /ɪ'səfe'dʒi:əl 'kænsər/ : 食管癌	pancreatic cancer /ˌpæŋkri'æɪtɪk 'kænsər/ : 胰腺癌
oncogenesis /ˌɒŋkəʊ'dʒenɪsɪs/ : 肿瘤发生	screening /'skri:nɪŋ/ : 筛查

Osteokines in Nonalcoholic Fatty Liver Disease.

非酒精性脂肪性肝病中的骨因子。

期刊: Current obesity reports 日期: 2024

Abstract

PURPOSE OF REVIEW: To critically summarize evidence on the potential role of osteokines in the pathogenesis and progression of nonalcoholic fatty liver disease (NAFLD). RECENT FINDINGS: There are emerging data supporting that certain osteokines, which are specific bone-derived proteins, may beneficially or adversely affect hepatic metabolism, and their alterations in the setting of osteoporosis or other bone metabolic diseases may possibly contribute to the development and progression of NAFLD. There is evidence showing a potential bidirectional association between NAFLD and bone metabolism, which may imply the existence of a liver-bone axis. In this regard, osteocalcin, osteoprotegerin, bone morphogenic protein 4 (BMP4) and BMP6 appear to have a positive impact on the liver, thus possibly alleviating NAFLD, whereas osteopontin, receptor activator of nuclear factor kappa Beta ligand (RANKL), sclerostin, periostin, BMP8B, and fibroblast growth factor 23 (FGF23) appear to have a negative impact on the liver, thus possibly exacerbating NAFLD. The potential implication of osteokines in NAFLD warrants further animal and clinical research in the field that may possibly result in novel therapeutic targets for NAFLD in the future.

中文翻译

综述目的：批判性总结骨因子在非酒精性脂肪性肝病（NAFLD）发病机制及进展中潜在作用的证据。最新发现：新兴数据表明，某些骨因子（即骨特异性蛋白）可能对肝脏代谢产生有益或不利影响，在骨质疏松或其他骨代谢疾病背景下，这些骨因子的改变可能促进NAFLD的发生发展。现有证据显示NAFLD与骨代谢之间存在潜在双向关联，提示可能存在肝-骨轴。其中，骨钙素、骨保护素、骨形态发生蛋白4（BMP4）和BMP6对肝脏具有积极影响，可能缓解NAFLD；而骨桥蛋白、核因子κB受体活化因子配体（RANKL）、硬骨素、骨膜蛋白、BMP8B及成纤维细胞生长因子23（FGF23）对肝脏具有负面影响，可能加重NAFLD。骨因子在NAFLD中的潜在作用需进一步开展动物及临床研究，未来或可为NAFLD提供新型治疗靶点。

核心词汇

osteokine /ˈɒstriːkain/ : 骨因子	nonalcoholic fatty liver disease /ˌnɒnælkeˈhɒlɪk ˈfæti ˈliːvə drɪˈziːz/ : 非酒精性脂肪性肝病
pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制	osteoporosis /ˌɒstriːpəˈrəʊsɪs/ : 骨质疏松症
hepatic metabolism /hɪˈpætrɪk məˈtæbəlɪzəm/ : 肝脏代谢	osteocalcin /ˌɒstriːˈkælsɪn/ : 骨钙素
osteoprotegerin /ˌɒstriːprəˈtɛdʒərɪn/ : 骨保护素	osteopontin /ˌɒstriːˈpɒntɪn/ : 骨桥蛋白
sclerostin /ˈskliːrəstɪn/ : 硬骨素	periostin /ˌperiˈɒstɪn/ : 骨膜蛋白
therapeutic target /ˌθɛrəˈpiːtɪk ˈtɑːrgɪt/ : 治疗靶点	

MAFLD: a multisystem disease.

MAFLD：一种多系统疾病

期刊: Therapeutic advances in endocrinology and metabolism 日期: 2023

Abstract

nonalcoholic fatty liver disease (NAFLD), affecting about 25% of general population and more than 50% of dysmetabolic patients, is an emerging cause of **chronic liver disease** and its complications. Recently, an international consensus of experts proposed to rename this disease as '**metabolic dysfunction**-Associated Fatty Liver Disease' (MAFLD) to focus on the bidirectional interplay between fatty liver and metabolic alterations and to stress the need of assessing fatty liver independently from alcohol consumption and other coexisting causes of liver disease. The peculiarity of NAFLD/MAFLD lies in the presence of a higher risk of not only - as expected - liver-related events but also of extrahepatic events, mostly **cardiovascular** and cancers. Available evidence suggests that these associations are not only the expression of sharing the same risk factors but shed light about the ability of NAFLD/MAFLD and particularly of its progressive form - nonalcoholic/**metabolic dysfunction**-associated **steatohepatitis** - to act as an independent **risk factor** via promotion of atherogenic **dyslipidemia** and a proinflammatory, profibrogenic, and procoagulant **systemic** environment. The present review summarizes available epidemiological and clinical evidence supporting the concept of NAFLD/MAFLD as a multisystemic disease, and highlights potential explanatory mechanisms underlying the association between NAFLD/MAFLD and extrahepatic disorders.

中文翻译

非酒精性脂肪性肝病（NAFLD）影响约25%的普通人群及超过50%的代谢紊乱患者，已成为慢性肝病及其并发症的新兴病因。近期国际专家共识建议将该病更名为“代谢功能障碍相关脂肪性肝病”（MAFLD），以强调脂肪肝与代谢异常的双向相互作用，并突出需独立于饮酒及其他共存肝病病因评估脂肪肝的必要性。NAFLD/MAFLD的特殊性在于其不仅增加预期的肝脏相关事件风险，更显著提升肝外事件（主要为心血管疾病和恶性肿瘤）的发生风险。现有证据表明，这种关联不仅源于共同危险因素的表达，更揭示了NAFLD/MAFLD（特别是其进展型——非酒精性/代谢功能障碍相关脂肪性肝炎）通过促进致动脉粥样硬化性血脂异常，营造促炎、促纤维化及促凝血系统性环境，从而作为独立危险因素的作用机制。本综述汇总支持NAFLD/MAFLD作为多系统疾病概念的流行病学与临床证据，并重点阐释NAFLD/MAFLD与肝外疾病关联的潜在机制。

核心词汇

nonalcoholic fatty liver disease /ˌnɒn.ælkəˈhɒlɪk ˈfæti ˈlɪvər dɪˈziːz/ : 非酒精性脂肪性肝病	steatohepatitis /ˌstiːətəʊˈhepəˈtaɪtɪs/ : 脂肪性肝炎
chronic liver disease /ˈkrɒnɪk ˈlɪvər dɪˈziːz/ : 慢性肝病	metabolic dysfunction /ˌmetəˈbɒlɪk dɪsˈfʌŋkʃən/ : 代谢功能障碍
dyslipidemia /dɪsˌlɪpiˈdiːmiə/ : 血脂异常	inflammation /ˌɪnfləˈmeɪʃən/ : 炎症
fibrosis /ˈfaɪˈbrəʊsɪs/ : 纤维化	cardiovascular /ˌkɑːrdiəvˈvæskjələr/ : 心血管的
cancer /ˈkænsər/ : 癌症	biopsy /ˈbaɪɒpsi/ : 活检
risk factor /rɪsk ˈfæktər/ : 危险因素	systemic /sɪˈstemɪk/ : 全身性的

Obesity and Dyslipidemia: A Review of Current Evidence.

肥胖与血脂异常：当前证据综述。

期刊: Current obesity reports 日期: 2023

Abstract

PURPOSE OF REVIEW: **obesity** is accompanied by atherogenic **dyslipidemia**, a specific lipid disorder characterized by both quantitative and qualitative changes of plasma lipoproteins. The main alterations in the lipid profile include **hypertriglyceridemia**, reduced high-density **lipoprotein** (HDL) **cholesterol** level, and elevated small dense low-density **lipoprotein** (LDL) particles. Epidemiological data show that **obesity** is more common in women and is a frequent risk factor for reproductive disorders, metabolic complications in **pregnancy**, and **cardiometabolic** disease later in life. The aim of this narrative review is to discuss recent advances in the research of **dyslipidemia** in **obesity**, with an emphasis on female-specific disorders and **cardiometabolic** risk. RECENT FINDINGS: The focus of current research on **dyslipidemia** in **obesity** is moving toward structurally and functionally modified plasma lipoproteins. Special attention is paid to the pro-atherogenic role of triglyceride-rich lipoproteins and their remnants. Introduction of advanced analytical techniques enabled identification of novel lipid biomarkers with potential clinical applications. In particular, proteomic and lipidomic studies have provided significant progress in the comprehensive research of HDL's alterations in **obesity**. **obesity**-related **dyslipidemia** is a widespread metabolic disturbance in **polycystic ovary syndrome** patients and high-risk pregnancies, but is seldom evaluated with respect to its impact on future **cardiometabolic** health. **obesity** and associated **cardiometabolic** diseases require a more depth insight into the quality of **lipoprotein** particles. Further application of omics-based techniques would enable a more comprehensive evaluation of **dyslipidemia** in order to reduce an excessive **cardiovascular risk** attributable to increased body weight. However, more studies on **obesity**-related female reproductive disorders are needed for this approach to be adopted in daily clinical practice.

中文翻译

综述目的：肥胖伴随致动脉粥样硬化性血脂异常，这是一种以血浆脂蛋白定量和定性改变为特征的特定脂质紊乱。血脂谱的主要改变包括高甘油三酯血症、高密度脂蛋白胆固醇水平降低以及小而密低密度脂蛋白颗粒升高。流行病学数据显示，肥胖在女性中更为常见，且是生殖障碍、妊娠期代谢并发症及晚年心血管代谢疾病的常见危险因素。本叙述性综述旨在探讨肥胖相关血脂异常研究的最新进展，重点关注女性特异性疾病及心血管代谢风险。近期发现：当前肥胖相关血脂异常的研究焦点正转向结构及功能修饰的血浆脂蛋白，尤其关注富含甘油三酯的脂蛋白及其残粒的促动脉粥样硬化作用。先进分析技术的引入使具有潜在临床应用价值的新型脂质生物标志物得以识别。特别是蛋白质组学和脂质组学研究在全面探索肥胖中高密度脂蛋白改变方面取得了显著进展。肥胖相关血脂异常是多囊卵巢综合征患者及高危妊娠中普遍存在的代谢紊乱，但其对未来心血管代谢健康的影响却鲜少被评估。肥胖及其相关心血管代谢疾病需更深入地洞察脂蛋白颗粒的质量。进一步应用组学技术将有助于更全面地评估血脂异常，以降低因体重增加导致的过度心血管风险。然而，要使该方法应用于日常临床实践，仍需开展更多关于肥胖相关女性生殖障碍的研究。

核心词汇

obesity /ou'bi:seɪti/ : 肥胖	dyslipidemia /dɪs,lɪpɪ'di:mɪə/ : 血脂异常
hypertriglyceridemia /,haɪpə'traɪglɪsərɪ'di:mɪə/ : 高甘油三酯血症	lipoprotein /,lɪpə'prəʊti:n/ : 脂蛋白
cholesterol /kə'lestəro:l/ : 胆固醇	cardiometabolic /,kɑ:rdɪəu'metə'bɒlɪk/ : 心脏代谢的
polycystic ovary syndrome /,pɒli'sɪstɪk 'oʊvəri 'sɪndroum/ : 多囊卵巢综合征	pregnancy /'prɛɡnənsɪ/ : 妊娠
proteomics /prəʊ'ti:əʊmɪks/ : 蛋白质组学	lipidomics /lɪ'pɪdəʊmɪks/ : 脂质组学
biomarker /'baɪəʊ'mɑ:rkər/ : 生物标志物	cardiovascular risk /,kɑ:rdɪəu'væskjələr rɪsk/ : 心血管风险

Semaglutide for the treatment of overweight and obesity: A review.

司美格鲁肽治疗超重和肥胖的综述。

期刊: Diabetes, obesity & metabolism 日期: 2023

Abstract

obesity is a chronic, relapsing disease associated with multiple complications and a substantial **morbidity**, **mortality** and health care burden. Pharmacological treatments for **obesity** provide a valuable adjunct to **lifestyle intervention**, which often achieves only limited **weight loss** that is difficult to maintain. The Semaglutide Treatment Effect in People with **obesity** (STEP) **clinical trial** programme is evaluating once-weekly subcutaneous semaglutide 2.4 mg (a glucagon-like peptide-1 analogue) in people with overweight or **obesity**. Across STEP 1, 3, 4 and 8, semaglutide 2.4 mg was associated with mean weight losses of 14.9%-17.4% in individuals with overweight or **obesity** without type 2 diabetes from baseline to week 68; 69%-79% of participants achieved $\geq 10\%$ **weight loss** with semaglutide 2.4 mg (vs. 12%-27% with **placebo**) and 51%-64% achieved $\geq 15\%$ **weight loss** (vs. 5%-13% with **placebo**). In STEP 5, mean **weight loss** was -15.2% with semaglutide 2.4 mg versus -2.6% with **placebo** from baseline to week 104. In STEP 2 (individuals with overweight or **obesity**, and type 2 diabetes), mean **weight loss** was -9.6% with semaglutide 2.4 mg versus -3.4% with **placebo** from baseline to week 68. Improvements in cardiometabolic risk factors, including high blood pressure, atherogenic lipids and benefits on physical function and quality of life were seen with semaglutide 2.4 mg. The safety profile of semaglutide 2.4 mg was consistent across trials, primarily **gastrointestinal** adverse events. The magnitude of **weight loss** reported in the STEP trials offers the potential for clinically relevant improvement for individuals with **obesity**-related diseases.

中文翻译

肥胖是一种慢性、复发性疾病，常伴随多种并发症，并带来显著的发病率、死亡率及医疗负担。肥胖的药物治疗为生活方式干预提供了宝贵的辅助手段，而单纯依靠生活方式干预往往只能实现有限的体重减轻，且难以长期维持。司美格鲁肽治疗肥胖效果（STEP）临床试验项目正在评估每周一次皮下注射2.4 mg司美格鲁肽（一种胰高血糖素样肽-1类似物）在超重或肥胖人群中的疗效。在STEP 1、3、4和8试验中，对于无2型糖尿病的超重或肥胖个体，从基线至第68周，2.4 mg司美格鲁肽组平均体重减轻14.9%-17.4%；69%-79%的参与者实现 $\geq 10\%$ 的体重减轻（安慰剂组为12%-27%），51%-64%实现 $\geq 15\%$ 的体重减轻（安慰剂组为5%-13%）。在STEP 5试验中，从基线至第104周，2.4 mg司美格鲁肽组平均体重减轻-15.2%，安慰剂组为-2.6%。在STEP 2试验（针对超重或肥胖且患有2型糖尿病的个体）中，从基线至第68周，2.4 mg司美格鲁肽组平均体重减轻-9.6%，安慰剂组为-3.4%。2.4 mg司美格鲁肽还改善了心血管代谢风险因素，包括高血压、致动脉粥样硬化性血脂异常，并提升了身体功能和生活质量。各试验中2.4 mg司美格鲁肽的安全性特征一致，主要为胃肠道不良事件。STEP试验中报告的体重减轻幅度，为肥胖相关疾病患者实现临床意义上的改善提供了可能。

核心词汇

obesity /ou'bi:səti/ : 肥胖

morbidity /mɔ:r'bidəti/ : 发病率

pharmacological treatment /,fɑ:rməke'la:dʒɪkəl 'tri:tment/ : 药物治疗

weight loss /weɪt lɔ:s/ : 体重减轻

hypertension /,haɪpə'tenʃən/ : 高血压

gastrointestinal /,gæstrouɪn'testɪnəl/ : 胃肠道的

clinical trial /'klɪnɪkəl 'traɪəl/ : 临床试验

complication /,kɑ:mplɪ'keɪʃən/ : 并发症

mortality /mɔ:r'tæləti/ : 死亡率

lifestyle intervention /'laɪfstɑɪl ,ɪntə'venʃən/ : 生活方式干预

cardiometabolic risk factor /,kɑ:rdiəu'metə'ba:lɪk rɪsk 'fæktə/ : 心血管代谢危险因素

adverse event /æd'vɜ:rs ɪ'vent/ : 不良事件

placebo /plə'si:bou/ : 安慰剂

Female fertility and pregnancy in autoimmune Addison's disease - a mini review.

自身免疫性艾迪生病的女性生育能力与妊娠——小型综述。

期刊: Frontiers in endocrinology 日期: 2025

Abstract

autoimmune Addison's disease (AAD) is by far the most common cause of **primary adrenal insufficiency** in developed countries, occurring more commonly in women compared with men. The condition is associated with a spectrum of disorders affecting **fertility** and reproductive health. **premature ovarian insufficiency** (POI) is a clinical condition defined by cessation of menstrual cycles and menopausal range gonadotrophins before the age of 40 years. This occurs with a prevalence of 1-2% in the general population, but has been estimated at 6-10% for women with AAD. One registry study demonstrated that one-third of those with AAD who develop POI, do so before the age of thirty. The onset of POI precedes or is contemporaneous with the diagnosis of AAD in the majority. It has also been demonstrated that women with AAD are more likely to use **hormone replacement therapy**. The **pathophysiology** of POI in this cohort is thought to be primarily through autoimmune mediated **inflammation** of the ovarian theca cells. In particular, cross-reacting autoantibodies to steroid-producing cells (StCA) have been identified which are present in AAD and POI. That said, when women with POI are excluded, **fertility** remains significantly reduced. Impaired **adrenal androgenesis** and resulting sex-hormone deficiency have also been implicated in subfertility in AAD. These lead to suboptimal follicular development. This, in turn, may also affect libido. Despite physiological **glucocorticoid replacement therapy**, patients with AAD consistently report reduced quality of life compared to matched controls. These factors may affect fecundity and likelihood of conception. Other autoimmune conditions such as hypothyroidism and type 1 diabetes occur with increased prevalence in those with AAD. These conditions have been shown to independently affect reproductive health. This review focuses on the current understanding of the factors and mechanisms impacting **fertility** in women with **autoimmune Addison's disease**.

中文翻译

自身免疫性艾迪生病（AAD）是目前发达国家原发性肾上腺皮质功能不全的最常见病因，女性发病率高于男性。该疾病与一系列影响生育及生殖健康的病症相关。早发性卵巢功能不全（POI）是一种临床病症，定义为40岁前出现月经周期停止及绝经期促性腺激素水平。该病在普通人群中的患病率为1-2%，但在AAD女性中估计达6-10%。一项登记研究显示，合并POI的AAD患者中，三分之一在30岁前发病。多数患者的POI发病时间早于或与AAD诊断同期。研究还表明，AAD女性更可能使用激素替代疗法。该群体POI的病理生理机制主要被认为通过自身免疫介导的卵巢膜细胞炎症。特别是，在AAD和POI中已发现针对类固醇生成细胞（StCA）的交叉反应性自身抗体。尽管如此，排除合并POI的女性后，生育能力仍显著降低。肾上腺雄激素生成障碍及其导致的性激素缺乏也被认为与AAD患者生育力低下相关。这些因素导致卵泡发育不良，进而可能影响性欲。尽管接受生理性糖皮质激素替代治疗，与匹配对照组相比，AAD患者始终报告生活质量下降。这些因素可能影响生育能力及受孕概率。其他自身免疫性疾病如甲状腺功能减退症和1型糖尿病在AAD患者中患病率升高，且已被证实可独立影响生殖健康。本综述聚焦当前对影响自身免疫性艾迪生病女性生育能力的因素及机制的认识。

核心词汇

autoimmune Addison's disease /ˌɔːtəʊɪ'mjuːn 'ædɪsənz dɪ 'ziːz/ : 自身免疫性艾迪生病	primary adrenal insufficiency /'praɪməri ə'driːnəl ,ɪnsə 'fɪʃənsi/ : 原发性肾上腺功能不全
premature ovarian insufficiency /'premətʃər əv'veəriən ,ɪnsə 'fɪʃənsi/ : 早发性卵巢功能不全	menstrual cycle /'menstruəl 'saɪkl/ : 月经周期
gonadotrophin /ˌɡɒnədəʊ'trəʊfɪn/ : 促性腺激素	hormone replacement therapy /'hɔːməʊn rɪ'pleɪsmənt 'θerəpi/ : 激素替代疗法
pathophysiology /ˌpæθəʊfɪzɪ'plədʒi/ : 病理生理学	inflammation /ˌɪnflə'meɪʃən/ : 炎症
theca cell /'θiːkə sel/ : 卵泡膜细胞	autoantibody /ˌɔːtəʊ'æntɪbɒdi/ : 自身抗体
adrenal androgenesis /ə'driːnəl ,ændrə'dʒenɪsɪs/ : 肾上腺雄激素生成	glucocorticoid replacement therapy /ˌɡluːkəʊ'kɔːtɪkɔɪd rɪ 'pleɪsmənt 'θerəpi/ : 糖皮质激素替代疗法
fertility /fɜː'tɪləti/ : 生育力	

Primary Aldosteronism: An Endocrine Society Clinical Practice Guideline.

原发性醛固酮增多症：美国内分泌学会临床实践指南。

期刊: The Journal of clinical endocrinology and metabolism 日期: 2025

Abstract

BACKGROUND: Primary **aldosteronism** (PA), a primary **adrenal** disorder leading to excessive **aldosterone** production by one or both **adrenal** glands, is a common cause of **hypertension**. It is associated with an increased risk of cardiovascular complications compared with primary **hypertension**. Despite effective methods for diagnosing and treating PA, it remains markedly underdiagnosed and undertreated. **OBJECTIVE:** To develop an updated guideline that provides a practical, clinical approach to identifying and managing PA to improve diagnosis rates and encourage targeted treatment. **METHODS:** The Guideline Development Panel (GDP), composed of a multidisciplinary panel of clinical experts and experts in systemic review methodology, used the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach to define 10 questions related to the diagnosis and treatment of PA. Systematic reviews were conducted for each question. The GDP used the GRADE Evidence to Decision (EtD) framework to consider contextual factors, such as stakeholder values and preferences, costs and required resources, cost-effectiveness, acceptability, feasibility, and the potential impact on health equity. **RESULTS:** We suggest that all individuals with **hypertension** be screened for PA by measuring **aldosterone** and **renin** and determining the **aldosterone** to **renin** ratio, and that subsequent clinical care be guided by the results. We suggest that individuals with PA receive PA-specific therapy, either medical or surgical. In individuals who screen positive for PA, we suggest (1) commencement of PA-specific medical therapy in individuals who do not desire or are not candidates for surgery and in situations where the probability of lateralizing PA (excess **aldosterone** produced by one **adrenal**) is low based on screening results; and (2) **aldosterone** suppression testing in situations when screening results indicate an intermediate probability for lateralizing PA and individualized decision making confirms a desire to pursue eligibility for surgical therapy. In those who test positive by **aldosterone** suppression testing, and in those in whom screening results show a high probability of lateralizing PA (obviating the need for **aldosterone** suppression testing), we suggest **adrenal lateralization** with **computed tomography** scanning and **adrenal venous sampling** prior to deciding the treatment approach (medical vs surgical). In all individuals with PA and an **adrenal adenoma**, we suggest performing a 1-mg overnight **dexamethasone suppression test**. We suggest the use of mineralocorticoid receptor antagonists (MRAs) over epithelial sodium-channel (ENaC) inhibitors in the medical treatment of PA. We suggest the use of **spironolactone** over other MRAs, given its lower cost and greater availability; however, all MRAs, when titrated to equivalent potencies, are anticipated to have similar efficacy in treating PA. Thus, MRAs with greater mineralocorticoid receptor specificity and fewer androgen/progesterone receptor-mediated side effects may be preferred in some situations. In individuals receiving MRA therapy, we suggest monitoring **renin** and, in those whose **hypertension** remains uncontrolled and **renin** is suppressed, titrating the MRA to increase **renin**. **CONCLUSION:** These recommendations provide a practical framework for the diagnosis and treatment of PA. They are based on currently available literature and take into consideration outcomes that are important to key stakeholders. The goal is to increase identification of individuals with PA and, by initiating PA-specific medical or surgical therapy, improve blood pressure control and reduce PA-associated adverse cardiovascular events. The guidelines also highlight important knowledge gaps in PA diagnosis and management.

中文翻译

背景：原发性醛固酮增多症（PA）是一种导致单侧或双侧肾上腺过度分泌醛固酮的原发性肾上腺疾病，是高血压的常见病因。与原发性高血压相比，PA患者发生心血管并发症的风险更高。尽管已有有效的诊断和治疗方法，但PA仍存在显著的漏诊和治疗不足问题。**目的：**制定更新的临床指南，提供识别和管理PA的实用临床方法，以提高诊断率并促进靶向治疗。**方法：**由多学科临床专家和系统评价方法学专家组成的指南制定小组（GDP）采用推荐分级的评估、制定与评价（GRADE）方法，确定了10个与PA诊断和治疗相关的问题。针对每个问题进行了系统评价。GDP使用GRADE证据到决策（EtD）框架，考虑了利益相关者的价值观和偏好、成本和所需资源、成本效益、可接受性、可行性以及对健康公平的潜在影响等背景因素。**结果：**我们建议所有高血压患者通过测量醛固酮和肾素并计算醛固酮与肾素比值来筛查PA，并根据结果指导后续临床治疗。我们建议PA患者接受针对PA的特异性治疗，包括药物或手术。对于筛查阳性的PA患者，我们建议：（1）对于不愿手术或不适合手术的患者，以及根据筛查结果判断单侧PA（单侧肾上腺过度分泌醛固酮）可能性较低的患者，开始PA特异性药物治疗；（2）当筛查结果显示单侧PA可能性中等，且个体化决策确认有接受手术治疗的意愿时，进行醛固酮抑制试验。对于醛固酮抑制试验阳性者，以及筛查结果显示单侧PA可能性高（无需进行醛固酮抑制试验）者，我们建议在决定治疗方案（药物或手术）前，通过计算机断层扫描和肾上腺静脉采血进行肾上腺侧别定位。对于所有合并肾上腺腺瘤的PA患者，我们建议进行1mg过夜地塞米松抑制试验。在PA的药物治疗中，我们建议优先使用盐皮质激素受体拮抗剂（MRA）而非上皮钠通道（ENaC）抑制剂。鉴于螺内酯成本较低且更易获得，我们建议优先使用

螺内酯而非其他MRA；然而，所有MRA在滴定至等效剂量时，预计对PA的治疗效果相似。因此，具有更高盐皮质激素受体特异性且雄激素/孕激素受体作用较弱的MRA

核心词汇

aldosteronism /æl'dɒstərəˌnɪzəm/ : 醛固酮增多症

aldosterone /æl'dɒstəˌroun/ : 醛固酮

adrenal /ə'driːnəl/ : 肾上腺

computed tomography /kəm'pjʊːtɪd tə'mɒɡrəfi/ : 计算机断层扫描

mineralocorticoid receptor antagonist /ˌmɪnərəloʊˈkɔːrtɪkɔɪd rɪ'septər æn'tæɡənɪst/ : 盐皮质激素受体拮抗剂

dexamethasone suppression test /ˌdekse'meθəsəʊn səˈpreʃən test/ : 地塞米松抑制试验

hypertension /ˌhaɪpər'tenʃən/ : 高血压

renin /'riːnɪn/ : 肾素

adenoma /ˌædɪ'noʊmə/ : 腺瘤

venous sampling /'viːnəs 'sæmplɪŋ/ : 静脉采血

spironolactone /ˌspaɪrənəʊˈlæktəʊn/ : 螺内酯

lateralization /ˌlætərəlaɪ'zeɪʃən/ : 侧化定位

云梦山-侵权必究

Outcomes after medical treatment for primary aldosteronism: an international consensus and analysis of treatment response in an international cohort.

原发性醛固酮增多症药物治疗后的结果：一项国际共识及国际队列中治疗反应的分析。

期刊: The lancet. Diabetes & endocrinology 日期: 2025

Abstract

BACKGROUND: primary aldosteronism can be treated medically but there is no standardised method to evaluate treatment outcomes. We aimed to develop criteria for assessing the outcomes of targeted medical treatment of primary aldosteronism, analyse outcomes across an international cohort, and identify factors associated with a complete treatment response. **METHODS:** An international panel of 31 primary aldosteronism experts used the Delphi method to reach consensus on the definition of complete, partial, or absent biochemical and clinical outcomes of medical treatment of primary aldosteronism. Clinical data at baseline and 6-12 months post-treatment were collected from patients with primary aldosteronism who started targeted medical treatment between 2016 and 2021 at 28 participating centres. **FINDINGS:** consensus was reached for defining complete, partial, or absent biochemical or clinical response. Of 1258 patients (with a mean age of 52 years [SD 11.5] and of whom 610 [48.5%] were female and 648 [51.5%] were male), 1057 (84.0%) had biochemical outcome data (559 [52.9%] had a complete biochemical response). The median daily dose of spironolactone was significantly higher for those with a complete biochemical response than for those without (40 mg [IQR 25-50] vs 25 mg [20-50]; $p=0.011$). Of the 1248 patients with clinical outcome data, 228 [18.3%] had a complete clinical response whereas 227 (18.2%) had an absent response. Patients with a complete clinical response were more likely than those with partial or absent clinical response to be women (OR 2.099, 95% CI 1.485-2.968; $p<0.001$), require lower doses of antihypertensive drugs at baseline (0.687, 0.603-0.782; $p<0.001$), and were less likely to have microalbuminuria or left ventricular hypertrophy (0.584, 0.391-0.873; $p=0.009$). **INTERPRETATION:** The primary aldosteronism Medical Treatment outcome (PAMO) criteria represent an internationally developed outcome standard that can guide clinical practice and research into primary aldosteronism. Efforts to optimise treatment intensity and minimise factors associated with an absent treatment response are needed to improve patient outcomes. **FUNDING:** None. **TRANSLATIONS:** For the Chinese (simple), Chinese (complex), Japanese, Korean, German, French, Spanish, Dutch, Swedish, Slovenian, Polish, Italian and Russian translations of the abstract see Supplementary Materials section.

中文翻译

背景：原发性醛固酮增多症可通过药物治疗，但目前缺乏标准化的疗效评估方法。本研究旨在制定评估原发性醛固酮增多症靶向药物治疗效果的标准化标准，分析国际队列的治疗结局，并识别与完全治疗应答相关的因素。**方法：**由31位原发性醛固酮增多症专家组成的国际专家组采用德尔菲法，就药物治疗后生化与临床结局的完全缓解、部分缓解及无缓解的定义达成共识。收集2016年至2021年间在28个参与中心启动靶向药物治疗的原发性醛固酮增多症患者基线及治疗后6-12个月的临床数据。**结果：**专家组就生化或临床应答的完全、部分及无缓解定义达成共识。在1258例患者（平均年龄52岁[标准差11.5]，其中女性610例[48.5%]，男性648例[51.5%]）中，1057例（84.0%）具有生化结局数据（其中559例[52.9%]达到完全生化应答）。完全生化应答组螺内酯中位日剂量显著高于非完全应答组（40 mg [四分位距25-50] vs 25 mg [20-50]； $p=0.011$ ）。在1248例具有临床结局数据的患者中，228例（18.3%）达到完全临床应答，而227例（18.2%）无应答。完全临床应答患者较部分应答或无应答患者更可能为女性（OR 2.099, 95% CI 1.485-2.968; $p<0.001$ ），基线时所需降压药剂量更低（0.687, 0.603-0.782; $p<0.001$ ），且较少出现微量白蛋白尿或左心室肥厚（0.584, 0.391-0.873; $p=0.009$ ）。**结论：**原发性醛固酮增多症药物治疗结局（PAMO）标准作为国际开发的结局评估标准，可指导原发性醛固酮增多症的临床实践与研究。需优化治疗强度并减少与治疗无应答相关的因素，以改善患者预后。**资金：**无。**翻译：**摘要的中文简体、中文繁体、日文、韩文、德文、法文、西班牙文、荷兰文、瑞典文、斯洛文尼亚文、波兰文、意大利文及俄文翻译见补充材料部分。

核心词汇

primary aldosteronism /ˈpraɪməri ˌældəˈstɛrənɪzəm/ : 原发性醛固酮增多症

biochemical response /ˌbaɪəʊˈkɛmɪkəl rɪˈsɒns/ : 生化反应

spironolactone /ˌspaɪrənəˈlæktəʊn/ : 螺内酯

hypertension /ˌhaɪpərˈtɛnʃən/ : 高血压

clinical response /ˈklɪnɪkəl rɪˈsɒns/ : 临床反应

antihypertensive drugs /ˌæntiˌhaɪpərˈtɛnsɪv drʌɡz/ : 抗高血压药物

microalbuminuria /ˌmaɪkrouˌælbjʊmɪˈnjuəriə/ : 微量白蛋白尿

Delphi method /ˈdelfaɪˈmeθəd/ : 德尔菲法

cohort /ˈkəʊhɔːrt/ : 队列

left ventricular hypertrophy /left venˈtrɪkjʊlər haɪˈpɜːrtroʊfi/ : 左心室肥厚

consensus /kənˈsensəs/ : 共识

outcome /ˈaʊtkʌm/ : 结局

云梦山-侵权必究

Pheochromocytoma: an updated scoping review from clinical presentation to management and treatment.

嗜铬细胞瘤：从临床表现到管理与治疗的最新范围综述

期刊: Frontiers in endocrinology 日期: 2024

Abstract

Pheochromocytomas and paragangliomas (PPGLs) are rare neuroendocrine tumors derived from chromaffin cells, with 80-85% originating in the **adrenal medulla** and 15-20% from extra-adrenal chromaffin tissues (paragangliomas). Approximately 30-40% of PPGLs have a hereditary component, making them one of the most genetically predisposed tumor types. Recent advances in genetic research have classified PPGLs into three molecular clusters: pseudohypoxia-related, kinase-signaling, and WNT-signaling pathway variants. Specifically, the detection of SDHB-related tumors indicates an increased risk of metastatic disease, which may impact decisions regarding **functional imaging** in patients with high suspicion of **metastasis** and influence targeted treatment strategies. Diagnosis of PPGLs primarily relies on **biochemical testing**, measuring catecholamines or their metabolites in plasma or urine. However, molecular testing, **functional imaging**, and targeted therapies have greatly enhanced diagnostic precision and management. Personalized treatment approaches based on genetic profiling are becoming integral to the clinical management of these tumors. In South American countries like Colombia, **functional imaging** techniques such as **positron emission tomography/computed tomography (PET/CT)** with tracers like 18F-DOPA, 18F-fluorodeoxyglucose (18F-FDG), and 68Ga-DOTA-conjugated somatostatin receptor-targeting peptides (68Ga-DOTA-SST) are used to guide follow-up and treatment strategies. **radionuclide therapy** with lutetium-177 DOTATATE is employed for patients showing uptake in 68Ga-DOTA-SST PET/CT scans, while access to 131-MIBG therapy remains limited due to high costs and availability. Recent clinical trials have shown promise for systemic therapies such as sunitinib and cabozantinib, offering potential new options for patients with slow or moderate progression of PPGLs. These advancements underscore the potential of personalized and targeted therapies to improve outcomes in this challenging patient population.

中文翻译

嗜铬细胞瘤和副神经节瘤（PPGLs）是源自嗜铬细胞的罕见神经内分泌肿瘤，其中80-85%起源于肾上腺髓质，15-20%来源于肾上腺外嗜铬组织（副神经节瘤）。约30-40%的PPGLs具有遗传成分，使其成为遗传易感性最高的肿瘤类型之一。近年遗传学研究进展将PPGLs分为三种分子簇：假缺氧相关、激酶信号通路和WNT信号通路变异。具体而言，SDHB相关肿瘤的检出提示转移性疾病风险增加，这可能影响对高度怀疑转移患者的显像策略选择，并指导靶向治疗方案制定。PPGLs诊断主要依赖生化检测，通过测定血浆或尿液中儿茶酚胺及其代谢产物。然而分子检测、功能显像和靶向治疗已显著提升诊断精准度与管理水平。基于基因分型的个体化治疗策略正逐步成为此类肿瘤临床管理的核心组成部分。在哥伦比亚等南美国家，采用18F-DOPA、18F-氟代脱氧葡萄糖（18F-FDG）和68Ga-DOTA-生长抑素受体靶向肽（68Ga-DOTA-SST）等示踪剂的正电子发射断层扫描/计算机断层扫描（PET/CT）功能显像技术，用于指导随访与治疗决策。对68Ga-DOTA-SST PET/CT显像显示摄取的病例采用镥-177 DOTATATE放射性核素治疗，而131-MIBG治疗因成本高昂和可及性受限应用有限。近期临床试验显示舒尼替尼和卡博替尼等全身治疗方案前景可期，为缓慢或中度进展的PPGLs患者提供了潜在新选择。这些进展凸显了个体化靶向治疗改善这一难治性患者群体预后的巨大潜力。

核心词汇

pheochromocytoma /ˌfiːoʊkroumɔːsaɪˈtoʊmə/ : 嗜铬细胞瘤	paraganglioma /ˌpærəɡæŋɡliˈoʊmə/ : 副神经节瘤
neuroendocrine tumor /ˌnjʊroʊˈɛndoʊkraɪnˈtuːmə/ : 神经内分泌肿瘤	adrenal medulla /əˈdriːnəl mɪˈdʌlə/ : 肾上腺髓质
chromaffin cell /ˈkrouməfɪn sɛl/ : 嗜铬细胞	metastasis /mɪˈtæstəsis/ : 转移
biochemical testing /ˌbaɪoʊˈkɛmɪkəlˈtɛstɪŋ/ : 生化检测	catecholamine /ˌkætiˈkoʊləmiːn/ : 儿茶酚胺
functional imaging /ˈfʌŋkʃənəlˈɪmɪdʒɪŋ/ : 功能成像	positron emission tomography /ˈpozɪtrɒn ɪˈmɪʃən təˈmɒɡrəfi/ : 正电子发射断层扫描
radionuclide therapy /ˌreɪdiəʊˈnuːklaɪdˈθɛrəpi/ : 放射性核素治疗	targeted therapy /ˈtɑːɡɪtɪdˈθɛrəpi/ : 靶向治疗

Clinical consensus guideline on the management of phaeochromocytoma and paraganglioma in patients harbouring germline SDHD pathogenic variants.

携带种系SDHD致病性变异患者的嗜铬细胞瘤和副神经节瘤管理临床共识指南

期刊: The lancet. Diabetes & endocrinology 日期: 2023

Abstract

Patients with **germline** SDHD pathogenic variants (encoding succinate dehydrogenase subunit D; ie, **paraganglioma** 1 syndrome) are predominantly affected by head and neck paragangliomas, which, in almost 20% of patients, might coexist with paragangliomas arising from other locations (eg, **adrenal medulla**, para-aortic, cardiac or thoracic, and pelvic). Given the higher risk of tumour **multifocality** and **bilaterality** for phaeochromocytomas and paragangliomas (PPGLs) because of SDHD pathogenic variants than for their **sporadic** and other **genotypic** counterparts, the management of patients with SDHD PPGLs is clinically complex in terms of **imaging**, treatment, and management options. Furthermore, locally aggressive disease can be discovered at a young age or late in the disease course, which presents challenges in balancing surgical intervention with various medical and **radiotherapeutic** approaches. The axiom-first, do no harm-should always be considered and an initial period of observation (ie, **watchful waiting**) is often appropriate to characterise tumour behaviour in patients with these pathogenic variants. These patients should be referred to specialised high-volume medical centres. This **consensus guideline** aims to help physicians with the clinical decision-making process when caring for patients with SDHD PPGLs.

中文翻译

携带SDHD胚系致病性变异（编码琥珀酸脱氢酶亚基D；即副神经节瘤1型综合征）的患者主要罹患头颈部副神经节瘤，其中近20%的患者可能合并其他部位（如肾上腺髓质、主动脉旁、心脏或胸腔、盆腔）的副神经节瘤。由于SDHD致病性变异导致的嗜铬细胞瘤和副神经节瘤（PPGLs）在肿瘤多灶性和双侧性方面的风险高于散发性及其他基因型病例，因此SDHD PPGLs患者在影像学检查、治疗及管理方案的选择上具有临床复杂性。此外，局部侵袭性疾病可能在年轻时期或病程晚期被发现，这给平衡手术干预与多种内科及放射治疗手段带来了挑战。应始终遵循"首要原则——不伤害"的准则，对携带这些致病性变异的患者，初始观察期（即主动监测）通常适用于评估肿瘤行为特征。此类患者应转诊至具备丰富经验的高容量医疗中心。本共识指南旨在帮助临床医生在管理SDHD PPGLs患者时进行临床决策。

核心词汇

germline /'dʒɜːrmlaɪn/ : 种系	pathogenic variant /ˌpæθə'dʒɛnɪk 'vɛriənt/ : 致病性变异
paraganglioma /ˌpæərəˌgæŋɡli'oumə/ : 副神经节瘤	phaeochromocytoma /ˌfiːoʊˌkroumouˈsaɪ'toumə/ : 嗜铬细胞瘤
adrenal medulla /ə'driːnəl mɪ'dʌlə/ : 肾上腺髓质	multifocality /ˌmʌltɪfoʊ'kælɪti/ : 多灶性
bilaterality /ˌbaɪləteɪ'rælɪti/ : 双侧性	sporadic /spə'rædɪk/ : 散发性
genotypic /dʒɛnə'tɪpɪk/ : 基因型的	imaging /'ɪmɪdʒɪŋ/ : 影像学检查
radiotherapeutic /ˌreɪdiəʊˌθɛrə'pjʊːtɪk/ : 放射治疗的	watchful waiting /'wɒtʃfəl 'weɪtɪŋ/ : 主动监测
consensus guideline /kən'sensəs 'ɡaɪdlaɪn/ : 共识指南	

Cushing Syndrome, Hypercortisolism, and Glucose Homeostasis: A Review.

库欣综合征、高皮质醇血症与葡萄糖稳态：综述。

期刊: Diabetes 日期: 2025

Abstract

Until recently, the prevalence of endogenous **Cushing syndrome** has been considered to be low. However, improved diagnostic strategies and increased awareness have broadened our understanding of **hypercortisolism** and its role in the pathophysiology of type 2 diabetes, **obesity**, **hypertension**, and **cardiovascular disease**. Recent studies from Europe, South America, and the U.S. have demonstrated that a significant percentage of individuals with difficult-to-control type 2 diabetes, despite treatment with multiple glucose-lowering agents, have **hypercortisolism** as a causative factor in their poorly managed diabetes. In this review, we examine the pathophysiologic mechanisms via which excess cortisol contributes to the impairment in **glucose homeostasis** and recommend that **hypercortisolism** be added to the Ominous Octet to form the Noxious Nine as the pathophysiologic foundation for the development of type 2 diabetes. ARTICLE HIGHLIGHTS: **hypercortisolism** as a causative factor in the development of type 2 diabetes has received scant attention. Studies from Europe, South America, and the U.S. have demonstrated that a significant percentage of individuals with poorly managed type 2 diabetes, despite treatment with multiple glucose-lowering agents, have endogenous hypersecretion of cortisol as a causative factor for their hyperglycemia. In vivo and in vitro studies in animals and humans have demonstrated that excess exposure to glucocorticoids can promote **insulin resistance** in muscle, liver, and adipocytes and impair **insulin secretion**. We propose a reverberating cycle in which **hypercortisolism** disrupts the normal **circadian rhythm** causing **insulin resistance** and hyperinsulinemia, which in turn further disrupts the hypothalamic-pituitary-adrenal axis.

中文翻译

过去，内源性库欣综合征的患病率一直被认为较低。然而，诊断策略的改进和认知度的提升，拓展了我们对皮质醇增多症及其在2型糖尿病、肥胖、高血压和心血管疾病病理生理学中作用的理解。近期来自欧洲、南美和美国的研究表明，在尽管使用多种降糖药物治疗但仍难以控制的2型糖尿病患者中，有相当比例的患者存在皮质醇增多症，这是其血糖控制不佳的致病因素。本综述探讨了过量皮质醇损害葡萄糖稳态的病理生理机制，并建议将皮质醇增多症纳入“致命八重奏”，形成“有害九重奏”，作为2型糖尿病发展的病理生理基础。文章亮点：皮质醇增多症作为2型糖尿病发展的致病因素一直未受重视。欧洲、南美和美国的研究表明，在血糖控制不佳的2型糖尿病患者中，尽管使用多种降糖药物治疗，仍有相当比例的患者存在内源性皮质醇分泌过多，这是其高血糖的致病因素。动物和人类的体内外研究表明，过量接触糖皮质激素可促进肌肉、肝脏和脂肪细胞的胰岛素抵抗，并损害胰岛素分泌。我们提出一个循环反馈机制：皮质醇增多症破坏正常昼夜节律，导致胰岛素抵抗和高胰岛素血症，进而进一步扰乱下丘脑-垂体-肾上腺轴。

核心词汇

Cushing syndrome /ˈkʊʃɪŋ ˈsɪndroʊm/ : 库欣综合征	hypercortisolism /ˌhaɪpərˈkɔːrtɪsəˌlɪzəm/ : 高皮质醇血症
obesity /ouˈbiːsəti/ : 肥胖	hypertension /ˌhaɪpərˈtɛnʃən/ : 高血压
cardiovascular disease /ˌkɑːrdiəvˈvæskjələr dɪˈziːz/ : 心血管疾病	glucose homeostasis /ˈgluːkəʊs ˌhoumiəʊˈsteɪsɪs/ : 葡萄糖稳态
insulin resistance /ˈɪnsəlɪn rɪˈzɪstəns/ : 胰岛素抵抗	insulin secretion /ˈɪnsəlɪn sɪˈkriːʃən/ : 胰岛素分泌
glucocorticoid /ˌgluːkəʊˈkɔːrtɪkɔɪd/ : 糖皮质激素	circadian rhythm /sɜːrˈkeɪdiən ˈrɪðəm/ : 昼夜节律

Update on clinical characteristics in the evaluation of pheochromocytoma and paraganglioma.

嗜铬细胞瘤和副神经节瘤评估中临床特征的更新

期刊: Best practice & research. Clinical endocrinology & metabolism 日期: 2024

Abstract

Pheochromocytomas and sympathetic paragangliomas (PPGL) are rare neuroendocrine tumors originating from **chromaffin tissue** of the **adrenal medulla** and extra-adrenal sympathetic paraganglia. Historically, many of these tumors were diagnosed postmortem, earning pheochromocytomas the moniker "great mimic" due to their diverse clinical manifestations that can resemble various other conditions. Over time, the clinical presentation of PPGL has evolved, with a shift from symptomatic or postmortem diagnoses to more frequent incidental discoveries or diagnoses through **screening**, with postmortem identification now being rare. The development of a clinical scoring system has improved the identification of patients at increased risk for PPGL. Notably, the proportion of PPGL patients with normal blood pressure ranges from 15 % to 40 %, varying based on the clinical context. Despite the tumor's reputation, PPGL is an exceedingly rare cause of resistant **hypertension**. Management of a **pheochromocytoma** crisis has advanced, with several classes of drugs available for treatment. However, PPGL during pregnancy remains a significant concern, associated with substantial maternal and fetal mortality rates. Additionally, PPGL can present as rare disorders, including **catecholamine**-induced **cardiomyopathy**, **Cushing syndrome**, and urinary bladder PGL. Given these varied presentations, heightened awareness and prompt recognition of PPGL are crucial for timely **diagnosis** and treatment, ultimately improving patient outcomes. In this article, we offer an in-depth analysis of the diverse clinical presentations of PPGL, highlighting their complexity and the associated diagnostic and treatment strategies.

中文翻译

嗜铬细胞瘤和交感神经副神经节瘤（PPGL）是起源于肾上腺髓质及肾上腺外交感神经副神经节嗜铬组织的罕见神经内分泌肿瘤。历史上，许多此类肿瘤在死后才被确诊，因其临床表现多样且可模拟多种其他疾病，嗜铬细胞瘤曾被称为“伟大的模仿者”。随着时间推移，PPGL的临床表现模式已发生演变，从症状性诊断或死后诊断转向更常见的偶然发现或筛查诊断，死后确诊现已罕见。临床评分系统的开发提高了对PPGL高风险患者的识别能力。值得注意的是，PPGL患者中血压正常者占比为15%至40%，具体比例因临床背景而异。尽管该肿瘤声名显赫，但PPGL作为难治性高血压病因的情况极为罕见。嗜铬细胞瘤危象的管理已取得进展，现有多种药物可用于治疗。然而，妊娠期PPGL仍是重大隐患，与较高的母婴死亡率相关。此外，PPGL可表现为罕见疾病，包括儿茶酚胺诱导性心肌病、库欣综合征及膀胱副神经节瘤。鉴于这些多样化的临床表现，提高警惕并快速识别PPGL对及时诊断和治疗至关重要，最终可改善患者预后。本文深入分析了PPGL的多样化临床表现，强调其复杂性及相关的诊断与治疗策略。

核心词汇

pheochromocytoma /ˈfiːoʊkroumouˌsaɪˈtoʊmə/ : 嗜铬细胞瘤	paraganglioma /ˌpærəˈgæŋɡliˈoʊmə/ : 副神经节瘤
neuroendocrine tumor /ˌnjʊroʊˈɛndoukraɪnˈtuːmə/ : 神经内分泌肿瘤	adrenal medulla /əˈdriːnəl mɪˈdʌlə/ : 肾上腺髓质
chromaffin tissue /ˈkrouməfɪnˈtɪʃuː/ : 嗜铬组织	hypertension /ˌhaɪpərˈtɛnʃən/ : 高血压
catecholamine /ˌkætiˈkoʊləmiːn/ : 儿茶酚胺	cardiomyopathy /ˌkɑːrdiəʊmaɪˈɒpiəθi/ : 心肌病
Cushing syndrome /ˈkʊʃɪŋˈsɪndroum/ : 库欣综合征	biopsy /ˈbaɪəpsi/ : 活检
screening /ˈskriːnɪŋ/ : 筛查	diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断

European Society of Endocrinology and Endocrine Society Joint Clinical Guideline: Diagnosis and Therapy of Glucocorticoid-induced Adrenal Insufficiency.

欧洲内分泌学会与内分泌学会联合临床指南：糖皮质激素诱导的肾上腺功能不全的诊断与治疗。

期刊: The Journal of clinical endocrinology and metabolism 日期: 2024

Abstract

Glucocorticoids are widely prescribed as anti-inflammatory and immunosuppressive agents. This results in at least 1% of the population using chronic glucocorticoid therapy, being at risk for glucocorticoid-induced adrenal insufficiency. This risk is dependent on the dose, duration and potency of the glucocorticoid, route of administration, and individual susceptibility. Once glucocorticoid-induced adrenal insufficiency develops or is suspected, it necessitates careful education and management of affected patients. tapering glucocorticoids can be challenging when symptoms of glucocorticoid withdrawal develop, which overlap with those of adrenal insufficiency. In general, tapering of glucocorticoids can be more rapidly within a supraphysiological range, followed by a slower taper when on physiological glucocorticoid dosing. The degree and persistence of HPA axis suppression after cessation of glucocorticoid therapy are dependent on overall exposure and recovery of adrenal function varies greatly amongst individuals. This first European Society of Endocrinology/endocrine Society joint clinical practice guideline provides guidance on this clinically relevant condition to aid clinicians involved in the care of patients on chronic glucocorticoid therapy.

中文翻译

糖皮质激素被广泛用作抗炎和免疫抑制剂。这导致至少1%的人群长期接受糖皮质激素治疗，面临糖皮质激素诱导的肾上腺功能不全的风险。该风险取决于糖皮质激素的剂量、疗程、效力、给药途径以及个体易感性。一旦发生或怀疑糖皮质激素诱导的肾上腺功能不全，需对受影响患者进行细致的教育和管理。当出现与肾上腺功能不全症状重叠的糖皮质激素撤药症状时，逐步减量糖皮质激素可能具有挑战性。通常，在超生理剂量范围内可较快减量，而进入生理剂量后需减缓减量速度。停用糖皮质激素后，下丘脑-垂体-肾上腺轴抑制的程度和持续时间取决于总体暴露量，且肾上腺功能恢复的个体差异显著。本首部欧洲内分泌学会/美国内分泌学会联合临床实践指南针对这一临床相关问题提供指导，以协助参与长期糖皮质激素治疗患者管理的临床医生。

核心词汇

glucocorticoid /ˌɡluːˈkɒrːtɪkɔɪd/ : 糖皮质激素	adrenal insufficiency /əˈdriːnəl ˌɪnsəˈfɪʃənsi/ : 肾上腺功能不全
immunosuppressive /ˌɪmjʊnəʊsəˈpresɪv/ : 免疫抑制的	HPA axis /eɪtʃ piː ɪˈæksɪs/ : 下丘脑-垂体-肾上腺轴
tapering /ˈteɪpərɪŋ/ : 逐渐减量	withdrawal /wɪðˈdrɔːəl/ : 戒断
supraphysiological /ˌsuːprəˈfɪziəˈlɒdʒɪkəl/ : 超生理剂量的	physiological /ˌfɪziəˈlɒdʒɪkəl/ : 生理剂量的
clinical practice guideline /ˈklɪnɪkəl ˈpræktɪs ˈɡaɪdlaɪn/ : 临床实践指南	endocrine /ˈendəkrɪn/ : 内分泌的
potency /ˈpəʊtənsi/ : 效力	

European Society of Endocrinology and Endocrine Society Joint Clinical Guideline: Diagnosis and therapy of glucocorticoid-induced adrenal insufficiency.

欧洲内分泌学会与内分泌学会联合临床指南：糖皮质激素诱导的肾上腺功能不全的诊断与治疗。

期刊: European journal of endocrinology 日期: 2024

Abstract

Glucocorticoids are widely prescribed as anti-inflammatory and immunosuppressive agents. This results in at least 1% of the population using chronic glucocorticoid therapy, being at risk for glucocorticoid-induced adrenal insufficiency. This risk is dependent on the dose, duration and potency of the glucocorticoid, route of administration, and individual susceptibility. Once glucocorticoid-induced adrenal insufficiency develops or is suspected, it necessitates careful education and management of affected patients. tapering glucocorticoids can be challenging when symptoms of glucocorticoid withdrawal develop, which overlap with those of adrenal insufficiency. In general, tapering of glucocorticoids can be more rapidly within a supraphysiological range, followed by a slower taper when on physiological glucocorticoid dosing. The degree and persistence of HPA axis suppression after cessation of glucocorticoid therapy are dependent on overall exposure and recovery of adrenal function varies greatly amongst individuals. This first European Society of Endocrinology/endocrine Society joint clinical practice guideline provides guidance on this clinically relevant condition to aid clinicians involved in the care of patients on chronic glucocorticoid therapy.

中文翻译

糖皮质激素被广泛用作抗炎和免疫抑制剂。这导致至少1%的人群长期接受糖皮质激素治疗，面临糖皮质激素诱导的肾上腺功能不全的风险。该风险取决于糖皮质激素的剂量、疗程、效力、给药途径以及个体易感性。一旦发生或怀疑糖皮质激素诱导的肾上腺功能不全，需对受影响患者进行细致的教育和管理。当出现与肾上腺功能不全症状重叠的糖皮质激素撤药症状时，逐步减量糖皮质激素可能具有挑战性。通常，在超生理剂量范围内可较快减量，而进入生理剂量阶段后需减缓减量速度。停用糖皮质激素后，下丘脑-垂体-肾上腺轴抑制的程度和持续时间取决于总体暴露量，且肾上腺功能恢复的个体差异显著。本首部欧洲内分泌学会/美国内分泌学会联合临床实践指南针对这一临床相关问题提供指导，以协助参与长期糖皮质激素治疗患者管理的临床医生。

核心词汇

glucocorticoid /ˌɡluːkɔːˈkɔɪd/ : 糖皮质激素	adrenal insufficiency /əˈdriːnəl ˌɪnsəˈfɪʃənsi/ : 肾上腺功能不全
immunosuppressive /ˌɪmjʊnəʊsəˈpresɪv/ : 免疫抑制的	HPA axis /eɪtʃ piː ˈæksɪs/ : 下丘脑-垂体-肾上腺轴
tapering /ˈteɪpərɪŋ/ : 逐渐减量	withdrawal /wɪðˈdrɔːəl/ : 戒断
supraphysiological /ˌsuːprəˈfɪziəˈlɒdʒɪkəl/ : 超生理剂量的	physiological /ˌfɪziəˈlɒdʒɪkəl/ : 生理剂量的
endocrine /ˈendəkrɪn/ : 内分泌的	clinical practice guideline /ˈklɪnɪkəl ˈpræktɪs ˈgaɪdlaɪn/ : 临床实践指南

A Contemporary Approach to the Diagnosis and Management of Adrenal Insufficiency.

肾上腺功能不全诊断与治疗的当代方法

期刊: Endocrinology and metabolism (Seoul, Korea) 日期: 2024

Abstract

adrenal insufficiency (AI) can be classified into three distinct categories based on its underlying causes: primary adrenal disorders, secondary deficiencies in **adrenocorticotropin**, or **hypothalamic suppression** from external factors, most commonly **glucocorticoid** medications used for anti-inflammatory therapy. The hallmark clinical features of AI include **fatigue**, appetite loss, unintentional weight loss, low blood pressure, and **hyponatremia**. Individuals with primary AI additionally manifest skin **hyperpigmentation**, **hyperkalemia**, and salt craving. The **diagnosis** of AI is frequently delayed due to the non-specific symptoms and signs early in the disease course, which poses a significant challenge to its early detection prior to an **adrenal crisis**. Despite the widespread availability of lifesaving **glucocorticoid** medications for decades, notable challenges persist, particularly in the domains of timely **diagnosis** while simultaneously avoiding misdiagnosis, patient education for averting adrenal crises, and the determination of optimal replacement therapies. This article reviews recent advancements in the contemporary diagnostic strategy and approaches to optimal treatment for AI.

中文翻译

肾上腺功能不全 (AI) 可根据其根本病因分为三类：原发性肾上腺疾病、继发性促肾上腺皮质激素缺乏，或由外部因素（最常见的是用于抗炎治疗的糖皮质激素药物）引起的下丘脑抑制。AI的典型临床特征包括疲劳、食欲减退、非自愿性体重下降、低血压和低钠血症。原发性AI患者还会出现皮肤色素沉着、高钾血症和嗜盐症状。由于疾病早期症状和体征缺乏特异性，AI的诊断常被延误，这给在肾上腺危象发生前实现早期发现带来了重大挑战。尽管数十年来救命性糖皮质激素药物已广泛可及，但在及时诊断同时避免误诊、患者教育以预防肾上腺危象、以及确定最佳替代治疗方案等领域仍存在显著挑战。本文综述了当代AI诊断策略和优化治疗方法的最新进展。

核心词汇

adrenal insufficiency /əˈdriːnəl ˌɪnsəˈfɪʃənsi/ : 肾上腺功能不全	glucocorticoid /ˌɡluːkɔːˈrɪtɪkɔɪd/ : 糖皮质激素
fatigue /ˈfeɪtiːɡ/ : 疲劳	hyponatremia /ˌhaɪpəʊnəˈtriːmiə/ : 低钠血症
hyperpigmentation /ˌhaɪpərˌpɪɡməntˈeɪʃən/ : 色素沉着过度	hyperkalemia /ˌhaɪpərkeˈliːmiə/ : 高钾血症
adrenal crisis /əˈdriːnəl ˈkraɪsɪs/ : 肾上腺危象	diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断
replacement therapy /rɪˈpleɪsmənt ˈθerəpi/ : 替代治疗	hypothalamic suppression /ˌhaɪpəʊθəˈlæmɪk səˈpreʃən/ : 下丘脑抑制
adrenocorticotropin /əˈdriːnɔːˌkɔːrtɪkəʊˈtrɒpɪn/ : 促肾上腺皮质激素	

Updates on renal phosphate transport.

肾磷酸盐转运的最新进展。

期刊: Current opinion in nephrology and hypertension 日期: 2025

Abstract

PURPOSE OF REVIEW: The kidneys control systemic **phosphate** balance by regulating **phosphate** transporters mediating the **reabsorption** of inorganic **phosphate** (Pi). At least three different Na⁺-driven Pi cotransporters are located in the **brush border membrane** (BBM) of **proximal tubule** cells, NaPi-IIa (SLC34A1), NaPi-IIc (SLC34A3) and PiT-2 (SLC20A2). This review will discuss novel aspects of their regulation, pharmacology, and genetics. RECENT FINDINGS: Renal NaPi transporters are not only acutely regulated by the phosphaturic hormones parathyroid **hormone** (PTH) and Fibroblast Growth Factor 23 (FGF23) but possibly also by further mechanisms. A role of inositol hexakisphosphate (IP6) kinases has been found and their deletion from kidneys causes **hypophosphatemia**, **hyperphosphaturia**, and bone demineralization. Inhibitors of NaPis elicit phosphaturia and may reduce levels of PTH and FGF23 in **chronic kidney disease** (CKD) models. The relevance of renal NaPi transporters is highlighted by loss-of-function mutations in SLC34 transporters and analysis of patients provides new insights into diseases caused by variants. Major manifestations include **nephrocalcinosis** and -lithiasis, **rickets**, and variants may predispose to an accelerated decline in **kidney** function. SUMMARY: Renal Pi transporters are regulated, may provide novel drug targets for prevention or treatment of **hyperphosphatemia**, and contribute to the genetic risk to develop **kidney** stones and CKD.

中文翻译

综述目的：肾脏通过调节介导无机磷酸盐（Pi）重吸收的磷酸盐转运蛋白来控制全身磷酸盐平衡。近端肾小管细胞刷状缘膜（BBM）上至少存在三种不同的钠离子驱动型磷酸盐协同转运蛋白：NaPi-IIa（SLC34A1）、NaPi-IIc（SLC34A3）和PiT-2（SLC20A2）。本综述将讨论其调控、药理学和遗传学的新进展。最新发现：肾脏NaPi转运蛋白不仅受磷酸盐调节激素甲状旁腺激素（PTH）和成纤维细胞生长因子23（FGF23）的急性调控，还可能受其他机制影响。研究发现肌醇六磷酸（IP6）激酶的作用，其在肾脏中的缺失会导致低磷血症、高磷尿症和骨脱矿。NaPi抑制剂可引发磷尿症，并可能降低慢性肾病（CKD）模型中PTH和FGF23的水平。SLC34转运蛋白功能缺失突变凸显了肾脏NaPi转运蛋白的重要性，对患者的分析为变异引起的疾病提供了新见解。主要表现包括肾钙质沉着症、肾结石、佝偻病，且变异可能增加肾功能加速下降的风险。总结：肾脏Pi转运蛋白受到调控，可能为预防或治疗高磷血症提供新的药物靶点，并构成肾结石和CKD遗传风险因素。

核心词汇

kidney /'kɪdni/ : 肾脏	phosphate /'fɒsfet/ : 磷酸盐
transporter /træns'pɔ:rtər/ : 转运体	reabsorption /,ri:əb'sɔ:rpʃən/ : 重吸收
proximal tubule /'prɒksɪməl 'tju:bjʊ:l/ : 近端小管	brush border membrane /brʌʃ 'bɔ:rdər 'membreɪn/ : 刷状缘膜
hormone /'hɔ:rmoun/ : 激素	hypophosphatemia /,haɪpəʊ'fɒsfə'ti:mɪə/ : 低磷血症
hyperphosphaturia /,haɪpər'fɒsfə'tjʊəriə/ : 高磷尿症	nephrocalcinosis /,nefrəʊ'kælsɪ'nəʊsɪs/ : 肾钙质沉着症
nephrolithiasis /,nefrəʊlɪ'thaɪəsɪs/ : 肾结石	rickets /'rɪkɪts/ : 佝偻病
chronic kidney disease /'krɒnɪk 'kɪdni dɪ'zi:z/ : 慢性肾脏病	hyperphosphatemia /,haɪpər'fɒsfə'ti:mɪə/ : 高磷血症

Update on the Genetics of Osteogenesis Imperfecta.

成骨不全症遗传学的最新进展

期刊: Calcified tissue international 日期: 2024

Abstract

osteogenesis imperfecta (OI) is a heterogeneous heritable **skeletal dysplasia** characterized by **bone fragility** and deformity, growth deficiency, and other secondary connective tissue defects. OI is now understood as a **collagen**-related disorder caused by defects of genes whose protein products interact with **collagen** for folding, post-translational modification, processing and trafficking, affecting **bone mineralization** and **osteoblast differentiation**. This review provides the latest updates on **genetics** of OI, including new developments in both dominant and rare OI forms, as well as the signaling pathways involved in OI **pathophysiology**. There is a special emphasis on discoveries of recessive mutations in **TENT5A**, **MESD**, **KDEL2** and **CCDC134** whose causality of OI types XIX, XX, XXI and XXII, respectively, is now established and expands the complexity of mechanisms underlying OI to overlap LRP5/6 and MAPK/ERK pathways. We also review in detail new discoveries connecting the known OI types to each other, which may underlie an eventual understanding of a final common pathway in OI cellular and **bone biology**.

中文翻译

成骨不全症 (OI) 是一种异质性遗传性骨骼发育不良，其特征为骨脆性、骨骼畸形、生长迟缓及其他继发性结缔组织缺陷。目前认为OI是一种胶原相关疾病，由基因缺陷引起，这些基因的蛋白产物通过与胶原蛋白相互作用参与其折叠、翻译后修饰、加工和转运过程，进而影响骨矿化和成骨细胞分化。本综述提供了OI遗传学的最新进展，包括显性和罕见OI类型的新发现，以及参与OI病理生理学的信号通路。特别强调了TENT5A、MESD、KDEL2和CCDC134基因隐性突变的新发现，这些突变分别被证实导致XIX型、XX型、XXI型和XXII型OI，并将OI的机制复杂性扩展至LRP5/6和MAPK/ERK通路的交叉重叠。我们还详细回顾了将已知OI类型相互关联的新发现，这些发现可能最终揭示OI细胞与骨生物学中共同终末通路的基础。

核心词汇

osteogenesis imperfecta /ˌɒstriəʊˈdʒɛnɪsɪs ɪmˈpɜːfɛktə/ : 成骨不全症	skeletal dysplasia /ˈskɛlɪtəl dɪsˈpleɪzə/ : 骨骼发育不良
bone fragility /bəʊn frəˈdʒɪlɪti/ : 骨脆性	collagen /ˈkɒlədʒən/ : 胶原蛋白
mutation /mjuːˈteɪʃən/ : 突变	bone mineralization /bəʊn ˌmɪnərəlaɪˈzeɪʃən/ : 骨矿化
osteoblast differentiation /ˌɒstriəʊblɑːst ˌdɪfəˈrenʃiˈeɪʃən/ : 成骨细胞分化	recessive mutation /rɪˈsesɪv mjuːˈteɪʃən/ : 隐性突变
signaling pathway /ˈsɪɡnəlɪŋ ˈpɑːθweɪ/ : 信号通路	pathophysiology /ˌpæθəʊˌfɪzɪˈɒlədʒi/ : 病理生理学
genetics /dʒɪˈnetɪks/ : 遗传学	bone biology /bəʊn baɪˈɒlədʒi/ : 骨生物学

Impact of osteoporosis and osteoporosis medications on fracture healing: a narrative review.

骨质疏松症及骨质疏松药物对骨折愈合的影响：一项叙述性综述。

期刊: Osteoporosis international : a journal established as result of cooperation between the European Foundation for Osteoporosis and the National Osteoporosis Foundation of the USA

日期: 2024

Abstract

antiresorptive medications do not negatively affect fracture healing in humans. teriparatide may decrease time to fracture healing. romosozumab has not shown a beneficial effect on human fracture healing. BACKGROUND: fracture healing is a complex process. Uncertainty exists over the influence of osteoporosis and the medications used to treat it on fracture healing. METHODS: Narrative review authored by the members of the Fracture Working Group of the Committee of Scientific Advisors of the International osteoporosis Foundation (IOF), on behalf of the IOF and the Societe Internationale de Chirurgie Orthopedique et de Traumatologie (SICOT). RESULTS: fracture healing is a multistep process. Most fractures heal through a combination of intramembranous and endochondral ossification. radiographic imaging is important for evaluating fracture healing and for detecting delayed or non-union. The presence of callus formation, bridging trabeculae, and a decrease in the size of the fracture line over time are indicative of healing. Imaging must be combined with clinical parameters and patient-reported outcomes. Animal data support a negative effect of osteoporosis on fracture healing; however, clinical data do not appear to corroborate with this. Evidence does not support a delay in the initiation of antiresorptive therapy following acute fragility fractures. There is no reason for suspension of osteoporosis medication at the time of fracture if the person is already on treatment. teriparatide treatment may shorten fracture healing time at certain sites such as distal radius; however, it does not prevent non-union or influence union rate. The positive effect on fracture healing that romosozumab has demonstrated in animals has not been observed in humans. CONCLUSION: Overall, there appears to be no deleterious effect of osteoporosis medications on fracture healing. The benefit of treating osteoporosis and the urgent necessity to mitigate imminent refracture risk after a fracture should be given prime consideration. It is imperative that new radiological and biological markers of fracture healing be identified. It is also important to synthesize clinical and basic science methodologies to assess fracture healing, so that a convergence of the two frameworks can be achieved.

中文翻译

抗骨吸收药物不会对人类骨折愈合产生负面影响。特立帕肽可能缩短骨折愈合时间。罗莫索单抗尚未显示出对人类骨折愈合的有利作用。背景：骨折愈合是一个复杂的过程。骨质疏松症及其治疗药物对骨折愈合的影响尚不确定。方法：由国际骨质疏松基金会（IOF）科学顾问委员会骨折工作组成员代表IOF和国际骨科与创伤外科学会（SICOT）撰写的叙述性综述。结果：骨折愈合是一个多步骤过程。大多数骨折通过膜内骨化和软骨内骨化相结合的方式愈合。影像学检查对于评估骨折愈合以及检测延迟愈合或不愈合至关重要。骨痂形成、骨小梁桥接以及骨折线随时间缩小均表明愈合。影像学必须与临床参数和患者报告结局相结合。动物数据支持骨质疏松症对骨折愈合有负面影响；然而，临床数据似乎并未证实这一点。证据不支持在急性脆性骨折后延迟启动抗骨吸收治疗。如果患者已在接受治疗，则没有理由在骨折时停用骨质疏松症药物。特立帕肽治疗可能缩短某些部位（如桡骨远端）的骨折愈合时间；然而，它不能预防不愈合或影响愈合率。罗莫索单抗在动物中表现出的对骨折愈合的积极作用尚未在人类中观察到。结论：总体而言，骨质疏松症药物似乎对骨折愈合没有不利影响。治疗骨质疏松症的益处以及骨折后降低即刻再骨折风险的迫切性应作为首要考虑因素。必须识别新的骨折愈合放射学和生物学标志物。同时，综合临床和基础科学方法来评估骨折愈合也很重要，以便实现两个框架的融合。

核心词汇

fracture healing /'fræktʃər 'hi:lɪŋ/ : 骨折愈合	osteoporosis /,ɒstiəʊpə'rousiːs/ : 骨质疏松症
antiresorptive /,æntɪrɪ'sɔ:rptɪv/ : 抗骨吸收的	teriparatide /,tɛrɪ'pærətɑɪd/ : 特立帕肽
romosozumab /,roumə'souzumæb/ : 罗莫索单抗	callus formation /'kæləs fɔ:r'meɪʃən/ : 骨痂形成
trabeculae /trə'bekjuli:/ : 骨小梁	intramembranous ossification /,ɪntrə'membrənəs ,ɒsɪfɪ'keɪʃən/ : 膜内成骨
endochondral ossification /,endəʊ'kɒndrəl ,ɒsɪfɪ'keɪʃən/ : 软骨内成骨	radiographic imaging /,reɪdɪəʊ'græfɪk 'ɪmɪdʒɪŋ/ : 放射影像学
fragility fracture /frə'dʒɪləti 'fræktʃər/ : 脆性骨折	

Pharmacologic Treatment of Primary Osteoporosis or Low Bone Mass to Prevent Fractures in Adults: A Living Clinical Guideline From the American College of Physicians.

成人原发性骨质疏松或低骨量的药物治疗以预防骨折：美国医师学会的活体临床指南

期刊: Annals of internal medicine 日期: 2023

Abstract

DESCRIPTION: This guideline updates the 2017 American College of Physicians (ACP) recommendations on **pharmacologic treatment** of primary **osteoporosis** or low bone mass to prevent fractures in adults. **METHODS:** The ACP Clinical Guidelines Committee based these recommendations on an updated systematic review of evidence and graded them using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) system. **AUDIENCE AND PATIENT POPULATION:** The audience for this guideline includes all clinicians. The patient population includes adults with primary **osteoporosis** or low bone mass. **RECOMMENDATION 1A:** ACP recommends that clinicians use bisphosphonates for initial **pharmacologic treatment** to reduce the risk of fractures in **postmenopausal** females diagnosed with primary **osteoporosis** (strong recommendation; high-certainty evidence). **RECOMMENDATION 1B:** ACP suggests that clinicians use bisphosphonates for initial **pharmacologic treatment** to reduce the risk of fractures in males diagnosed with primary **osteoporosis** (conditional recommendation; low-certainty evidence). **RECOMMENDATION 2A:** ACP suggests that clinicians use the RANK ligand inhibitor (**denosumab**) as a second-line **pharmacologic treatment** to reduce the risk of fractures in **postmenopausal** females diagnosed with primary **osteoporosis** who have contraindications to or experience adverse effects of bisphosphonates (conditional recommendation; moderate-certainty evidence). **RECOMMENDATION 2B:** ACP suggests that clinicians use the RANK ligand inhibitor (**denosumab**) as a second-line **pharmacologic treatment** to reduce the risk of fractures in males diagnosed with primary **osteoporosis** who have contraindications to or experience adverse effects of bisphosphonates (conditional recommendation; low-certainty evidence). **RECOMMENDATION 3:** ACP suggests that clinicians use the **sclerostin inhibitor** (romosozumab, moderate-certainty evidence) or recombinant PTH (**teriparatide**, low-certainty evidence), followed by a **bisphosphonate**, to reduce the risk of fractures only in females with primary **osteoporosis** with very high risk of **fracture** (conditional recommendation). **RECOMMENDATION 4:** ACP suggests that clinicians take an individualized approach regarding whether to start **pharmacologic treatment** with a **bisphosphonate** in females over the age of 65 with low bone mass (**osteopenia**) to reduce the risk of fractures (conditional recommendation; low-certainty evidence).

中文翻译

描述：本指南更新了2017年美国内科医师学会（ACP）关于原发性骨质疏松症或低骨量成人骨折预防的药物治疗建议。
方法：ACP临床指南委员会基于最新系统证据综述制定建议，并采用GRADE（建议评估、制定与分级系统）进行分级。
受众与患者群体：本指南受众为全体临床医生，患者群体包括原发性骨质疏松症或低骨量成人。
建议1A：ACP推荐临床医生使用双膦酸盐类药物作为初始药物治疗，以降低确诊原发性骨质疏松症的绝经后女性骨折风险（强推荐；高确定性证据）。
建议1B：ACP建议临床医生使用双膦酸盐类药物作为初始药物治疗，以降低确诊原发性骨质疏松症的男性骨折风险（条件性推荐；低确定性证据）。
建议2A：ACP建议临床医生对存在双膦酸盐禁忌症或不良反应的确诊原发性骨质疏松症绝经后女性，使用RANK配体抑制剂（地舒单抗）作为二线药物治疗以降低骨折风险（条件性推荐；中等确定性证据）。
建议2B：ACP建议临床医生对存在双膦酸盐禁忌症或不良反应的确诊原发性骨质疏松症男性，使用RANK配体抑制剂（地舒单抗）作为二线药物治疗以降低骨折风险（条件性推荐；低确定性证据）。
建议3：ACP建议临床医生仅对骨折风险极高的原发性骨质疏松症女性，使用硬骨抑制素抑制剂（罗莫索单抗，中等确定性证据）或重组甲状旁腺激素（特立帕肽，低确定性证据）后序贯双膦酸盐治疗以降低骨折风险（条件性推荐）。
建议4：ACP建议临床医生对65岁以上低骨量（骨量减少）女性，采取个体化决策是否启动双膦酸盐药物治疗以降低骨折风险（条件性推荐；低确定性证据）。

核心词汇

osteoporosis /ˌɒstiəpəˈrəʊsɪs/：骨质疏松症

bisphosphonate /ˈbaɪsˈfɒsfəneɪt/：双膦酸盐

sclerostin inhibitor /ˈskliərəstɪn ɪnˈhɪbɪtər/：硬骨素抑制剂

osteopenia /ˌɒstiəˈpiːniə/：骨量减少

contraindication /ˌkɒntrəˌɪndɪˈkeɪʃən/：禁忌症

fracture /ˈfræktʃər/：骨折

denosumab /deˈnəʊsəməb/：地舒单抗

teriparatide /ˌtɛrɪˈpærətɪd/：特立帕肽

pharmacologic treatment /ˌfɑːrməkəˈlɒdʒɪk ˈtriːtmənt/：药物治疗

adverse effect /ædˈvɜːrs ɪˈfekt/：不良反应

fracture risk /'fræktʃər rɪsk/ : 骨折风险

postmenopausal /ˌpoustmene'pɔ:zəl/ : 绝经后的

云梦山-侵权必究

Recent advances in fibroblast growth factor 23-related hypophosphatemic disorders.

成纤维细胞生长因子23相关低磷血症疾病的最新进展。

期刊: Current opinion in endocrinology, diabetes, and obesity 日期: 2024

Abstract

PURPOSE OF REVIEW: Fibroblast growth factor 23 (FGF23) is a **hormone** to reduce blood **phosphate** concentration. Excessive actions of FGF23 induce FGF23-related **hypophosphatemic disorders**, such as X-linked hypophosphatemic **rickets** (XLH) and tumor-induced **osteomalacia** (TIO). We will summarize recent advances in the **diagnosis** and **treatment** of FGF23-related **hypophosphatemic disorders**. RECENT FINDINGS: The measurement of blood FGF23 is useful to make a **diagnosis** of FGF23-related **hypophosphatemic disorders**. It was reported that many patients with FGF23-related **hypophosphatemic disorders**, especially TIO, were misdiagnosed, therefore, it is necessary to enhance the awareness of these diseases. A novel method to inhibit excessive actions of FGF23 by a human **monoclonal antibody** for FGF23, burosumab, has been approved in several countries. In more long-term observation than clinical trials, burosumab has also been shown to improve **biochemical abnormalities** and symptoms of **rickets/osteomalacia**. Following these advances, several registries and **consensus recommendations** on FGF23-related **hypophosphatemic disorders**, especially XLH, have been established in each country or region. SUMMARY: Further long-term effects of burosumab and the precise mechanism of FGF23 **overproduction** in patients with FGF23-related **hypophosphatemic disorders** need to be clarified in the future studies.

中文翻译

综述目的：成纤维细胞生长因子23（FGF23）是一种降低血磷浓度的激素。FGF23过度作用可导致FGF23相关性低磷血症性疾病，如X连锁低磷性佝偻病（XLH）和肿瘤性骨软化症（TIO）。本文将总结FGF23相关性低磷血症性疾病诊断与治疗的最新进展。最新发现：血液FGF23检测有助于诊断FGF23相关性低磷血症性疾病。研究报道许多FGF23相关性低磷血症性疾病患者（尤其是TIO患者）曾被误诊，因此需提高对此类疾病的认识。通过人源化FGF23单克隆抗体布洛索单抗抑制FGF23过度作用的新方法已在多国获批。相较于临床试验，更长期的观察显示布洛索单抗可改善生化异常及佝偻病/骨软化症症状。基于这些进展，各国或地区已建立针对FGF23相关性低磷血症性疾病（尤其是XLH）的登记系统和共识建议。总结：布洛索单抗的远期疗效及FGF23相关性低磷血症性疾病患者FGF23过度产生的精确机制仍需未来研究阐明。

核心词汇

hypophosphatemic disorders /ˌhaɪpəʊˈfæːsfeɪˈtiːmɪk dɪsˈɔːrdəz/ : 低磷血症性疾病	rickets /ˈrɪkɪts/ : 佝偻病
osteomalacia /ˌɒstiəʊməˈleɪʃə/ : 骨软化症	monoclonal antibody /ˌmɒnəˈklaʊnəl ˈæntɪˌbɒdi/ : 单克隆抗体
diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断	treatment /ˈtriːtmənt/ : 治疗
hormone /ˈhɔːrmoun/ : 激素	phosphate /ˈfɑːsfeɪt/ : 磷酸盐
biochemical abnormalities /ˌbaɪəʊˈkɛmɪkəl ˌæbnɔːrˈmælɪtiz/ : 生化异常	consensus recommendations /ˌkənˈsɛnsəs ˌrɛkəməˈnɛɪʃənz/ : 共识推荐
overproduction /ˌoʊvəprəˈdʌkʃən/ : 过度产生	